

LIVER AND BILE DUCTS

The normal adult liver weighs 1400 to 1600 g. It has a dual blood supply, with the portal vein providing 60% to 70% of hepatic blood flow and the hepatic artery supplying the remaining 30% to 40%. The portal vein and the hepatic artery enter the inferior aspect of the liver through the hilum, or *porta hepatis*. Within the liver, the branches of the portal veins, hepatic arteries, and bile ducts travel in parallel within portal tracts.

The lobule model is one useful way to consider the anatomic organization of the liver (Fig. 18.1). In this model, the liver is divided into 1- to 2-mm hexagonal lobules that are oriented around the terminal tributaries of the hepatic vein. The veins lie at the center of the lobules and hence are termed central veins, and portal tracts are located at the periphery. Each lobule can be further subdivided into six triangular acini. The hepatocytes in the vicinity of the central vein are called *pericentral* and are said to be in “zone 3,” while those near the portal tract are *periportal* and belong to “zone 1.” Notably, certain types of hepatic injury preferentially affect hepatocytes in particular zones (see Fig. 18.1). These differing sensitivities result from variation in hepatocyte oxygenation (highest in zone 1, lowest in zone 3) and metabolic activities from the periphery to the center of the lobule.

Within the lobule, hepatocytes are organized into anastomosing sheets or “plates” extending from portal tracts to the terminal hepatic veins. Between the trabecular plates of hepatocytes are vascular sinusoids. Blood traverses the sinusoids and exits into the terminal hepatic veins through numerous orifices in the vein wall. Hepatocytes are thus bathed on two sides by a mixture of portal venous and hepatic arterial blood. The sinusoids are lined by fenestrated endothelium. Beneath the endothelial cells lies the *space of Disse*, into which protrude abundant hepatocyte microvilli. Scattered macrophages called *Kupffer cells* are attached to the luminal face of endothelial cells, and lipid-containing *hepatic stellate cells* are found in the space of Disse. Between abutting hepatocytes are bile canaliculi, which are channels 1 to 2 μm in diameter

formed by grooves in the plasma membranes of facing hepatocytes and separated from the vascular space by tight junctions. These channels drain into the *canals of Hering* that, in turn, connect to bile ductules. The ductules empty into the interlobular bile ducts within portal tracts. Large numbers of lymphocytes (mostly gamma delta T cells, but also natural killer [NK] cells) are also present in normal liver, mainly adjacent to biliary tracts and in liver sinusoids.

GENERAL FEATURES OF LIVER DISEASE

The liver is vulnerable to a wide variety of primary metabolic, toxic, microbial, circulatory, and neoplastic disorders. Hepatic damage also occurs secondary to other diseases, such as heart failure, disseminated cancer, and extrahepatic infections. The enormous functional reserve of the liver masks the clinical impact of mild liver damage, but with progression of diffuse disease or disruption of bile flow, the consequences of deranged liver function may become life threatening.

Apart from acute liver failure, liver disease is an insidious process in which symptoms of hepatic decompensation may first appear weeks, months, or many years after the onset of injury. The ebb and flow of hepatic injury may be imperceptible to the patient and detectable only by abnormal laboratory tests (Table 18.1).

Mechanisms of Injury and Repair

Hepatocyte and Parenchymal Responses

Injured or dysfunctional hepatocytes in a variety of disorders may demonstrate several potentially reversible morphologic changes. These include accumulation of fat (steatosis) and bilirubin (cholestasis), as well as ballooning of hepatocytes, a change marked by cell swelling, cytoplasmic clearing, and clumping of intermediate filaments, which when prominent, form *Mallory hyaline*. Ballooned hepatocytes are a hallmark of alcohol-associated liver disease or metabolic dysfunction–associated steatohepatitis (see later discussion), but may also occur with ischemic or toxic injury, or with cholestasis.

When injury is irreversible, hepatocytes may die by necrosis or apoptosis. In hepatocyte necrosis, ion imbalances due to defective plasma membrane transporter function cause water to flow into the cell, which swells and ruptures. Membrane abnormalities also lead to accumulation of intracellular calcium and several events that trigger mitochondrial dysfunction. Even before membrane rupture, loss of membrane integrity leads to the release of cytoplasmic constituents into the extracellular compartment, including substances that alert innate immune cells to an ongoing threat (damage-associated molecular patterns; see Chapter 3). The remnants of the necrotic cells are rapidly phagocytosed by macrophages, which tend to cluster and mark sites of hepatocyte necrosis (Fig. 18.2). This form of injury is the predominant mode of death in ischemic/hypoxic injury and a significant part of hepatic injury in the setting of oxidative stress.

Hepatocyte apoptosis is an active form of “programmed” cell death resulting in hepatocyte shrinkage, nuclear chromatin condensation (*pyknosis*), fragmentation (*karyorrhexis*), and cellular fragmentation into acidophilic *apoptotic bodies* (Chapter 2). Apoptotic hepatocytes were first clearly described in yellow fever by William Thomas Councilman and therefore have been referred to as *Councilman bodies*; while apoptosis occurs in many forms of liver disease, by convention this eponym is restricted to that disease. In other settings in which apoptotic hepatocytes are seen (e.g., acute and chronic hepatitis), the term *acidophil bodies* is used, due to their deeply eosinophilic staining characteristics (Fig. 18.3).

If parenchymal injury is widespread, it may lead to **confluent necrosis, a zonal loss of contiguous hepatocytes**. This may be seen in acute toxic or ischemic injuries or in severe viral or autoimmune

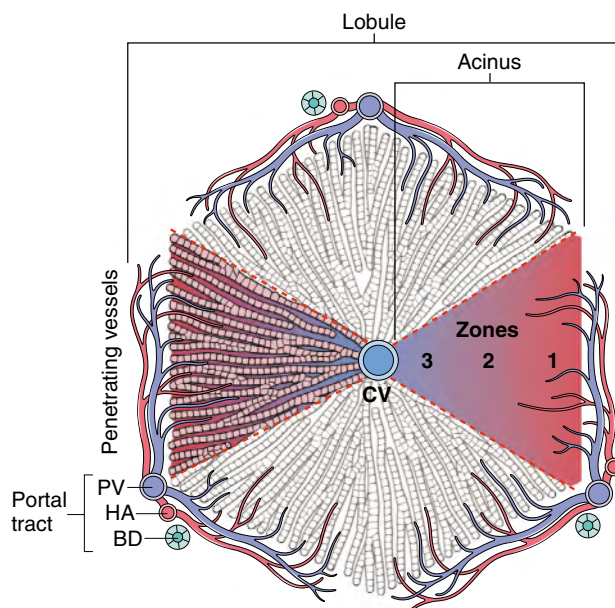
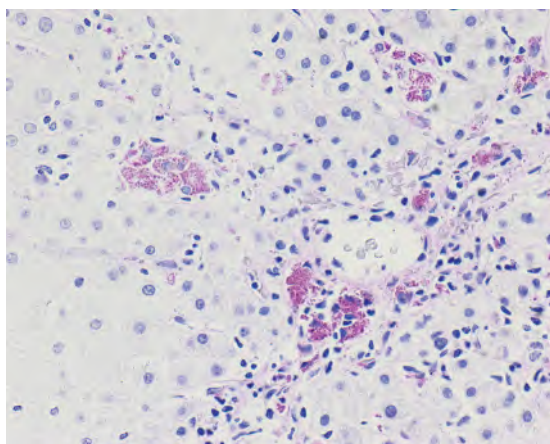
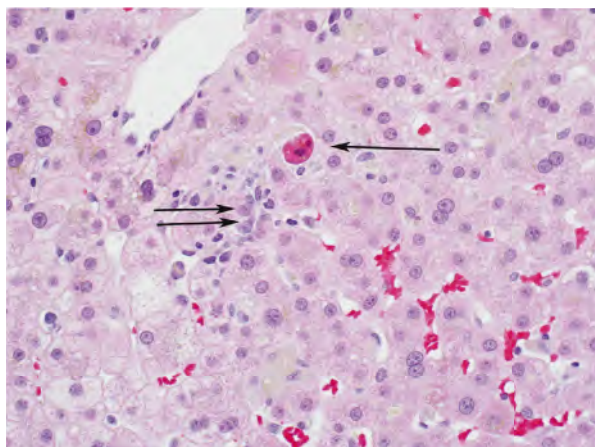


Fig. 18.1 Model of liver anatomy. In the hexagonal lobular model, the terminal hepatic vein (CV) is at the center of a “lobule,” while the portal tracts are at the periphery; these landmarks serve to identify “periportal” and “pericentral” parenchyma. In the acinar model, based on blood flow, three zones can be defined, zone 1 being the closest to the blood supply and zone 3 being the farthest. BD, Bile duct; HA, hepatic artery; PV, portal vein.

TABLE 18.1 Laboratory Evaluation of Liver Disease

Test Category	Serum Measurement	Expected Change in Disease
Hepatocyte integrity	Cytosolic hepatocellular enzymes Serum aspartate aminotransferase (AST) Serum alanine aminotransferase (ALT) Serum lactate dehydrogenase (LDH)	Increased
Biliary excretory function	Substances normally secreted in bile Serum bilirubin Total: unconjugated plus conjugated Direct: conjugated only Urine bilirubin Serum bile acids Plasma membrane enzymes (from damage to bile canaliculus) Serum alkaline phosphatase Serum γ -glutamyl transpeptidase (GGT)	Increased
Hepatocyte synthetic function	Proteins secreted into the blood Serum albumin	Decreased
	Coagulation factors Prothrombin time (PT) and partial thromboplastin time (PTT): fibrinogen, prothrombin, factors V, VII, IX, and X	Decreased
	Hepatocyte metabolism Serum ammonia	Increased
	Aminopyrine breath test (hepatic demethylation)	Decreased

**Fig. 18.2** Clusters of macrophages with periodic acid–Schiff-positive intracellular material derived from necrotic hepatocytes.**Fig. 18.3** Foci of lobular hepatitis in chronic hepatitis C show an apoptotic hepatocyte ("acidophil body," arrow) and a focus of mononuclear infiltration surrounding a more darkly stained, injured hepatocyte (double arrows).

hepatitis. Confluent necrosis may begin with hepatocyte dropout in zone 3 near the central vein. The resulting space is filled by cellular debris, macrophages, and remnants of the reticulin meshwork. In *bridging necrosis*, the area of necrosis extends from the central veins to the portal tracts, or across adjacent portal tracts (often with an inapparent central vein within the area of injury). In pan-acinar necrosis, the entire lobule is obliterated. Even in diseases such as viral hepatitis, in which hepatocytes are the principal targets of attack, secondary vascular insults—due to inflammation or thrombosis—may produce large areas of confluent hepatocyte necrosis. This process occurs in many types of liver disease in which there is extensive hepatocyte loss and collapse of the supporting reticulin framework. The resultant scarring (known in the liver as *cirrhosis*) in which hepatocytes are surrounded by fibrosis represents a common end point for chronic liver diseases.

Replacement of lost hepatocytes occurs primarily by replication of mature hepatocytes adjacent to those that have died, even when there is significant confluent necrosis. Hepatocytes are almost stem cell-like in their ability to replicate. Eventually, however, in individuals with chronic disease, hepatocytes enter replicative senescence; once this occurs, resident tissue stem cell populations may begin to expand and differentiate, an event marked by *ductular reactions*. These ductlike structures contain multipotent cells that may contribute significantly to parenchymal regeneration.

Scar Formation and Regression

The principal cell type involved in scar deposition in the liver is the hepatic stellate cell. When quiescent, the main function of stellate cells is lipid storage, including vitamin A. However, in several forms of acute and chronic injury, stellate cells become activated and differentiate into highly fibrogenic myofibroblasts (Fig. 18.4). The stimuli for stellate cell activation are varied and include: (1) inflammatory cytokines, such as tumor necrosis factor alpha (TNF- α), produced by Kupffer cells, macrophages, and other cell types; (2) altered interactions with extracellular matrix (ECM); and (3) toxins and reactive oxygen species (ROS). Following activation, the conversion of stellate cells into myofibroblasts is stimulated by signals transmitted by platelet-derived growth factor

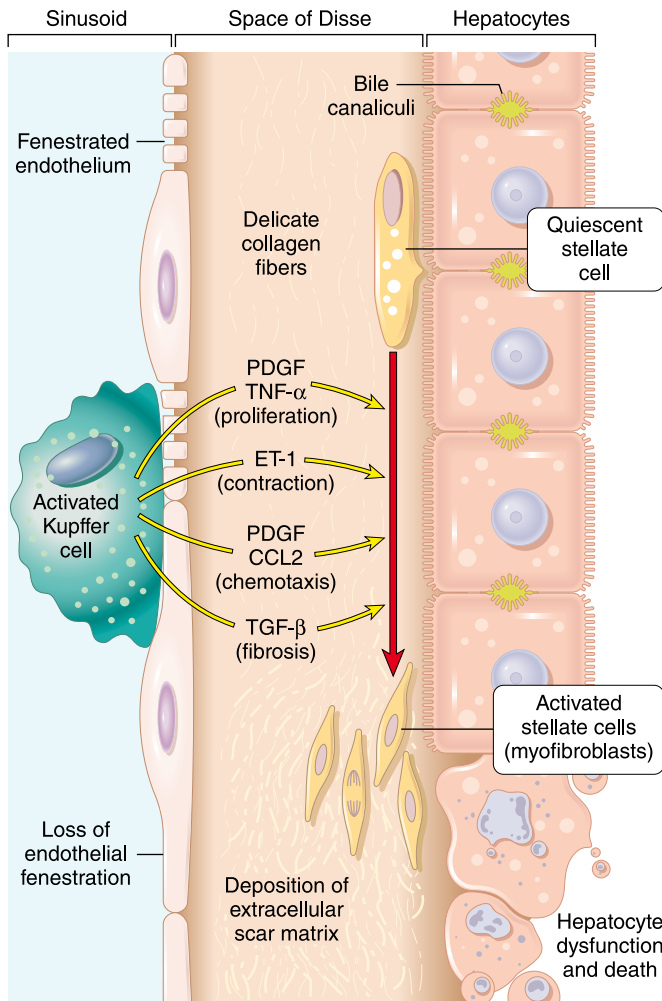


Fig. 18.4 Schematic of stellate cell activation. In several forms of acute and chronic injury, stellate cells become activated and differentiate into highly fibrogenic myofibroblasts. The conversion of stellate cells into myofibroblasts is stimulated by growth factors and cytokines, including platelet-derived growth factor (*PDGF*), and transforming growth factor β (*TGF β*). The activated stellate cells in turn also release cytokines, growth factors, and/or chemotactic and vasoactive factors. If the injury and inflammatory stimuli persist, then extracellular matrix deposition and scarring commence, often in the space of Disse. There is also concurrent loss of sinusoidal endothelial cell fenestration, a change termed *sinusoidal capillarization*. *ET*, Endothelin-1; *CCL2*, CC motif chemokine ligand 2; *TNF*, tumor necrosis factor.

receptor β and by cytokines, such as transforming growth factor β (*TGF- β*), as well as chemokines, which may be released from resident Kupffer cells or recruited macrophages and lymphocytes. The activated stellate cells in turn also release cytokines, growth factors, and/or chemotactic and vasoactive factors. If the injury and inflammatory stimuli persist, then extracellular matrix deposition and scarring commence, often in the space of Disse (see Fig. 18.4); there is also concurrent loss of sinusoidal endothelial cell fenestration, a change termed *sinusoidal capillarization* that is particularly prominent in metabolic dysfunction-associated steatohepatitis and also seen in hepatocellular carcinoma (both described later).

Areas of hepatocyte loss in chronic liver disease, perhaps related to vascular compromise, are replaced by dense fibrous septa through collapse of the underlying reticulin framework and deposition of collagen by myofibroblasts. Portal fibroblasts may also play a role in the

scarring that accompanies chronic liver injury in some disorders. Eventually, the fibrous septa encircle surviving hepatocytes and give rise to diffuse scarring (cirrhosis). In chronic liver disease, surviving hepatocytes replicate in an effort to restore the parenchyma, forming regenerative nodules that are a predominant feature in most cirrhotic livers.

If the chronic injury leading to scar formation is interrupted (e.g., upon clearance of hepatitis virus infection or cessation of alcohol use), then stellate cell activation and scarring ceases and fibrous septae may begin to be broken down by metalloproteinases produced by hepatocytes, leading to partial resolution and an appearance termed *incomplete septal cirrhosis*. Unfortunately, vascular remodeling and other architectural changes that occur in cirrhosis may not revert to normal, even with extensive scar resorption, which may explain why vascular abnormalities such as portal hypertension fail to improve in some patients.

Liver Failure

The most severe form of liver disease is liver failure, which may be acute (due to sudden and massive hepatic destruction) or (more commonly) chronic, following years of insidious, progressive liver injury. In some cases, individuals with chronic liver disease develop *acute-on-chronic liver failure*, in which an unrelated acute injury is superimposed on late-stage chronic disease or there is a “flare” of the chronic disease that leads to liver decompensation. Whatever the sequence, 80% to 90% of functional capacity must be lost before hepatic failure appears. When the liver can no longer maintain homeostasis, liver transplantation offers the best hope for survival; without transplantation, the mortality rate in persons with hepatic failure is about 80%.

Acute Liver Failure

Acute liver failure is defined as an acute liver illness associated with encephalopathy and coagulopathy that occurs within 26 weeks of the initial liver injury in the absence of preexisting liver disease. Usually, the syndrome manifests within 8 weeks of injury, and many patients progress to coma within only a week. Within this 26-week window, it is useful to know the interval between the onset of symptoms and liver failure since this may provide helpful clues to the etiology. Very rapid onset acute liver failure is most often induced by drugs or toxins and is typically the result of *massive hepatic* necrosis. This form of liver failure has been referred to as *fulminant liver failure* in the past, a term that remains entrenched in the literature and is still used interchangeably. Accidental or deliberate ingestion of acetaminophen (Chapter 9) accounts for almost 50% of adult cases in the United States, while autoimmune hepatitis, other drugs/toxins, and acute hepatitis A and B infections account for most remaining cases. In Asia, acute hepatitis B and E are the predominant causes. With acetaminophen toxicity, liver failure occurs within 1 week of the onset of symptoms, whereas liver failure due to hepatitis viruses takes longer to develop. The mechanism of hepatocellular necrosis may be direct toxic damage (as with acetaminophen), but more often is a combination of toxicity and immune-mediated hepatocyte destruction (e.g., hepatitis virus infection). Rare causes of acute liver failure include abnormalities of blood flow, metabolic disorders, and malignancies, most commonly leukemia or lymphoma (33%), followed by metastatic breast cancer (30%) and colon cancer (7%). An etiology cannot be established in about 15% of adult and 50% of pediatric cases.

MORPHOLOGY

Acute liver failure usually is associated with **massive hepatic necrosis**, with broad regions of parenchymal loss surrounding islands of preserved or regenerating hepatocytes (Fig. 18.5). Affected livers are small and shrunken. The prominence of hepatocellular dropout and of ductular reactions in these

livers depends on the nature and duration of the insult. Toxic injuries, such as acetaminophen overdoses, usually take place within hours to days, too brief a period to allow for scar formation or regeneration. Acute viral infections may cause failure over weeks to a few months, so that although hepatocyte injury continues to outpace repair, regeneration is often evident along with scarring.

Rarely, acute liver failure is associated with widespread dysfunction of liver cells without obvious cell death, such as in **diffuse microvesicular steatosis** related to fatty liver of pregnancy or in idiosyncratic reactions to toxins (e.g., valproate, tetracycline). In these settings, hepatocyte metabolism is severely affected, usually due to mitochondrial dysfunction, preventing the liver from carrying out its normal functions. Furthermore, non-hepatotropic viruses, particularly cytomegalovirus, herpes simplex viruses, and adenovirus, can cause acute liver failure, in states of immunodeficiency, such as untreated infection with human immunodeficiency virus (HIV), posttransplant immunosuppression, and certain lymphoid malignancies. With better treatments for HIV infection, these forms of viral hepatitis are declining in incidence.

Clinical Features

Acute liver failure manifests first with nausea, vomiting, and jaundice, followed by life-threatening encephalopathy and coagulation defects. Typically, serum liver transaminases are markedly elevated. The liver is initially enlarged due to hepatocyte swelling, inflammatory infiltrates, and edema; as parenchyma is destroyed, however, the liver

shrinks dramatically. Decline of serum transaminases (as hepatocytes die) is often not a sign of improvement but rather an indication that few viable hepatocytes remain; this suspicion is confirmed if jaundice, coagulopathy, and encephalopathy worsen as enzyme levels fall. With unabated progression, multiorgan system failure occurs and, if transplantation is not possible, death ensues. Other manifestations of acute liver failure are:

- **Cholestasis** occurs due to alterations of bile formation and flow, leading to retention of bilirubin and other solutes that are eliminated in bile. Yellow discoloration of the skin and conjunctiva (*jaundice* and *icterus*, respectively) occur with bilirubin retention, and there is increased risk of potentially life-threatening bacterial infection.
- **Hepatic encephalopathy** is a spectrum of disturbances in consciousness, ranging from subtle behavioral abnormalities, to marked confusion and stupor, to deep coma and death. Encephalopathy may progress over days, weeks, or months following acute liver injury. Associated fluctuating neurologic signs include rigidity and hyperreflexia. *Asterixis*, a particularly characteristic sign, is manifested as nonrhythmic, rapid extension-flexion movements of the head and extremities, best seen when the arms are held in extension with dorsiflexed wrists. Hepatic encephalopathy is believed to be caused by elevated ammonia levels, which correlate with impaired neuronal function and cerebral edema. The principal source of the ammonia is the gastrointestinal tract, where it is produced by microorganisms and enterocytes during glutamine metabolism. Normally, ammonia is transported in the portal vein to the liver, where it is metabolized in the urea cycle; in severe liver disease, this detoxification mechanism fails. Several mechanisms of CNS toxicity for ammonia have been proposed, including direct effects on neurons and indirect effects mediated through its metabolism in astrocytes to glutamine, which appears to contribute to edema by serving as an osmolyte.
- **Coagulopathy.** Hepatocytes are responsible for synthesis of clotting factors II (prothrombin), V, VII, IX, X, XI, and XII, as well as fibrinogen (Chapter 4). Thus, with liver failure, factor deficiencies and coagulopathy develop. Easy bruisability is an early sign, which can progress to life-threatening or fatal intracranial bleeding. The liver is also responsible for helping to remove activated coagulation factors from the circulation, and loss of this function in some instances leads to disseminated intravascular coagulation (Chapter 14), further exacerbating the bleeding tendency.
- **Portal hypertension** arises when there is diminished flow through the portal venous system, which may occur because of obstruction at the prehepatic, intrahepatic, or posthepatic level. Although it can occur in acute liver failure, portal hypertension is more commonly seen with chronic liver failure and is described later. In acute liver failure, if portal hypertension develops within days to weeks, obstruction is predominantly intrahepatic, and the major clinical consequence is *ascites*. In chronic liver disease, portal hypertension develops over months to years, and its effects are more complex and widespread (discussed later).
- **Hepatorenal syndrome** is a form of renal failure occurring in individuals with liver failure in whom there is no intrinsic morphologic or functional cause for kidney dysfunction. Its onset is marked by a drop in urine output and increasing levels of urea and creatinine in the blood. Its pathophysiology is incompletely understood, but the triggering event is hypothesized to be portal hypertension and secondary increased production of vasodilators such as nitric oxide by endothelial cells in the splanchnic vasculature. This in turn leads to systemic vasodilation and diminished renal perfusion, which is sensed by the kidney, provoking activation of

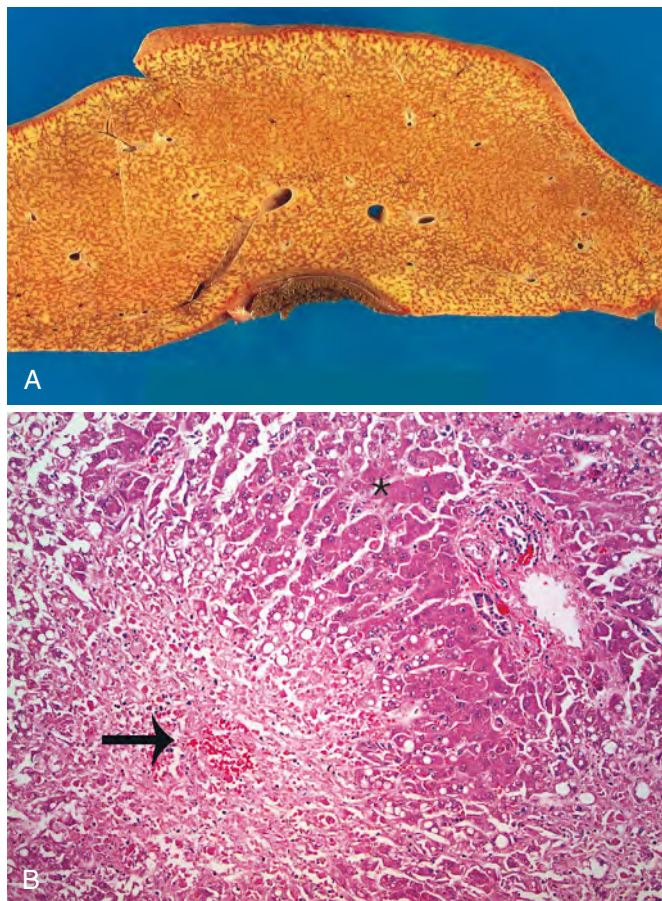


Fig. 18.5 (A) Massive necrosis, cut section of liver. The liver is small (700 g), bile-stained, soft, and congested. (B) Hepatocellular necrosis caused by acetaminophen overdose. Confluent necrosis is seen in the perivenular region (zone 3) (*large arrow*). Residual normal tissue is indicated by an *asterisk*. (Courtesy Dr. Matthew Yeh, University of Washington, Seattle.)

the renin/angiotensin axis. In the setting of portal hypertension and persistent vasodilator production, the principal effect of renin/angiotensin activation is to further diminish renal perfusion and glomerular filtration rate, leading to renal failure. The decline in renal function is reversible if liver function is restored, for example, by liver transplantation. Patients who develop hepatorenal syndrome usually have portal hypertension due to cirrhosis, severe alcoholic hepatitis, or (less often) metastatic tumors. However, patients with fulminant hepatic failure from any cause may develop hepatorenal syndrome.

Chronic Liver Failure and Cirrhosis

Liver failure in chronic liver disease is most often associated with advanced fibrosis/cirrhosis, a condition marked by diffuse remodeling of the liver into parenchymal nodules (often regenerative) surrounded by fibrous bands and a variable degree of vascular (often portosystemic) shunting. The leading causes of chronic liver failure worldwide are chronic hepatitis B, chronic hepatitis C, metabolic dysfunction–associated steatotic liver disease, and alcoholic-associated liver disease. In the United States, chronic liver disease is the twelfth most common cause of mortality and accounts for most liver-related deaths.

Although cirrhosis and chronic liver failure are often associated, they are not synonymous; not all cirrhosis leads inexorably to chronic liver failure, and not all end-stage chronic liver disease is cirrhotic. For example, chronic diseases such as primary biliary cholangitis, primary sclerosing cholangitis, nodular regenerative hyperplasia, chronic schistosomiasis, and fibropolycystic liver disease are often not accompanied by fully established cirrhosis, even at end stage. On the other hand, patients with well-treated autoimmune hepatitis or those with suppressed hepatitis B or cured hepatitis C often do not progress to end-stage liver disease, even if cirrhosis is present.

In some diseases that give rise to cirrhosis, such as untreated viral hepatitis, alcohol-associated liver disease, metabolic dysfunction–associated steatotic liver disease, and metabolic diseases, the morphology and pathophysiology of cirrhosis show some distinctive features (described in subsequent sections of this chapter). Thus, while the term cirrhosis implies the presence of severe chronic disease, it is not a specific diagnosis and has variable prognostic implications. There are also some instances in which cirrhosis arises without any clear cause; the term *cryptogenic cirrhosis* is sometimes applied to such cases.

MORPHOLOGY

Cirrhosis is marked by the presence of parenchymal nodules surrounded by dense bands of fibrosis throughout the liver, converting the normally smooth liver capsule into a bumpy surface with depressed areas of scarring and bulging regenerative nodules (Fig. 18.6). Microscopically, the extent of the fibrosis is highlighted using special stains for collagen (Fig. 18.7A). The size of the nodules, the pattern of scarring (linking portal tracts to each other versus linking portal tracts to central veins), the degree of parenchymal collapse, and the extent of vascular thrombosis (particularly of the portal vein) vary between diseases and, to some degree, between individuals with the same disease. Morphologic features of regression include thin incomplete scars (see Fig. 18.7B) associated with variable ductular reaction and architectural changes. As mentioned earlier, not all end-stage livers are cirrhotic, but all reveal varying degrees of injury, repair, regeneration, and fibrosis. Thus, end-stage liver disease is best viewed as a spectrum of ineffective attempts to repair ongoing, intermittent, or past injury.



Fig. 18.6 Cirrhosis resulting from chronic viral hepatitis. Note the depressed areas of dense scar separating bulging regenerative nodules over the liver surface.

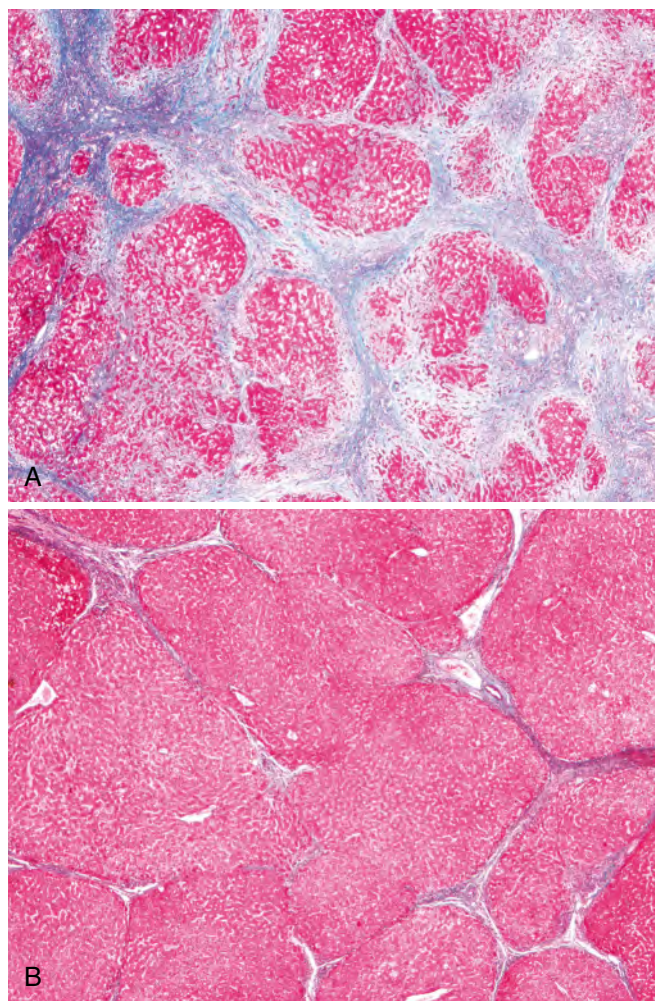


Fig. 18.7 Cirrhosis associated with alcohol overuse (A) and after long-term abstinence (B). (A) Thick bands of collagen separate rounded cirrhotic nodules. (B) After 1 year of abstinence, most scars are gone. (Masson trichrome stain). (Courtesy Drs. Hongfa Zhu and Isabel Fiel, Mount Sinai School of Medicine, New York.)

Clinical Features

About 40% of individuals with cirrhosis are asymptomatic until the most advanced stages of the disease. When symptomatic, they present with nonspecific manifestations: anorexia, weight loss, weakness, and, in advanced disease, symptoms and signs of liver failure discussed earlier. The common causes of death are hepatic encephalopathy, bleeding from esophageal varices, bacterial infections (resulting from damage to the gut mucosal barrier and Kupffer cell dysfunction), and hepatocellular carcinoma.

In addition to jaundice, encephalopathy, and coagulopathy (all also seen with acute liver failure), chronic liver failure is associated with several other significant features.

- *Persistent cholestasis can lead to pruritus* (itching), the intensity of which can be profound. Some patients scratch their skin raw, risking repeated bouts of potentially life-threatening infection. Relief may only be found after liver transplantation.
- *Impaired estrogen metabolism leads to hyperestrogenemia, which has several effects.* It produces vascular changes that may lead to palmar erythema (a reflection of local vasodilation) and spider angiomas of the skin. Each angioma is a central, pulsating, dilated arteriole from which small vessels radiate. In males, hyperestrogenemia may also produce hypogonadism and gynecomastia.
- *Hypogonadism can also occur in females due to disruption of hypothalamic-pituitary axis function,* either through nutritional deficiencies associated with the chronic liver disease or primary hormonal alterations.

Portal Hypertension

Increased resistance to portal blood flow may develop in a variety of circumstances, which can be divided into *prehepatic, intrahepatic, and posthepatic* (Table 18.2). The major prehepatic conditions are obstructive thrombosis, narrowing of the portal vein before it ramifies within the liver, or massive splenomegaly with increased splenic vein blood flow. The main posthepatic causes are severe right-sided heart failure, constrictive pericarditis, and hepatic vein outflow obstruction. The dominant intrahepatic cause is cirrhosis, accounting for most cases of portal hypertension. Far less frequent intrahepatic causes are schistosomiasis, massive fatty change, diffuse fibrosing granulomatous disease such as sarcoidosis, and diseases affecting the portal microcirculation such as nodular regenerative hyperplasia (discussed later).

TABLE 18.2 Location and Causes of Portal Hypertension

Prehepatic Causes
Obstructive thrombosis of portal vein
Structural abnormalities such as narrowing of the portal vein before it ramifies in the liver
Splenomegaly with increased splenic blood flow
Intrahepatic Causes
Cirrhosis from any cause
Nodular regenerative hyperplasia
Primary biliary cholangitis (even in the absence of cirrhosis)
Schistosomiasis
Massive fatty change
Diffuse, fibrosing granulomatous disease (e.g., sarcoid)
Infiltrative malignancy, primary or metastatic
Focal malignancy with invasion into portal vein (particularly hepatocellular carcinoma)
Amyloidosis
Posthepatic Causes
Severe right-sided heart failure
Constrictive pericarditis
Hepatic vein outflow obstruction

The pathophysiology of portal hypertension is complex and involves resistance to portal flow at the level of sinusoids and an increase in portal flow caused by hyperdynamic circulation. The increased resistance to portal flow at the level of the sinusoids is caused by contraction of vascular smooth muscle cells and myofibroblasts, and disruption of blood flow by scarring and the formation of parenchymal nodules. Alterations in sinusoidal endothelial cells that contribute to the intrahepatic vasoconstriction associated with portal hypertension include a decrease in nitric oxide (NO) production and increased release of endothelin-1, angiotensinogen, and eicosanoids. Sinusoidal remodeling and anastomoses between the arterial and portal system in the fibrous septa contribute to portal hypertension by imposing arterial pressures on the low-pressure portal venous system. Sinusoidal remodeling and intrahepatic shunts also interfere with the metabolic exchange between sinusoidal blood and hepatocytes.

Another major factor in the development of portal hypertension is an increase in portal venous blood flow resulting from a hyperdynamic circulation. This is caused by arterial vasodilation, primarily in the splanchnic circulation. The increased splanchnic arterial blood flow in turn leads to increased venous efflux into the portal venous system. While various mediators such as prostacyclin and TNF- α have been implicated in the causation of the splanchnic arterial vasodilation, NO has emerged as the most significant.

The four major consequences of portal hypertension are (1) hepatic encephalopathy (described under liver failure), (2) ascites, (3) the formation of portosystemic venous shunts, and (4) congestive splenomegaly. These are illustrated in Fig. 18.8 and are described next.

Ascites. The accumulation of fluid in the peritoneal cavity is called ascites, which in 85% of cases is caused by cirrhosis. Ascites usually becomes clinically detectable when at least 500 mL of fluid have accumulated. The fluid is generally serous, having less than 3 g/dL of protein (largely albumin), and a serum-to-ascites albumin difference of ≥ 1.1 g/dL. The fluid may contain a scant number of mesothelial cells and mononuclear leukocytes. The presence of neutrophils suggests infection, whereas the presence of red cells points to possible disseminated intraabdominal cancer. With longstanding ascites, seepage of peritoneal fluid through transdiaphragmatic lymphatics may produce hydrothorax, usually on the right side. The pathogenesis of ascites is complex, involving sinusoidal hypertension, hypoalbuminemia, increased hepatic lymph flow, and splanchnic vasodilation and hyperdynamic circulation.

Portosystemic Shunts. In the setting of chronic portal hypertension, vascular dilation and remodeling often leads to the development of thin-walled venous shunts between the portal and systemic circulations that bypass the liver. These shunts may appear wherever the systemic and portal circulations share capillary beds (see Fig. 18.8). The principal sites are veins around and within the rectum (manifest as hemorrhoids), the esophagogastric junction (producing varices), the retroperitoneum, and the falciform ligament of the liver (involving periumbilical and abdominal wall collaterals). Abdominal wall collaterals appear as dilated subcutaneous veins extending from the umbilicus toward the rib margins (caput medusae) and constitute a clinical hallmark of portal hypertension. Although hemorrhoidal bleeding may occur, it is rarely massive or life threatening. Much more important are esophagogastric varices, which appear in about 40% of individuals with advanced cirrhosis of the liver and cause massive hematemesis and death in about half of those affected. Each episode of bleeding is associated with a mortality rate of approximately 30%.

Splenomegaly. Longstanding portal hypertension may cause congestive splenomegaly. The degree of splenic enlargement varies widely,

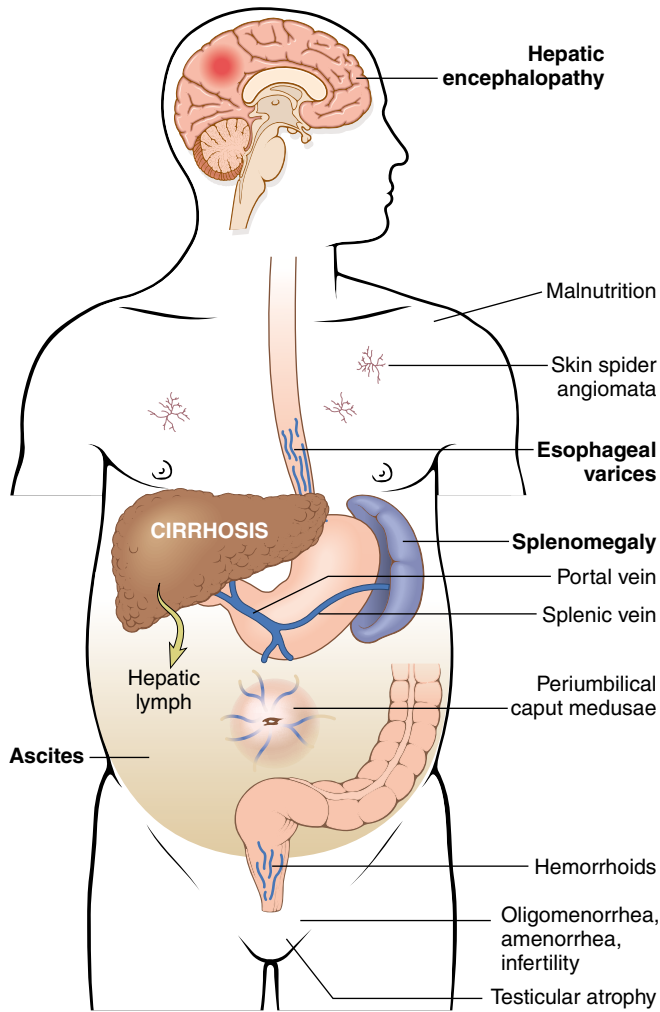


Fig. 18.8 Major clinical consequences of portal hypertension in the setting of cirrhosis, shown for the male. In females, oligomenorrhea, amenorrhea, and infertility as a result of hypogonadism are frequent. Clinically significant findings are in boldface.

and the splenic weight may reach as much as 1000 g (five- to six-times normal), but it is not necessarily correlated with other features of portal hypertension. Splenomegaly may secondarily induce hematologic abnormalities attributable to “hypersplenism,” particularly thrombocytopenia or even pancytopenia, in large part because of sequestration of blood elements in the expanded splenic red pulp.

Pulmonary Complications of Liver Failure and Portal Hypertension. We conclude this discussion with two pulmonary syndromes that occur in the setting of chronic liver failure and portal hypertension:

- **Hepatopulmonary syndrome** is seen in up to 30% patients with cirrhosis of the liver and portal hypertension. The syndrome is caused by the dilation of intrapulmonary capillary and precapillary vessels up to 500 μ M in size. Right-to-left shunting of blood through the dilated vessels produces ventilation-perfusion mismatch and impairs oxygenation of blood, leading to hypoxemia. The resultant dyspnea is worse in the upright position, because gravity exacerbates the ventilation-perfusion mismatch. The pathogenesis is unclear, although it has been postulated that the diseased liver may not clear factors such as endothelin-1 that stimulate endothelial cells to produce vasodilators such as nitric oxide.

- **Portopulmonary hypertension** refers to pulmonary arterial hypertension arising in liver disease. It also is not well understood, but it seems to depend on concomitant portal hypertension and excessive pulmonary vasoconstriction and vascular remodeling. The most common clinical manifestations are dyspnea on exertion and clubbing of the fingers.

Acute-on-Chronic Liver Failure

Some individuals with stable but well-compensated, advanced chronic liver disease suddenly develop signs of acute liver failure. In such patients, there is often established cirrhosis with extensive vascular shunting. Thus, large volumes of functioning liver parenchyma have a borderline vascular supply, leaving the liver highly vulnerable to superimposed insults. The short-term mortality of patients with this form of liver failure is around 50%.

Patients with chronic hepatitis B infection who become superinfected with hepatitis D may suddenly decompensate, as may patients with medically suppressed hepatitis B infection in whom viral mutants arise that are resistant to therapy, both due to acute flares of disease. Ascending cholangitis in a patient with primary sclerosing cholangitis or fibropolycystic liver disease (described later) may also cause rapid decompensation of liver function. Rare patients with metabolic dysfunction-associated steatohepatitis may develop severe hepatic dysfunction following rapid weight loss or malnutrition; the mechanism of injury in such cases is unknown.

Other causes of decompensation are systemic rather than intrahepatic insults. For example, sepsis and its attendant hypotension may undermine hepatic parenchyma with a borderline vascular supply. Likewise, acute cardiac failure or a superimposed drug or toxic injury might tip a well-compensated cirrhotic patient into failure. Finally, the possibility of malignancy, either a primary liver tumor, particularly hepatocellular carcinoma or cholangiocarcinoma, or liver metastases from some other site (e.g., colon), should always be kept in mind as a possible cause of acute liver failure in previously well-compensated chronic liver disease.

KEY CONCEPTS

Liver Failure

- Liver failure may follow acute injury or chronic injury but may also occur as an acute insult superimposed on an otherwise well-compensated chronic liver disease.
- The mnemonic for causes of acute liver failure is:
A: Acetaminophen, hepatitis A, autoimmune hepatitis
B: Hepatitis B
C: Hepatitis C, cryptogenic
D: Drugs/toxins, hepatitis D
E: Hepatitis E, esoteric causes (e.g., Wilson disease, Budd-Chiari syndrome, lymphoma, carcinoma)
F: Fatty change of the microvesicular type (e.g., fatty liver of pregnancy, valproate, tetracycline, Reye syndrome)
- Serious and sometimes fatal sequelae of liver failure include coagulopathy, encephalopathy, portal hypertension, bleeding esophageal varices, hepatorenal syndrome, and portopulmonary hypertension.

INFECTIOUS DISORDERS

Viral Hepatitis

The term “viral hepatitis” is most commonly used in the context of liver involvement by hepatotropic viruses, which include hepatitis A, B, C, D,

and E. Nonhepatotropic viruses such as Epstein-Barr virus, cytomegalovirus, herpes simplex virus, adenovirus, and yellow fever virus can also cause hepatitis, often in association with systemic infection. Liver involvement can occur in a wide variety of other systemic viral infections including SARS-CoV-2, but it is generally mild and often subclinical.

In the sections that follow, we will first discuss individual hepatotropic viruses, and then review certain clinicopathologic syndromes that are common to all.

Hepatitis A Virus

Hepatitis A virus (HAV) infection is a self-limited disease and does not lead to chronic hepatitis or a carrier state. It accounts for about 25% of clinically evident acute hepatitis worldwide. There has been a nearly five-fold surge in cases in the United States in the last few years with an estimated 5000 to 10,000 new cases per year.

Epidemiology. HAV is spread by ingestion of contaminated water and foods and is endemic in countries with poor hygiene and sanitation. In high resource countries, the prevalence of seropositivity (indicative of previous exposure) increases gradually with age, reaching 50% by 50 years of age in the United States. HAV is shed in the stool for 2 to 3 weeks before and 1 week after the onset of jaundice. Close personal contact with an infected individual or fecal-oral contamination during this period accounts for most cases and can result in outbreaks in institutional settings such as schools and nurseries. Water-borne epidemics can occur in places with overcrowded, unsanitary conditions. In high-income countries, sporadic infections may be contracted by the consumption of raw or steamed shellfish (oysters, mussels, clams), which concentrate the virus from seawater contaminated with human sewage. Sexual transmission may occur, but maternal-fetal transmission does not. Since HAV viremia is transient, blood-borne transmission of HAV is rare, and donated blood is not screened for this virus.

Pathogenesis

Discovered in 1973, HAV is a small, nonenveloped, positive-strand RNA picornavirus that occupies its own genus, *Hepatovirus*. Ultrastructurally, HAV is an icosahedral capsid that is 27 nm in diameter. The receptor for HAV on hepatocytes is HAVcr-1 (also known as TIM-1), a glycoprotein that is a member of a family of proteins that serve as receptors for several other viruses, such as dengue and ebola. HAV is not cytopathic, and the hepatocellular injury is inflicted by cytotoxic T lymphocytes and NK cells that recognize and kill hepatocytes infected by the virus.

Clinical Features

The incubation period for HAV is 2 to 6 weeks. IgM antibody against HAV appears with the onset of symptoms and persists for 3 to 6 months (Fig. 18.9). IgG anti-HAV appears during recovery from the acute infection and persists for years, conferring lifelong immunity against reinfection. Affected individuals have nonspecific symptoms such as fatigue and loss of appetite, and they often develop jaundice. Most patients recover within 3 months, and disease resolution occurs in nearly all patients by 6 months. Acute liver failure occurs in 0.1% to 0.3% of patients, especially those with chronic liver disease due to another etiology. Other uncommon complications include prolonged cholestasis and relapse of disease within 6 months of the original onset. Extrahepatic manifestations include rash; arthralgia; and immune complex-mediated complications like leukocytoclastic vasculitis, glomerulonephritis, and cryoglobulinemia. The HAV vaccine is effective in preventing infection.

Hepatitis B Virus

Hepatitis B virus (HBV) infection has varied clinical outcomes, which depend on the age of exposure, comorbid conditions (including exposure

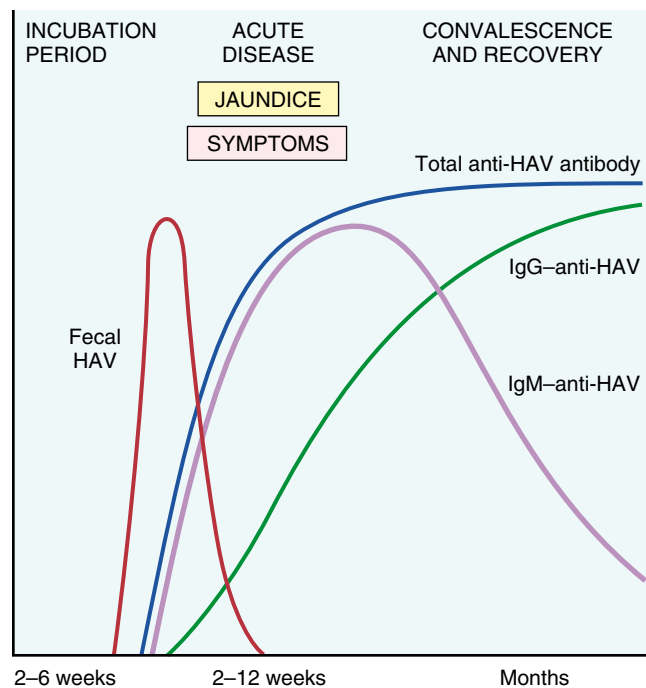


Fig. 18.9 Temporal changes in serologic markers in acute hepatitis A infection (HAV). IgA, Immunoglobulin A; IgM, immunoglobulin M.

to other infectious agents), and host immunity. **The major clinical consequences include (1) acute hepatitis followed by recovery and clearance of the virus, (2) acute hepatic failure with massive liver necrosis, (3) chronic hepatitis with or without progression to cirrhosis, and (4) an asymptomatic, “healthy” carrier state (Fig. 18.10).**

Epidemiology. According to WHO, 254 million people were living with chronic HBV infection in 2022, with 1.2 million new infections every year and 1.1 million deaths yearly. There are approximately 14,000 new cases of acute HBV infections in the US every year. The highest prevalence (>8%) is in Africa, Asia, and the Western Pacific rim, also relatively high (2% to 7%) in southern and eastern Europe, and lower (<2%) in western Europe, North America, and Australia. The virus has a parenteral mode of transmission through unprotected sex, blood transfusion, and sharing of needles and syringes for intravenous drug use. Transfusion-related transmission has been sharply curtailed by screening of donated blood for HBsAg and exclusion of paid blood donors. In high-prevalence regions, transmission during childbirth accounts for 90% of cases. Horizontal transmission can also occur in children through minor breaks in the skin or mucous membranes or with close bodily contact.

Pathogenesis

HBV was first linked to hepatitis in the 1960s when Australia antigen (now known as hepatitis B surface antigen) was identified. The virus is a member of the *Hepadnaviridae*, a family of DNA viruses that cause hepatitis in multiple animal species. The mature virion has an outer surface envelope composed of viral proteins and host-derived lipids that surrounds a core consisting of nucleocapsid proteins, viral polymerase, and viral DNA. The HBV genome is a partially double-stranded circular DNA molecule with several open reading frames encoding the following proteins:

- **Hepatitis B surface antigen (HBsAg)**, which refers to three related viral envelope glycoproteins, large, middle, and small HBsAg. Large HBsAg is usually associated with complete virions, while noninfectious

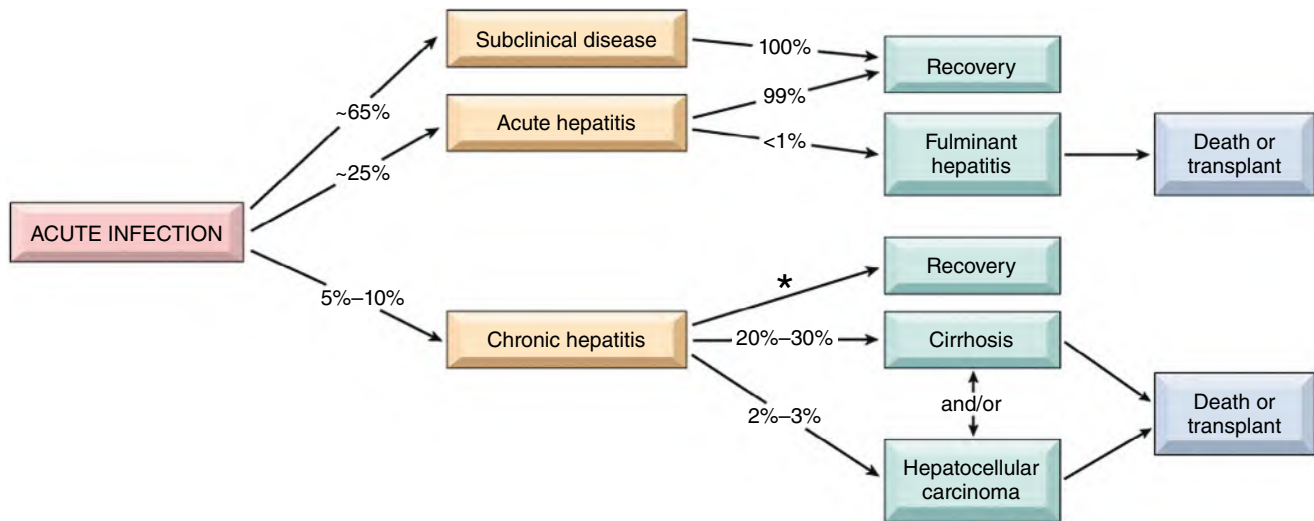


Fig. 18.10 Potential outcomes of hepatitis B infection in adults and their approximate frequencies in the United States. *Spontaneous hepatitis B surface antigen clearance occurs during chronic hepatitis B viral infection at an estimated annual incidence of 1% to 2% in Western countries.

viral envelope proteins (mainly small HBsAg) are released in large quantities by infected hepatocytes free of viral core elements.

- *Hepatitis B core antigen (HBcAg)*, the nucleocapsid protein, which plays a role in assembly of complete virions; and a longer polypeptide transcript with a precore and core region, designated *hepatitis Be antigen (HBeAg)*.
- *HBV polymerase (Pol)*, which exhibits both DNA polymerase and reverse transcriptase activities.
- *Hepatitis B X protein (HBx)*, which is not necessary for virus replication, but may act as a transcriptional activator of both viral genes and a subset of host genes. It has been implicated in the pathogenesis of HBV-related hepatocellular carcinoma.

Entry of HBV into hepatocytes is enabled by binding of large HBsAg to the bile salt transporter known as sodium taurocholate cotransporting polypeptide (NTCP). The viral genome enters the nucleus, where the plus strand is synthesized to form covalently closed circular DNA (ccc DNA). HBV replication occurs through reverse transcription via an RNA intermediate.

The host immune response to the virus is the main determinant of the outcome of the infection. High-level viral replication and high production of viral proteins can produce cytopathic changes in infected cells; however, most hepatocyte injury is caused by an attack of CD8+ cytotoxic T cells on infected cells. A strong response by virus-specific CD4+ and CD8+ interferon (IFN)- γ -producing cells is associated with resolution of acute infection.

Clinical Features

HBV has a prolonged incubation period (4 to 26 weeks). The course of the infection is followed by measuring viral antigens and serologic responses in the blood, as shown in Fig. 18.11.

- *HBsAg* appears before the onset of symptoms and peaks in acute, symptomatic disease. In those who clear the infection, it often declines to undetectable levels in 12 weeks, but can be present for as long as 24 weeks. By contrast, HBsAg persists in cases that progress to chronicity.
- *Anti-HBs antibody* begins to rise following the resolution of acute disease, generally after the disappearance of HBsAg (see

Fig. 18.11A). In some cases, the appearance of anti-HBs antibody is delayed until weeks to months after disappearance of HBsAg; in such instances, the serologic diagnosis can be made by detection of IgM anti-HBc antibody. Anti-HBs antibodies tend to persist for life, conferring protection; this is the basis for current vaccination strategies using HBsAg. By contrast, anti-HBs antibodies are not produced in cases that progress to chronic liver disease, which is also associated with persistent, but variable, elevations of serum transaminases in most (but not all) cases (see Fig. 18.11B).

- *HBeAg*, *HBV DNA*, and *HBV DNA polymerase* are detectable in the serum soon after HBsAg and signify active viral replication. Persistence of HBeAg is an important indicator of continued virus replication, infectivity, and probable progression to chronic hepatitis. One caveat is that mutated strains of HBV can emerge that do not produce HBeAg but are replication competent and express HBcAg. In such patients, the HBeAg may be low or undetectable despite the presence of serum HBV DNA.
- *Anti-HBe antibody* signifies that acute infection has peaked and is on the wane, whereas in cases that progress to chronic infection, anti-HBe antibody is not produced or appears only late in the disease course.

Acute HBV infection is mild or subclinical in nearly two-thirds of adults, while others have nonspecific constitutional symptoms such as anorexia, fever, jaundice, and upper right quadrant pain. Immune complex-mediated phenomena such as glomerulonephritis may occur, as well as other immunologic disorders, such as *polyarteritis nodosa* (Chapter 11). Acute liver failure is rare, occurring in approximately 0.1% to 0.5% of acutely infected individuals in 4 to 8 weeks. In most cases, the infection is self-limited and resolves without treatment. The infection persists and becomes chronic in 5% to 10% of infected individuals. The risk of chronic infection is inversely related to age and is greatest (approximately 90%) in infants who are exposed to the virus through transmission from their mothers at the time of birth. Individuals who acquire HBV at birth and who subsequently develop chronic hepatitis are also at greatest risk of developing hepatocellular carcinoma (see Fig. 18.10), which is one of the most common and deadliest cancers in parts of the world such as China where perinatal transmission occurs frequently.

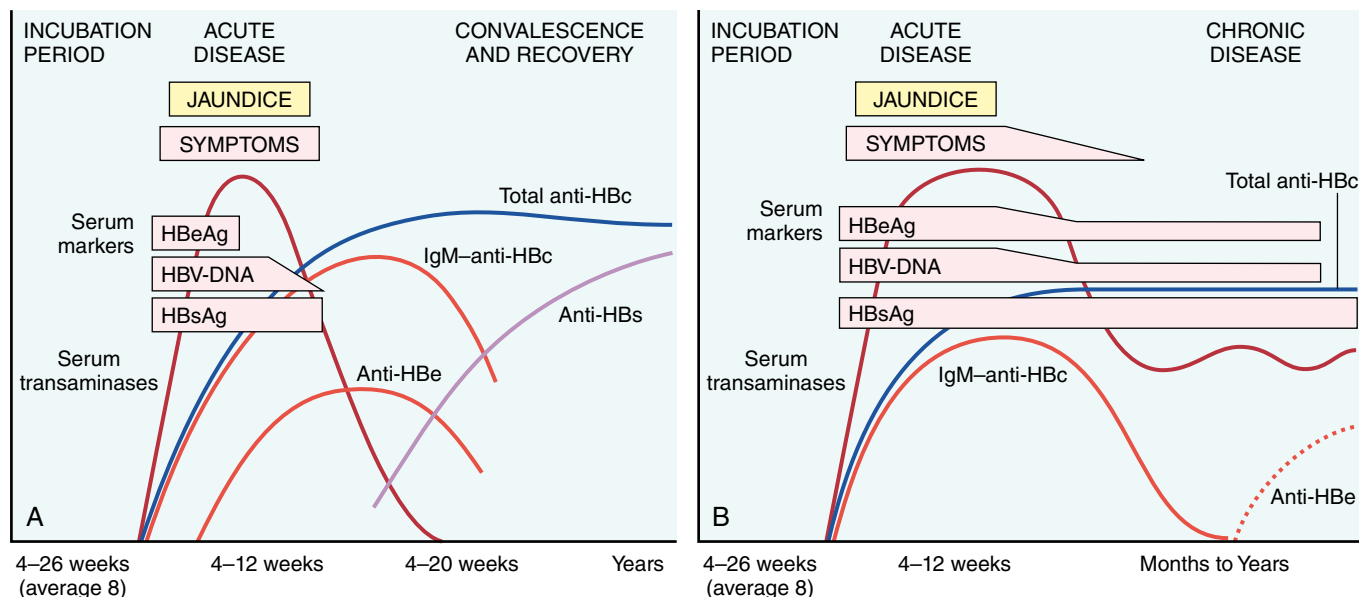


Fig. 18.11 Temporal changes in serologic markers in hepatitis B viral infection. (A) Acute infection with resolution. (B) Progression to chronic infection. In some cases of chronic hepatitis B virus, serum transaminases may become normal.

Vaccination is highly effective, as it induces a protective anti-HBs antibody response in 95% of infants, children, and adolescents. In those with chronic HBV, treatment with interferon and antiviral agents such as reverse transcriptase inhibitors (entecavir, tenofovir) can slow disease progression, reduce liver damage, and lower the risk of cirrhosis and hepatocellular carcinoma, but complete cure is difficult.

Hepatitis C Virus

Hepatitis C Virus (HCV) rarely causes symptomatic acute hepatitis but is the most common cause of chronic viral hepatitis.

Epidemiology. HCV affects approximately 50 to 60 million people worldwide, and there are 2.4 million chronic HCV cases in the United States. Screening of donor blood has decreased the annual incidence of infection from 230,000 new infections per year in the mid-1980s to 17,000 new infections per year currently. The mode of transmission is parenteral; intravenous drug abuse, multiple sexual partners, needle-stick injury, and multiple contacts with an HCV-infected person are the most common risk factors. Transfusion-related transmission has largely disappeared in the United States. Perinatal transmission can occur in children. One-third of infected individuals have no identifiable risk factors, an enduring hepatologic mystery.

Pathogenesis

HCV, discovered in 1989, is a member of the *Flaviviridae* family. It is a small, enveloped, single-stranded RNA virus with one open reading frame that codes for a single polypeptide, which is subsequently processed into several functional proteins. These include two envelope proteins, E1 and E2, and five core proteins: p7, NS2, NS3/4a (protease), NS5A (replication complex), and NS5B (RNA polymerase). The role of these proteins in the life cycle of HCV is summarized in Fig. 18.12.

The inability of the host immune response to eliminate HCV is related to rapid emergence of genetic variants, both within the population and within infected individuals. HCV RNA polymerase has low fidelity, giving rise to variants that fall into seven genotypes and many subtypes. As the disease progresses, multiple “personal”

genetic variants, known as *quasispecies*, emerge in infected individuals. Anti-HCV antibodies directed against E2 envelope protein are common and have neutralizing activity, but emergent variants with altered E2 epitopes escape from this host defense. NS3/NS4A protease also impairs the interferon-mediated cellular antiviral response, defeating another potential means of host defense.

Clinical Features

The incubation period for HCV hepatitis ranges from 4 to 26 weeks, with a mean of 9 weeks. HCV RNA is detectable in blood for 1 to 3 weeks in acute infection and coincides with an increase in transaminases. Anti-HCV antibodies emerge 3 to 6 weeks after infection. Spontaneous clearance of the virus after 4 to 6 months is rare; hence, persistent detection of HCV RNA after this period is an indicator of chronic HCV infection (Fig. 18.13). Assay for anti-HCV antibodies is used as a screening test for HCV, but it can be negative in early infection and in immunosuppressed individuals. After successful treatment, or in cases with spontaneous viral clearance, anti-HCV antibodies may be detected in the absence of HCV RNA, but may disappear with time.

Acute HCV infection is asymptomatic in about 85% of individuals, and severe acute hepatitis is rare. Except for a fortunate few, persistent infection and chronic hepatitis are the usual outcome. In contrast with HBV, chronic disease occurs in 80% to 90% of HCV-infected individuals, and approximately 20% of these individuals progress to cirrhosis over 20 to 30 years. Fluctuating elevations of serum aminotransferases are typically present during the chronic phase of the disease. Even those with normal transaminases are at risk for developing permanent liver damage, and hence all those with detectable serum HCV RNA levels need treatment and long-term follow-up. Older age, male sex, alcohol use, immunosuppressive drugs, hepatitis B/HIV co-infection, and diseases associated with insulin resistance, including obesity, type 2 diabetes, and metabolic syndrome have been associated with progression. Those who develop cirrhosis are at risk for development of hepatocellular carcinoma. Although the overall risk is small, in the United States, HCV is responsible for about one-third of the cases of liver cancer.

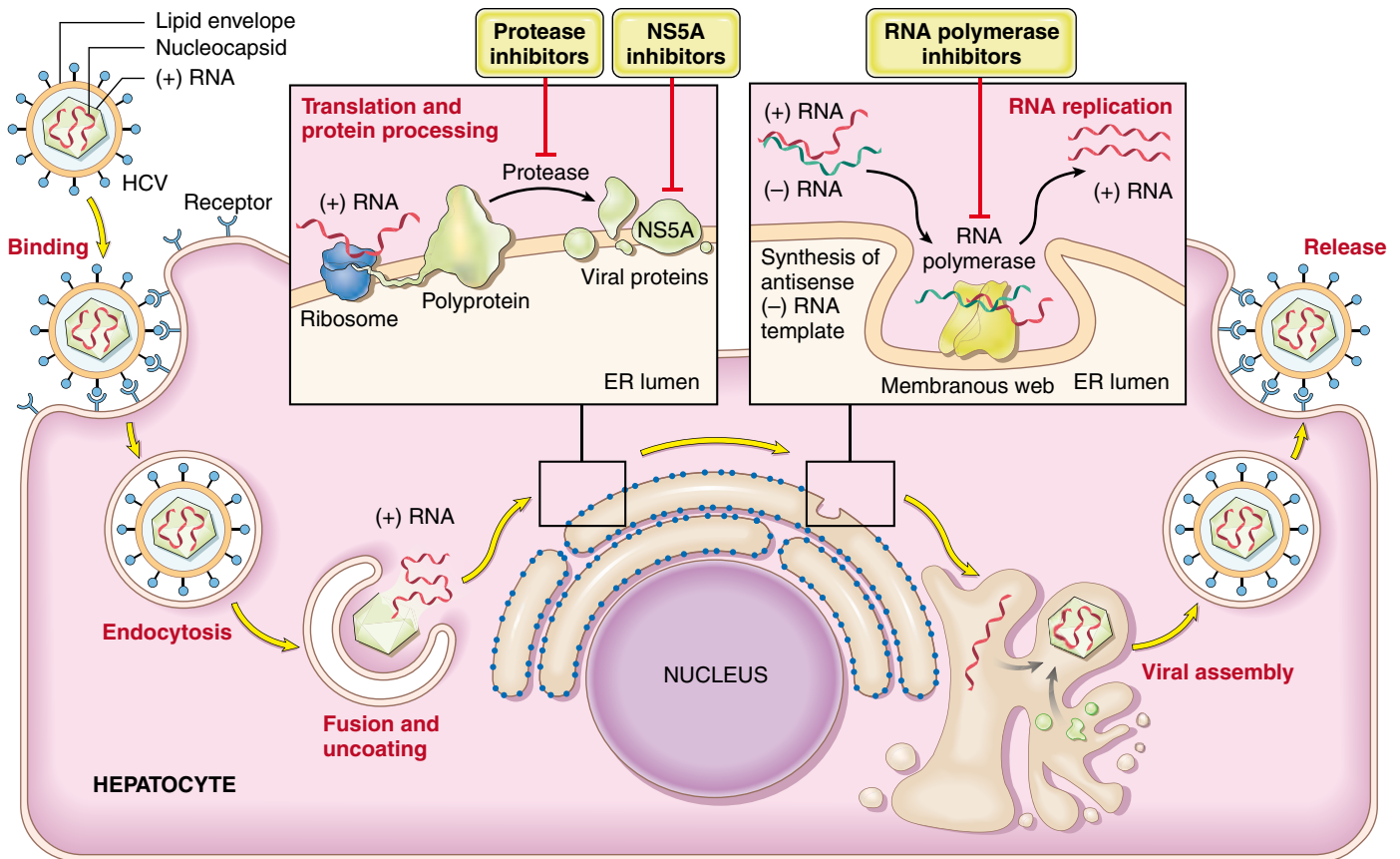


Fig. 18.12 Life cycle of hepatitis C virus (HCV). Viral entry, replication, assembly, and budding are shown, emphasizing steps that can be effectively targeted with antiviral drugs.

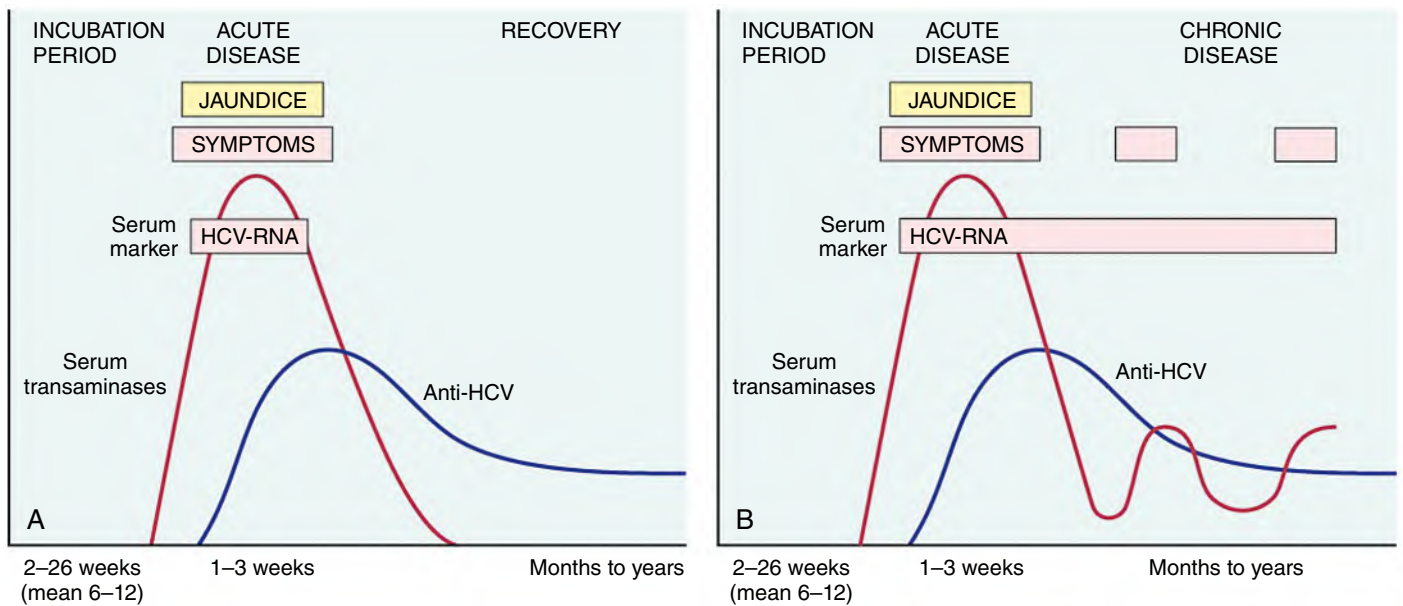


Fig. 18.13 Temporal changes in serologic markers in hepatitis C viral (HCV) infection. (A) Acute infection with resolution. (B) Progression to chronic infection.

Fortunately, the treatment of hepatitis C has been revolutionized in the past few years by the availability of highly effective antiviral agents. This remarkable advance has been made possible by understanding the role of viral proteins in the virus' life cycle, which in turn has enabled the development of specific drug antagonists (see Fig. 18.12). As in HIV infection, these drugs are used in combination to overcome resistance, and include agents that target the NS3/4a protease, the NS5A replication complex, and the NS5B polymerase. Unlike HIV, HCV does not integrate into the genome, and as a result true cures are often achieved. Sustained therapeutic response is defined as undetectable HCV RNA after 12 to 24 weeks of drug therapy; overall, such responses are obtained in 80% to 90% of cases, with lower response rates in patients infected with genotype 3 HCV. HCV appears to be cured in >99% of patients who achieve a sustained viral response.

Hepatitis D Virus

Hepatitis D virus (HDV) or "the delta agent," is a unique RNA virus that is dependent for its life cycle on HBV.

Epidemiology. It is estimated that 5% of HBV-infected individuals are co-infected with HDV, which amounts to approximately 15 million people worldwide. The highest prevalence is in the Amazon basin, central Africa, the Middle East, and the Mediterranean basin. HDV is uncommon in Southeast Asia and China. Disease transmission occurs through a parenteral route, often in association with intravenous drug use and multiple blood transfusions.

Pathogenesis

HDV, discovered in 1977, is a 35-nm, double-shelled particle. The external coat antigen of HDV surrounds an internal polypeptide assembly, designated delta antigen (HDAg), the only protein produced by the virus. Associated with the virus particle is a circular single-stranded RNA that is the smallest genome of any known animal virus. Replication of the virus is through RNA-directed RNA synthesis by host RNA polymerase. The basis for disease flares associated with HDV superinfection is not well understood; both cytopathic effects and enhanced cytotoxic host responses have been proposed.

Clinical Features

Infection with HDV arises in two settings, each with somewhat different clinical courses.

- **Co-infection** occurs following exposure to serum containing both HDV and HBV. Co-infection can result in acute hepatitis that is indistinguishable from acute hepatitis B. It is self-limited and is usually followed by clearance of both viruses. However, there is a higher rate of acute hepatic failure in intravenous drug users.
- **Superinfection** occurs when a chronic carrier of HBV is exposed to a new inoculum of HDV. This results in disease 30 to 50 days later, presenting either as severe acute hepatitis in an asymptomatic HBV carrier, or as an exacerbation of preexisting chronic hepatitis B infection. Chronic HDV infection occurs in more than 80% of superinfections and may have two phases, an acute phase with active HDV replication and suppression of HBV replication associated with high transaminase levels, and a chronic phase in which HDV replication decreases, HBV replication increases, and transaminase levels fluctuate.

HDV RNA is detectable in the blood and liver just before and in the early days of acute symptomatic disease. IgM anti-HDV antibody is the most reliable indicator of recent HDV exposure, although its appearance is late and frequently short-lived. With chronic delta hepatitis arising from HDV superinfection, HBsAg is present in serum, and anti-HDV antibodies persist for months or longer.

Co-infection with HDV and HBV increases the risk of progression to cirrhosis and hepatocellular carcinoma (HCC). Chronic HDV infection is treated with interferon- γ , but clearance of the virus occurs in only a minority of cases. Newer agents targeting viral entry and replication are in clinical trials. Vaccination for HBV also prevents HDV infection.

Hepatitis E Virus

Hepatitis E virus (HEV) is an enterically transmitted, water-borne infection that occurs primarily in young to middle-aged adults.

Epidemiology. HEV is a zoonotic disease with animal reservoirs, such as monkeys, cats, pigs, and dogs. Epidemics have been reported in Asia and the Indian subcontinent, sub-Saharan Africa, the Middle East, China, and Mexico. HEV infection accounts for more than 30% of cases of sporadic acute hepatitis in India, exceeding the frequency of HAV. Cases can occur in travelers to these regions and are also seen in high-income nations in association with pig farming or consumption of organ meat.

Pathogenesis

Discovered in 1983, HEV is an unenveloped, positive-stranded RNA virus in the genus *Orthohepevirus* of the Hepeviridae family. The RNA genome is 7.3 kb in length and contains four open reading frames that encode multiple proteins, including a viral protease and viral RNA polymerase. HEV is not cytopathic, and hepatic damage is thought to stem from the host response to cells infected with virus.

Clinical Features

Virions are shed in stool during the acute illness, and infection is typically by the fecal-oral route. The average incubation period following exposure is 4 to 5 weeks, followed by a self-limited acute hepatitis in most cases, with resolution in 2 to 4 weeks. Before the onset of clinical illness, HEV RNA and HEV virions can be detected by PCR in stool and serum. The onset of clinical symptoms, rise of serum aminotransferases, and appearance of IgM anti-HEV are virtually simultaneous. IgM is replaced with persistent IgG anti-HEV antibodies during recovery.

Chronic liver disease or persistent viremia in immunocompetent patients is not observed. However, acute HEV infection is associated with a mortality rate approaching 20% in pregnant females. Chronic HEV infection can occur in the setting of immunosuppression, such as in those with AIDS and in transplant recipients.

Clinicopathologic Syndromes of Viral Hepatitis

Table 18.3 provides a summary of the salient features of infection by various hepatitis viruses. Acute infections by all the hepatotropic viruses can either be asymptomatic or symptomatic. Although HBV and HCV are responsible for most cases of chronic hepatitis, many other disorders have similar clinicopathologic features, especially autoimmune and drug/toxin-induced hepatitis, described later. Therefore, serologic and molecular studies are essential for the diagnosis of viral hepatitis and for distinguishing among the various types.

Major features of the main clinicopathologic syndromes associated with hepatotropic virus infections are:

- **Acute asymptomatic infection with recovery.** Here, infection is identified due to minimally elevated serum transaminases or, after recovery, by the presence of antiviral antibodies. HAV and HBV infection can be subclinical events, verified only by the presence of anti-HAV or anti-HBV antibodies.
- **Acute symptomatic infection with recovery.** Symptomatic disease can be divided into four phases: (1) incubation period, (2) symptomatic preicteric phase, (3) symptomatic icteric phase, and (4) convalescence. The incubation period for the different viruses is given in Table 18.3. Peak infectivity occurs during the last

TABLE 18.3 Hepatitis Viruses

Virus	Hepatitis A	Hepatitis B	Hepatitis C	Hepatitis D	Hepatitis E
Type of virus	ssRNA	Partially dsDNA	ssRNA	Circular defective ssRNA	ssRNA
Route of transmission	Fecal-oral (contaminated food or water)	Parenteral, sexual contact, perinatal	Parenteral; intranasal cocaine use	Parenteral	Fecal-oral
Mean incubation period	2–6 weeks	2–26 weeks	4–26 weeks	Same as HBV	4–5 weeks
Frequency of chronic liver disease	Never	5%–10%	>80%	10% (co-infection); 90%–100% for superinfection	In immunocompromised hosts only
Diagnosis	Serum IgM antibodies	HBsAg or HBeAg antibodies; PCR for HBV DNA	ELISA for HCV antibodies; PCR for HCV RNA	Serum IgM and IgG antibodies; PCR for HDV RNA	Serum IgM and IgG antibodies; PCR for HEV RNA

dsDNA, Double-stranded DNA; ELISA, enzyme-linked immunosorbent assay; HBsAg, hepatitis B core antigen; HBsAg, hepatitis B surface antigen; HBV, hepatitis B virus; HCV, hepatitis C virus; HDAg, hepatitis D antigen; HDV, hepatitis D virus; HEV, hepatitis E virus; IgG, immunoglobulin G; IgM, immunoglobulin M; IV, intravenous; PCR, polymerase chain reaction; ssRNA, single-stranded RNA.

From Washington K. Inflammatory and infectious diseases of the liver. In: Iacobuzio-Donahue CA, Montgomery EA, eds. *Gastrointestinal and Liver Pathology*. Churchill Livingstone; 2005.

asymptomatic days of the incubation period and the early days of acute symptoms.

- **Acute liver failure.** Viral hepatitis accounts for approximately 10% of cases of acute hepatic failure. Hepatitis A and E are the most common causes worldwide, while HBV is more common in Asia and the Mediterranean. HEV can cause liver failure in pregnant females. The treatment is to provide supportive care and allow for replication of residual hepatocytes to lead to restitution of the liver. Liver transplantation is the only option if the disease does not resolve before secondary infection and failure of other organs develops.
- **Chronic hepatitis.** This is defined as symptomatic, biochemical, or serologic evidence of continuing or relapsing hepatic disease for more than 6 months and/or by morphologic presence of fibrosis. In some patients, persistent elevation of serum transaminases may be the only clinical evidence of chronicity. Prolongation of the prothrombin time, hyperglobulinemia, hyperbilirubinemia, and mild elevations in alkaline phosphatase level can occur. In symptomatic individuals, the most common finding is fatigue; less common symptoms are malaise, loss of appetite, and occasional bouts of mild jaundice. Immune complex disease can develop due to circulating antibody-antigen complexes in chronic HBV and HCV infection, and can manifest as vasculitis, glomerulonephritis, and cryoglobulinemia.
- **Carrier state.** A “carrier” is an individual who harbors and can transmit an organism, but has no symptoms. For hepatotropic viruses, carrier state has been used to describe two separate scenarios: (1) individuals who harbor the virus but have no liver disease; and (2) individuals who harbor the virus and have asymptomatic nonprogressive liver damage. In both cases, particularly the latter, affected individuals constitute reservoirs of infection. For HBV infection, the term “healthy carrier” has been used for an individual with HBsAg and anti-HBe, but without HBeAg. These patients have normal aminotransferases, low or undetectable serum HBV DNA, and lack of significant inflammation or liver injury on biopsy. For HCV, a state equivalent to the HBV “healthy carrier” is not recognized.

HIV is an important comorbid factor in hepatotropic viral infections. Co-infection of HIV and hepatitis viruses is common in the United States, as between 10% and 25% of HIV-infected individuals are also infected with HBV and HCV, respectively. Chronic HBV and HCV infection are leading causes of morbidity and mortality in HIV-infected individuals, although the severity and progression in immunocompetent HIV patients

is similar to those who are HIV negative. Untreated HIV infection significantly exacerbates the severity of liver disease caused by HBV or HCV.

MORPHOLOGY

The general morphologic features of viral hepatitis are depicted schematically in Fig. 18.14. The morphologic changes in acute and chronic viral hepatitis due to hepatotropic viruses are largely similar and overlap with those associated with autoimmune hepatitis, adverse drug reactions, and Wilson disease.

In acute viral hepatitis, the liver may be normal in size, enlarged (due to inflammation), or shrunken in those cases that are associated with acute liver failure and massive liver necrosis (see Fig. 18.5). Microscopically, there is a **portal and lobular inflammatory infiltrate** comprised predominantly of lymphocytes and variably admixed with plasma cells and eosinophils. The hepatocyte injury may result in necrosis or apoptosis (see Figs. 18.2 and 18.3), often with pigmented macrophages scavenging the dead cell debris. Necrosis of groups of hepatocytes (i.e., **confluent necrosis**) may be seen in severe cases and can progress to necrosis of the entire lobule (i.e., panlobular or panacinar necrosis) or connect vascular structures (i.e., **bridging necrosis**). Liver failure can develop with massive hepatic necrosis.

The defining histologic feature of chronic viral hepatitis is portal lymphocytic, or lymphoplasmacytic, inflammation with fibrosis. The inflammatory cells often cross the limiting plate and lead to injury of periportal hepatocytes (**interface activity**). This may be accompanied by a variable degree of lobular inflammation. Fibrosis develops with increasing liver damage manifesting initially as portal and periportal fibrosis followed by portoportal bridging fibrosis and eventually cirrhosis.

There are some morphologic features that are distinctive for particular subtypes of chronic viral hepatitis. In chronic hepatitis B, the endoplasmic reticulum of hepatocytes is swollen and filled with HBsAg, leading to a **“ground-glass” appearance**. Immunostaining for hepatitis B surface and core antigens can confirm the HBV infection (Fig. 18.15). Chronic hepatitis C typically shows prominent lymphoid aggregates or fully formed lymphoid follicles in the portal tracts (Fig. 18.16). Steatosis is common in chronic hepatitis C and can be marked with genotype 3 infection. Bile duct injury can be seen in hepatitis C, mimicking biliary disease, but it is typically a focal finding.

Liver biopsy is often performed to confirm the diagnosis of chronic hepatitis and to assess inflammatory activity (grade) and fibrosis (stage). Stage of the disease is used along with other clinical parameters to determine the therapeutic approach.

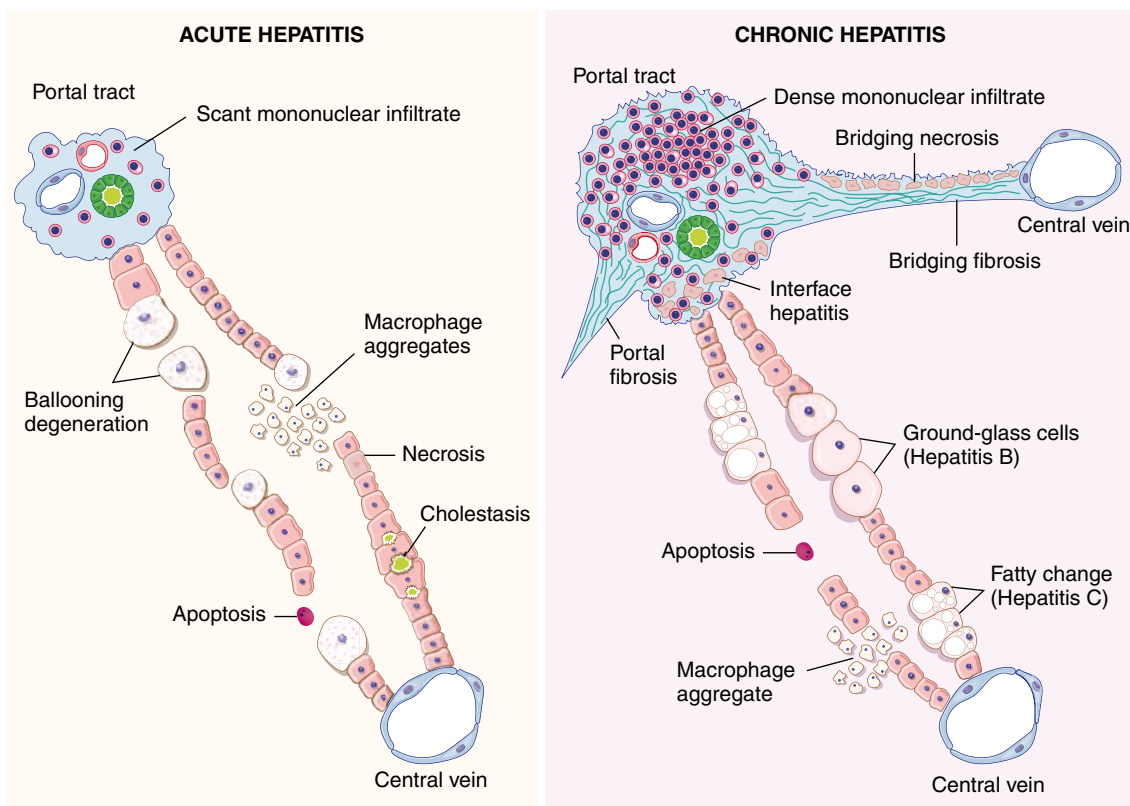


Fig. 18.14 Diagrammatic representation of the morphologic features of acute and chronic hepatitis. Acute hepatitis is characterized by lobular inflammation and hepatocellular injury, and chronic hepatitis shows dense portal inflammation. Bridging necrosis can occur in severe acute hepatitis, and fibrosis is seen in chronic hepatitis. Ductular reaction is often present in areas of fibrosis in chronic hepatitis.

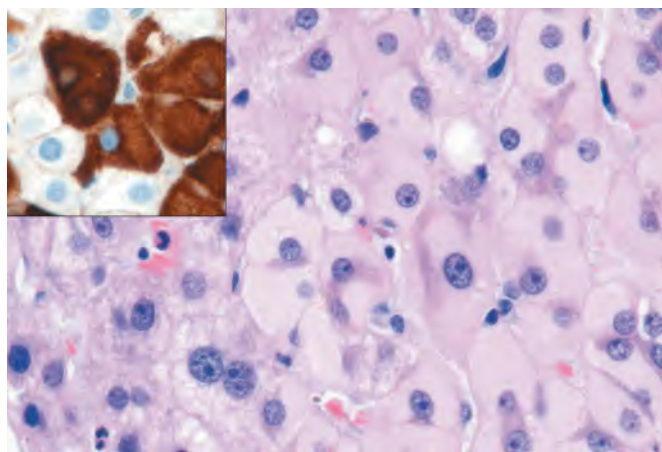


Fig. 18.15 Ground-glass hepatocytes in chronic hepatitis B caused by accumulation of hepatitis B surface antigen in the endoplasmic reticulum. The cytoplasmic inclusions are light pink and finely granular on hematoxylin and eosin staining; immunostaining (inset) confirms that they contain hepatitis B surface antigen (brown).

KEY CONCEPTS

Viral Hepatitis

- Hepatitis A is transmitted through the fecal-oral route, causes acute hepatitis, and does not lead to chronic liver disease.

- Hepatitis B is transmitted parenterally with most infections being subclinical, but can cause acute hepatitis, chronic hepatitis, and cirrhosis.
- Hepatitis C is most often associated with progression to chronic liver disease (80% or more), while acute infections are almost always subclinical.
- Hepatitis D is a defective virus, requiring hepatitis B co-infection for replication and infection.
- Hepatitis E is endemic in equatorial regions and frequently epidemic; it causes acute disease, which can be severe in pregnancy and can lead to chronic hepatitis in immunocompromised individuals.
- The inflammatory cells in both acute and chronic viral hepatitis are mainly T cells; the morphologic findings overlap with other hepatic disorders such as autoimmune hepatitis, drug-induced liver injury, and Wilson disease.
- Biopsy assessment in chronic viral hepatitis reveals the extent of fibrosis (stage), which can determine the therapeutic course of action.
- Patients with HBV or HCV are at increased risk for development of HCC, especially in the setting of cirrhosis.

Bacterial, Fungal, and Parasitic Infections

Bacteria, fungi, helminths, and other parasites/protozoa can involve the liver and biliary tree as localized infections or as part of a systemic disease. Examples of bacterial infections include *Staphylococcus aureus* in toxic shock syndrome, *Salmonella typhi* in typhoid fever, *Treponema pallidum* in secondary or tertiary syphilis, and *Bartonella henselae* in cat scratch disease. Biliary obstruction creates a milieu for bacterial proliferation leading to infection of the biliary tree, often called ascending cholangitis. When severe, the infection may extend to the liver and produce intrahepatic abscesses. Spread of bacteria

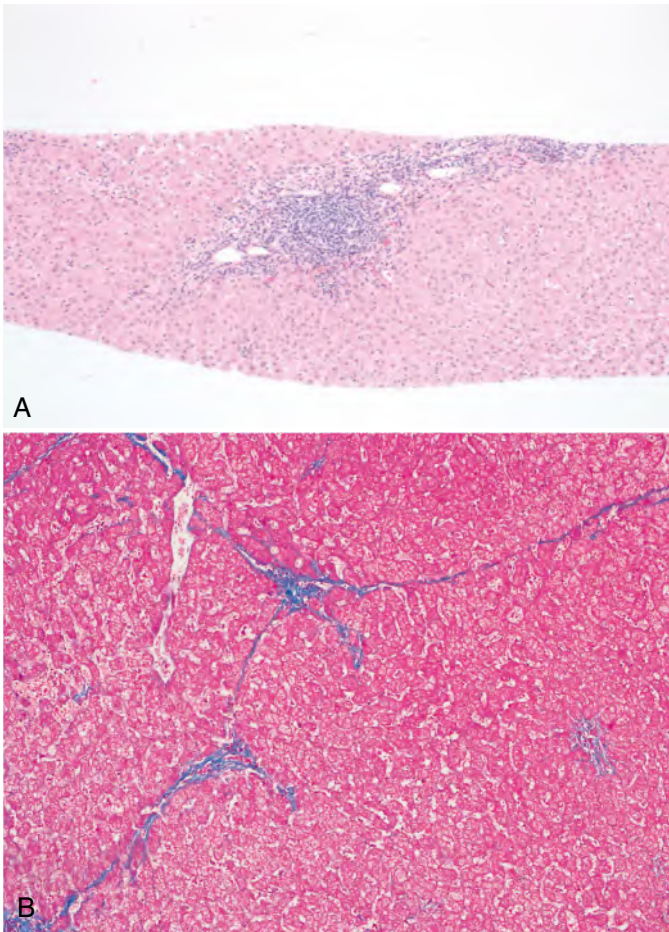


Fig. 18.16 Chronic viral hepatitis due to hepatitis C virus. (A) Characteristic portal tract expansion by a lymphoid aggregate (H&E stain). (B) Delicate bridging fibrosis, seen later in the disease course (trichrome stain with collagen fibers stained blue).

through the hematogenous route or direct spread from adjacent infected tissues can also lead to abscess formation. Liver abscesses are associated with fever, right upper quadrant pain, and tender hepatomegaly. Jaundice may result from extrahepatic biliary obstruction. Antibiotic therapy is generally effective, but surgical drainage may be necessary for large lesions. Extrahepatic bacterial infections, particularly intraabdominal infections, may lead to nonspecific inflammatory changes in the liver. Sepsis can be associated with ductular reaction and cholestasis presenting as bile plugs in ductules (ductular or *cholangiolar cholestasis*).

The liver can be involved in disseminated fungal (e.g., histoplasma) and mycobacterial infections. Liver histology shows epithelioid granulomas, with or without necrosis, at sites of involvement. The diagnosis rests on serologic findings, blood cultures, or visualization of the organisms in liver biopsies.

Parasitic and helminthic infections that involve the liver include malaria, schistosomiasis, strongyloidiasis, cryptosporidiosis, leishmaniasis, echinococcosis, amebiasis, and flukes (*Fasciola hepatica*, *Opisthorchis* species, and *Clonorchis sinensis*; Chapter 8). Schistosomiasis is most commonly found in Asia, Africa, and South America in areas where the water contains numerous freshwater snails as a vector, and it is particularly likely to cause chronic liver disease. Liver fluke infections, most common in Southeast Asia, are notorious for increasing the risk of cholangiocarcinoma (discussed later). Hydatid cysts are

usually caused by echinococcal infections (Chapter 8). They often have characteristic calcifications in the cyst walls, which may enable radiologic diagnosis. Hydatid cysts are uncommon in high resource countries. Cystic liver degeneration or abscesses can be caused by amebic and other protozoal and helminthic infections. The incidence of amebic infections is low in high resource countries and is usually found in immigrants from endemic regions.

AUTOIMMUNE HEPATITIS

Autoimmune hepatitis is a chronic, progressive hepatitis associated with genetic predisposition, autoantibodies, and therapeutic response to immunosuppression. There is a female predominance (78%).

Pathogenesis

Autoimmune hepatitis has a strong association with specific HLA alleles in European (DR3), Japanese (DR4), and South American populations (DRB1). Proposed triggers for the immune reaction include viral infections, drugs/toxins, and vaccination, but the antigens that are the target of autoimmunity are unknown, and as with other autoimmune disorders, that basis for the immune attack on hepatocytes remains obscure. The lymphocytic infiltrate in the liver is composed predominantly of CD4⁺ helper T cells with CD8⁺ cytotoxic T cells at the interface. CD4⁺ T cells play an important role in activating B lymphocytes and their differentiation into plasma cells, which are responsible for production of autoantibodies. The mechanism by which the interplay of lymphocytes, autoantibodies, and HLA types lead to liver injury is unclear. A small number of drugs such as minocycline, nitrofurantoin, and α -methyldopa can trigger the formation of autoantibodies and hepatocellular injury that mimics autoimmune hepatitis.

MORPHOLOGY

The features of autoimmune hepatitis overlap with acute and chronic hepatitis of other etiologies. Extensive inflammation and hepatocellular injury at the interface, as well as in the hepatic parenchyma, is characteristic of autoimmune hepatitis. Numerous plasma cells in clusters are typical (Fig. 18.17). Lymphocytes and plasma cells may be seen within the cytoplasm of hepatocytes, a phenomenon called emperipolesis. Necrosis of groups of hepatocytes (confluent necrosis) can involve the perivenular areas, the entire acinus (panacinar necrosis), or connect vascular structures (bridging necrosis). The resultant regenerative features can manifest as “rosettes,” a circular arrangement of hepatocytes around a dilated canaliculus. Most cases have some degree of fibrosis at initial presentation, which increases with disease progression. Cirrhosis with limited inflammatory activity (i.e., “burnt out cirrhosis”) is seen at initial presentation in some cases, the end result of subclinical disease.

Clinical Features

Untreated autoimmune hepatitis leads to death or progresses to cirrhosis in most cases. It has a wide range of presentations ranging from asymptomatic disease detected by elevated transaminases during screening, to acute and chronic presentations. Acute presentations may be indistinguishable from acute viral or drug-induced hepatitis and may lead to acute liver failure. More commonly, the presentation is insidious with nonspecific symptoms such as fatigue, anorexia, nausea, and abdominal pain. Patients without symptoms or apparently acute presentations often prove to have fibrosis and even cirrhosis on biopsy, indicating the existence of longstanding subclinical disease that went unrecognized.

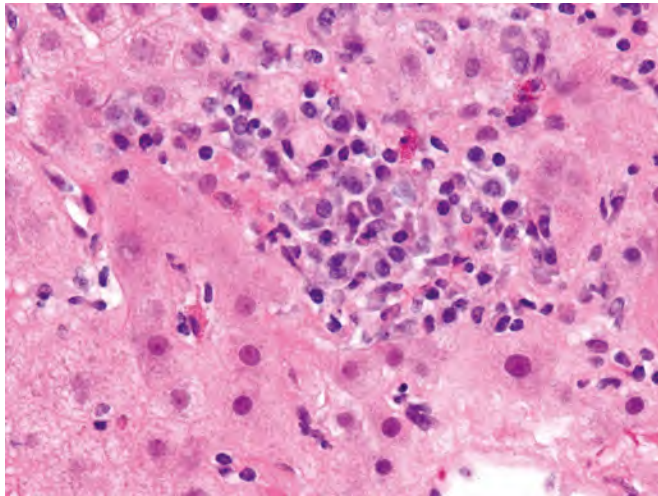


Fig. 18.17 Autoimmune hepatitis. A focus of lobular hepatitis with prominent plasma cells typical for this disease is shown.

In a small subset of patients, there may be overlapping features with primary biliary cholangitis or primary sclerosing cholangitis, the latter occurring most often in children. The diagnosis of overlap syndromes requires the unequivocal presence of serologic, biochemical, and/or histologic evidence of both diseases. In some instances, the second component of the overlap syndrome becomes evident months or years after treatment of the first component. Autoimmune hepatitis may also be associated with other autoimmune diseases like type 1 diabetes mellitus, thyroiditis, and celiac sprue. Autoimmune hepatitis-like injury induced by drugs is important to recognize because discontinuation of the drug typically leads to clinical recovery.

The diagnosis of autoimmune hepatitis is based on the combination of four features: autoantibodies, elevation of serum IgG, exclusion of other etiologies (e.g., viral hepatitis, drugs), and supportive histologic findings on liver biopsy. These features have been combined to create a scoring scheme, which enables categorization of cases into definite and probable autoimmune hepatitis (Table 18.4). Based on the types of autoantibodies present, autoimmune hepatitis is subclassified as types 1 and 2. *Type 1* is more common and is typically characterized by the presence of antinuclear antibodies (ANAs) and anti-smooth muscle actin (SMA) antibodies. Antibodies against soluble liver antigen/liver-pancreas antigen (SLA/LPA) and, less commonly, antimitochondrial antibodies (AMA, more typical of primary biliary cholangitis) also may be seen. *Type 2* disease is more common in children and is characterized by anti-liver kidney microsome-1 (LKM-1) antibodies, which are directed against CYP2D6, and anti-liver cytosol-1 (ACL-1) antibodies. The other clinical features, natural history, and pathologic findings are similar in the two types. Serum IgG is often elevated in autoimmune hepatitis and may provide another clue to the diagnosis.

Autoantibody titers do not correlate well with disease severity or outcome and may not be present in a small subset of cases (i.e., seronegative autoimmune hepatitis). Autoantibodies also may be present in other disease processes such as steatohepatitis and chronic viral hepatitis (e.g., with HBV and HCV); hence, it is important to make the diagnosis based on overall clinical, serologic, and histologic features.

The treatment of choice is immunosuppression with prednisone, with or without azathioprine, leading to remission in 80% to 90% of patients, generally within 12 months of treatment. Other immunosuppressants are used if the side effects of steroids cannot be tolerated. Patients with incomplete responses or with multiple relapses are

TABLE 18.4 Simplified Diagnostic Criteria of the International Autoimmune Hepatitis Group (with updated histologic criteria)

		Points ^a
Autoantibodies	ANA or ASMA or LKM >1:80	2
	ANA or ASMA or LKM >1:40	1
	SLA/LP Positive (>20 units)	0
IgG (or gamma-globulins)	>1.10 times normal limit	2
	Upper normal limit	1
Liver histology ^b	Typical for autoimmune hepatitis	2
	Compatible with autoimmune hepatitis	1
	Atypical for autoimmune hepatitis	0
Absence of viral hepatitis	Yes	2
	No	0

^aDefinite autoimmune hepatitis (AIH): N = 7; probable AIH: N = 6.

^bLikely (score 2): Portal and/or lobular lymphocytic/lymphoplasmacytic inflammation with more than mild interface/lobular inflammatory activity and no features suggestive of another liver disease.

Possible (score 1): Portal and/or lobular lymphocytic/lymphoplasmacytic inflammation with mild interface/lobular inflammatory activity, or histologic features of likely AIH are represented along with features suggestive of another liver disease.

Unlikely (score 0): Typical features of AIH are absent and there are features suggestive of another liver disease.

AIH, Autoimmune hepatitis; ANA, antinuclear antibody; ASMA, anti-smooth muscle actin; IgG, immunoglobulin G; LKM, anti-liver kidney microsomal antibodies; LP, liver pancreas; SLA, soluble liver antigen.

Modified from Hennes EM, Zeniya M, Czaja AJ, et al. Simplified criteria for the diagnosis of autoimmune hepatitis. *Hepatology*. 2008;48(1):169-176; and Lohse AW, Sebode M, Bhatthal PS, et al. Consensus recommendations for histological criteria of autoimmune hepatitis from the International AIH Pathology Group: results of a workshop on AIH histology hosted by the European Reference Network on Hepatological Diseases and the European Society of Pathology. *Liver Int*. 2022;42(5):1058-1069.

at higher risk of progression to cirrhosis and at risk for developing hepatocellular carcinoma. Liver transplantation may be necessary for cirrhotic patients. The 10-year survival rate after liver transplant is 75%, with disease recurrence in 20% of patients.

KEY CONCEPTS

Autoimmune Hepatitis

- The diagnosis of autoimmune hepatitis is based on a combination of four features: autoantibodies, elevated serum IgG, pathologic findings, and exclusion of viral/drug etiologies.
- The most common autoantibodies in type 1 autoimmune hepatitis are ANAs and anti-smooth muscle antibodies (ASMA), while type 2 autoimmune hepatitis is characterized by anti-LKM1 autoantibodies.
- Autoimmune hepatitis can have varied presentations: asymptomatic liver enzyme elevation, acute liver failure, chronic hepatitis, and cirrhosis.
- Typical histologic features of autoimmune hepatitis include a high degree of hepatocyte necrosis and inflammation with plasma cells.

DRUG- AND TOXIN-INDUCED LIVER INJURY

Drug- or toxin-induced liver injury is a major cause of acute liver failure in the United States. Due to its central role in metabolism, the liver is susceptible to injury by a wide array of compounds found in medicines (Table 18.5), herbal products, dietary supplements, poisonous plants and fungi (e.g., *Amanita phalloides* mushrooms), and household and industrial products (e.g., ointments, perfumes, shampoo, cleaning solvents, pesticides). Based on US Drug Induced Liver Injury Network data, 10% of the patients with drug-induced liver injury die or require liver transplantation, and 17% go on to develop chronic liver disease.

TABLE 18.5 Patterns of Drug- and Toxin-Induced Hepatic Injury

Pattern of Injury	Morphologic Findings	Examples of Associated Agents
Cholestatic	Bland hepatocellular cholestasis, without inflammation	Contraceptive and anabolic steroids, antibiotics, ART
Hepatitis with cholestasis	Cholestasis with lobular necroinflammatory activity; may show bile duct destruction	Antibiotics, phenothiazines, statins
Acute hepatitis: inflammation dominant pattern	Lymphocytic inflammation with variable admixture of plasma cells and eosinophils. Single cell or confluent necrosis can be present	Antimicrobials such as antibiotics and isoniazid, anticonvulsants, methyldopa, CNS drugs like phenytoin, immunotherapeutic agents like PD-1/PD-L1 inhibitors
Acute hepatitis: necrosis dominant pattern	Confluent necrosis with sparse inflammation	Acetaminophen, halothane
Chronic hepatitis	Portal-based lymphocytic or lymphoplasmacytic inflammation with variable fibrosis	Nitrofurantoin, NSAIDs
Steatotic liver disease	Large and small fat droplets "Microvesicular steatosis" (diffuse small fat droplets) Steatohepatitis with Mallory hyaline	Ethanol, corticosteroids, methotrexate, total parenteral nutrition Valproate, tetracycline, aspirin (Reye syndrome), ART Ethanol, amiodarone, irinotecan
Fibrosis and cirrhosis	Periportal and pericellular fibrosis	Alcohol, methotrexate, enalapril, vitamin A and other retinoids
Granulomas	Noncaseating epithelioid granulomas Fibrin ring granulomas	Sulfonamides, amiodarone, isoniazid Allopurinol
Vascular lesions	Sinusoidal obstruction syndrome (veno-occlusive disease): obliteration of central veins Budd-Chiari syndrome Peliosis hepatis: blood-filled cavities, not lined by endothelial cells	High-dose chemotherapy, bush teas Oral contraceptives Anabolic steroids, tamoxifen
Neoplasms	Hepatocellular adenoma Hepatocellular carcinoma Cholangiocarcinoma Angiosarcoma	Oral contraceptives, anabolic steroids Alcohol, Thorotrast Thorotrast Thorotrast, vinyl chloride

ART, Antiretroviral therapy; NSAIDs, non-steroidal anti-inflammatory drugs.

Modified from Washington K. Metabolic and toxic conditions of the liver. In: Iacobuzio-Donahue CA, Montgomery EA, eds. *Gastrointestinal and Liver Pathology*. Churchill Livingstone; 2005.

Pathogenesis

Principles of drug and toxic injury are discussed in Chapter 9. Liver injury may develop immediately after exposure to the responsible agent or manifest after weeks or even months. In most instances, the injury is mediated by reactive metabolites generated in the liver. The cytochrome family of enzymes, particularly cytochrome P-450, is involved in most of these metabolic reactions. Since these enzymes are more active in the central zone of the lobule, necrosis of perivenular hepatocytes is a typical feature of drug-induced liver injury. Agents that induce the cytochrome system, such as rifampicin, phenytoin, isoniazid, tobacco smoke, and ethanol, may exacerbate the toxicity of other drugs.

Drug-induced liver injury can be idiosyncratic (unpredictable) or dose-dependent (predictable). Idiosyncratic reactions are the most common form of drug-induced liver injury and usually occur after 1 to 3 months of exposure (see Table 18.5). In some instances, they appear to stem from a hypersensitivity response to the drug or its metabolite(s). It is thought that there is genetic susceptibility in those who have idiosyncratic reactions. For example, individuals with a variant of N-acetyltransferase (NAT2) have decreased enzymatic activity, raising drug levels in the liver and leading to susceptibility to isoniazid hepatotoxicity. Antimicrobial drugs are the most common culprits in idiosyncratic reactions, accounting for nearly one-half of cases. Other commonly implicated agents include cardiovascular drugs, central nervous system therapeutics, antineoplastic drugs, immunotherapy drugs used in cancer like PD-1/PD-L1/CTLA4 inhibitors, and analgesics like nonsteroidal

antiinflammatory drugs. A wide variety of herbal and nutritional agents also have been implicated in idiosyncratic reactions.

A classic, dose-dependent, predictable hepatotoxin is acetaminophen, now the most common cause of acute liver failure necessitating transplantation in the United States. The mechanisms for this toxicity are detailed in Chapter 9. Other examples of direct-acting hepatotoxins that may produce severe liver injury include organic solvents and toxins in mushrooms.

MORPHOLOGY

Drugs can lead to one or more patterns of injury. Hepatocellular injury accounts for almost one-half of cases, and the remainder are divided approximately equally between cholestatic and mixed hepatocellular/cholestatic patterns. Idiosyncratic reactions with a hepatocellular pattern of injury show typical features of acute hepatitis dominated by inflammation and varying degrees of necrosis. The picture is similar to viral and autoimmune hepatitis, and centrilobular necrosis is a common feature. Progression to chronic hepatitis, and even cirrhosis, can be seen in a small minority of cases. Injury caused by intrinsic hepatotoxins is dominated by necrosis with minimal inflammation. Injury centered on bile ducts is characterized by varying combinations of cholestasis and ductular reaction, which can progress to chronic cholestasis and bile duct loss in a minority of cases.

Some agents produce other patterns of injury. Drugs like amiodarone (antiarrhythmic), tamoxifen (antiestrogen), irinotecan (antineoplastic), and methotrexate (immunosuppressive) can cause a **steatohepatitis-like** pattern of injury. Mitochondrial dysfunction caused by drugs such as tetracycline (antibiotic),

valproic acid (anticonvulsant), and zidovudine (antiretroviral) can result in **microvesicular steatosis**. Endothelial injury to sinusoids and central veins can be caused by cytotoxic agents (azathioprine, oxaliplatin), leading to **sinusoidal obstruction syndrome** (formerly called veno-occlusive disease).

Clinical Features

Drug-induced liver injury can have a wide variety of presentations and should always be included in the differential diagnosis of liver disease. Since there are no specific clinical or pathologic features, the diagnosis is established based on temporal association of drug or toxin exposure with onset of liver injury. Liver enzyme tests can be used to gauge whether the injury is primarily (1) hepatocellular (alanine aminotransferase [ALT] ≥ 5 times the upper limit of normal, or ALT/alkaline phosphatase [ALP] ratio > 5), (2) cholestatic (ALP ≥ 2 times the upper limit of normal, or ALT/ALP ratio < 2), or (3) mixed (increased ALT and ALP with ALT/ALP ratio between 2 and 5). Exclusion of other etiologies and disease recovery (in most cases) following withdrawal of the offending agent support the presumptive diagnosis.

KEY CONCEPTS

Drug-Induced Liver Injury

- There are two mechanisms of liver injury:
 - Direct hepatotoxicity, a dose-dependent phenomenon of the drug, or its metabolite, which predictably affects exposed individuals, a typical example being acetaminophen, which is the most common cause of acute liver failure in the United States.
 - Idiosyncratic (hypersensitivity) response, which is not dose-dependent, accounts for most drug-induced liver injury, but typically occurs in a minority of individuals.
- Drugs can mimic any clinical or histologic pattern of injury and should be included in the differential diagnosis of liver diseases in diverse clinical settings.
- Correlation of temporal profile of drug intake and onset of disease is necessary for diagnosis.

STEATOTIC LIVER DISEASE

Alcohol-associated (alcohol-related) liver disease and metabolic dysfunction-associated steatotic liver disease (MASLD) share many similarities and are considered together in this section.

Alcohol-Associated Liver Disease

Excessive alcohol (ethanol) consumption is a major cause of liver disease in most Western countries; it accounts for 5.9% of deaths globally and more often leads to death and disability earlier in life than other forms of chronic liver injury. There are three distinctive, albeit overlapping, forms of alcohol-induced liver injury: (1) steatosis, or fatty change, (2) alcoholic steatohepatitis, and (3) fibrosis, which leads to cirrhosis.

Pathogenesis

The pharmacokinetics and metabolism of alcohol are described in Chapter 9. Pertinent to this discussion are the detrimental effects of alcohol and its byproducts on hepatocellular function. Short-term ingestion of as little as 80 g of alcohol (six beers or 8 ounces of 80-proof liquor) over one to several days generally produces mild, reversible, hepatic steatosis. The risk of severe hepatic injury becomes significant with intake of 80 g or more of ethanol per day, and daily ingestion of 160 g or more for 10 to 20 years is frequently

associated with severe liver injury. However, only 10% to 15% of patients with alcohol use disorder develop cirrhosis. Thus, other factors also influence the development and severity of alcoholic liver disease, including:

- **Sex.** Although most patients with alcohol-associated liver disease are men, on a dose-for-dose basis, females are more susceptible to alcohol-induced hepatic injury. Sex differences in alcohol pharmacokinetics and metabolism may contribute, as well as estrogen-dependent responses of the liver to gut-derived endotoxin (lipopolysaccharide [LPS]). Although exact mechanisms are not known, it appears that estrogen increases gut permeability to endotoxins. These are transported in the portal system to the liver, where they bind the LPS receptor CD14 expressed on Kupffer cells, which is an event that stimulates Toll-like receptor 4 (TLR4). Toll-like receptor signaling triggers the release of cytokines and chemokines, which may contribute to the inflammation that accompanies alcohol-associated liver disease.
- **Ethnic and genetic differences.** Studies with twins suggest that there is a genetic component in alcohol-associated liver disease, although it remains difficult to separate genetic from environmental factors. Genetic variation in alcohol detoxifying enzymes and cytokine promoters may play significant roles and contribute to differences across populations. *ALDH*2*, a variant of aldehyde dehydrogenase (*ALDH*) found in 50% of Asians, has a very low enzyme activity. Individuals homozygous for *ALDH*2* are unable to oxidize acetaldehyde and are intolerant of alcohol, which produces upper body flushing and variable levels of nausea and lethargy.
- **Comorbid conditions.** Iron overload, metabolic dysfunction-associated steatohepatitis, and infection with HCV and HBV synergize with alcohol to increase the severity of liver disease. In recognition of the overlap between MASLD and alcohol-associated liver disease, a category termed “MASLD and increased alcohol intake” (MetALD), describes individuals with MASLD who consume greater than 140 g/week (females) or 210 g/week (males) of alcohol.

Excessive alcohol intake causes steatosis, dysfunction of mitochondria, microtubules and cellular membranes, and oxidative stress, and the resulting injury leads to varying degrees of inflammation and hepatocyte death. Several factors appear to contribute to steatosis. Alcohol metabolism by alcohol dehydrogenase and acetaldehyde dehydrogenase generates large amounts of reduced nicotinamide adenine dinucleotide (NADH). This alters the redox balance in hepatocytes and has a plethora of effects that favor lipogenesis, including suppression of fatty acid oxidation and increased expression of enzymes that carry out fatty acid synthesis. The accumulation of intrahepatic lipids may be further exacerbated by impaired lipoprotein assembly and secretion. Alcohol also enhances peripheral catabolism of fat, increasing the circulating pool of lipids that are available for uptake by hepatocytes.

The exact mechanisms underlying hepatocyte injury and alcohol-associated hepatitis are uncertain, but several likely contributory factors have been identified.

- **Acetaldehyde** (the product of alcohol dehydrogenase) induces lipid peroxidation and acetaldehyde-protein adduct formation, disrupting cytoskeleton and membrane function and possibly producing neoantigens.
- **CYP2E1 induction.** High levels of alcohol consumption induce liver microsomes that contain CYP2E1, a component of the cytochrome P-450 system. CYP2E1 metabolism of alcohol produces ROS that damage cellular protein, membranes, and mitochondria, effects that may promote apoptosis.

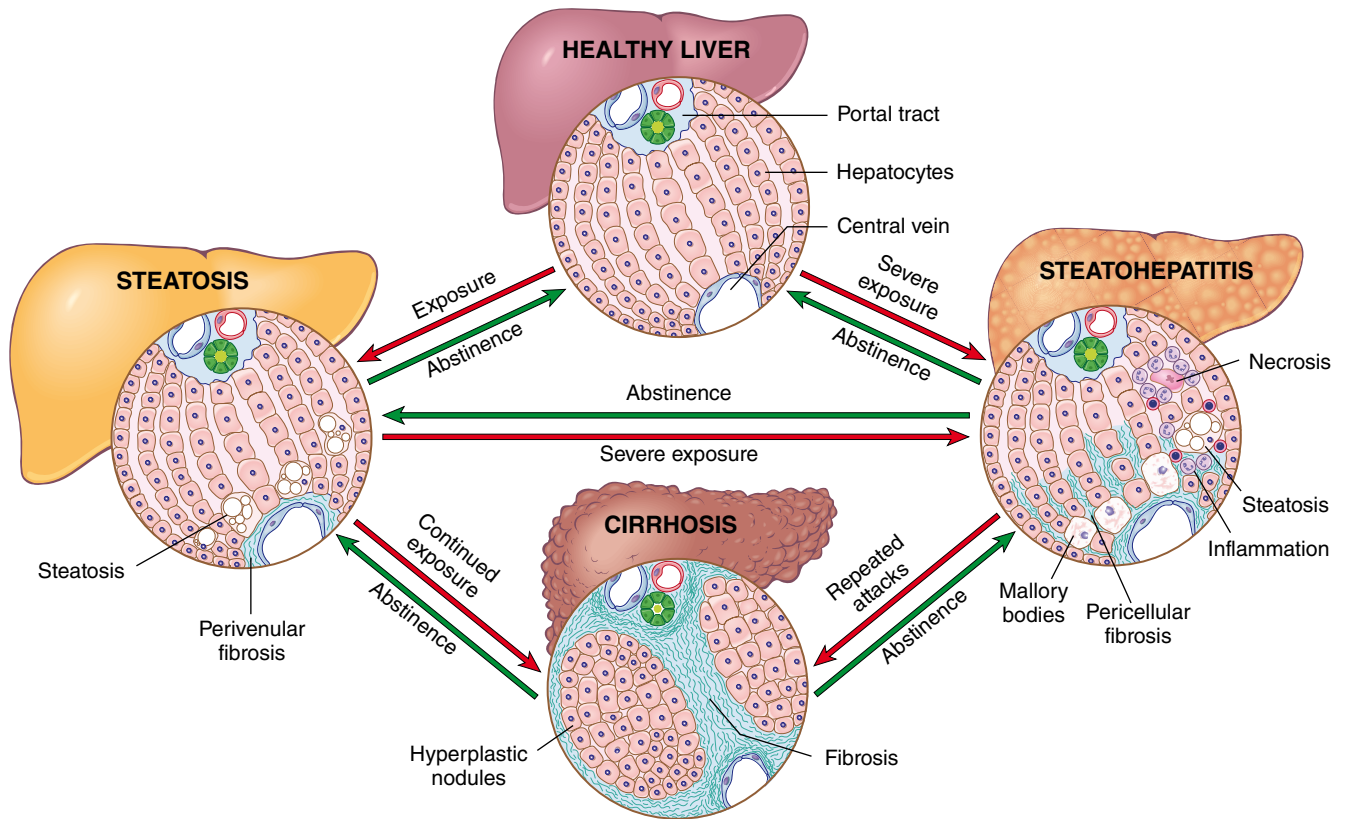


Fig. 18.18 Alcohol-associated liver disease. The interrelationships among hepatic steatosis, alcoholic hepatitis, and alcoholic cirrhosis are shown, along with depictions of key morphologic features. It should be noted that some patients present initially with cirrhosis without any of the other forms of alcoholic liver disease.

- **Methionine metabolism.** Alcohol impairs hepatic metabolism of methionine, which decreases glutathione levels, thereby sensitizing the liver to oxidative injury, and contributes to homocysteine production, which may induce the endoplasmic reticulum stress response.

The induction of cytochrome P-450 enzymes in the liver by alcohol also enhances the conversion of other drugs (e.g., acetaminophen) to toxic metabolites. Furthermore, as already mentioned, alcohol has been linked to increased uptake of bacterial endotoxin from the gut. The resulting release of pathogen-associated molecular patterns (PAMPs) from the gut and damage-associated molecular patterns (DAMPs) from dying hepatocytes results in inflammasome activation (see Chapter 6), evidenced by increased levels of IL-1 β and caspase-1 in the livers of patients with alcohol-associated liver disease. Damage caused by these and other factors lead to inflammation and, with chronicity, liver fibrosis and deranged vascular perfusion (summarized in Fig. 18.18). In essence, alcohol-associated liver disease can be regarded as a maladaptive state in which cells in the liver respond in an increasingly pathologic manner to a stimulus (alcohol) that originally was only marginally harmful.

MORPHOLOGY

Characteristic changes in alcohol-associated liver disease begin in centrilobular zone 3 and extend outward toward the portal tracts as the injury increases in severity. **Hepatic steatosis (fatty liver)** is an early, predictable effect of alcohol consumption. After even moderate intake of alcohol, lipid droplets accumulate in hepatocytes. Lipid accumulation begins as small droplets that coalesce into large droplets, which distend the hepatocyte and push the

nucleus aside (Fig. 18.19). Macroscopically, the fatty liver of chronic alcohol use disorder is enlarged (as heavy as 4 to 6 kg), soft, yellow, and greasy. Steatosis may be separated into microvesicular and macrovesicular forms. **Macrovesicular steatosis** is the predominant form in alcohol-associated liver disease. An unusual exception is **alcoholic foamy degeneration**, a form of diffuse small droplet macrovesicular to microvesicular steatosis sometimes seen with chronic heavy alcohol use that is associated with endoplasmic reticulum and mitochondrial damage. In general, fatty change is completely reversible if there is abstinence from further intake of alcohol.

In a subset of patients with alcohol-associated liver disease, liver inflammation (**alcohol-associated hepatitis**) and fibrosis (see Fig. 18.19) are prominent features. The morphologic findings in alcohol-associated hepatitis include:

- **Ballooned hepatocytes** (Fig. 18.20). These are injured, swollen hepatocytes with cleared-out cytoplasm and cytoskeletal damage, which when extensive results in formation of **Mallory hyaline**. It is composed of skeins of intermediate filaments such as keratin 8 and keratin 18, which are partially degraded and ubiquitinated, and can be identified on immunohistochemical staining (see Fig. 18.20B). Mallory hyaline may reflect a failed attempt to sequester and degrade damaged cytoplasmic proteins. Other ballooned hepatocytes lack Mallory hyaline and contain fat droplets instead. While ballooned hepatocytes are not entirely specific for alcohol-associated steatohepatitis, they are essential for its diagnosis. Other conditions with ballooned hepatocytes include metabolic dysfunction-associated steatohepatitis, Wilson disease, and chronic biliary tract diseases.
- **Inflammation and necrosis.** Neutrophils are generally more prominent in alcohol-associated hepatitis than in metabolic dysfunction-associated steatohepatitis and can localize around ballooned hepatocytes, particularly

when Mallory hyaline is present. Lobular lymphocytic infiltrates are common, and portal lymphocytic infiltrates also may occur, along with increases in macrophages in both portal tracts and lobular parenchyma. Necrosis or apoptosis is usually spotty, but more prominent hepatocellular injury and confluent necrosis can occur in some cases.

- **Perivenular/pericellular fibrosis.** Steatohepatitis is often accompanied by fibrosis. This usually starts in acinar zone 3 as pericellular, or perisinusoidal, fibrosis with a “chicken wire” appearance (see Fig. 18.19). With continued damage, this progresses to portal/periportal fibrosis, and then to bridging fibrosis and cirrhosis, which in many cases is micronodular (“Laennec cirrhosis”) (Fig. 18.21). Early stages of scarring can regress with cessation of alcohol use, but with the development of cirrhosis and its vascular derangements (related to perivenular fibrosis and fibrous obliteration, termed phleboscrosis, and veno-occlusive lesions), the chances of full restoration of normal function diminish. Complete regression of alcohol-associated cirrhosis, although reported, is rare.

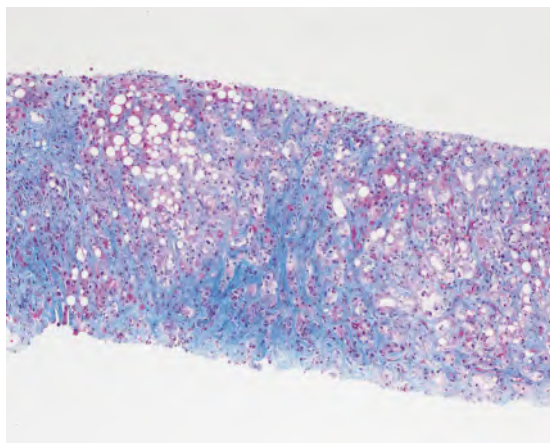


Fig. 18.19 Alcohol-associated steatosis and fibrosis. A mix of small and large fat droplets (seen as clear vacuoles) is most prominent around the central vein and extends outward to the portal tracts. Some fibrosis (stained blue) is present in a characteristic perisinusoidal chicken wire pattern. (Masson trichrome stain).

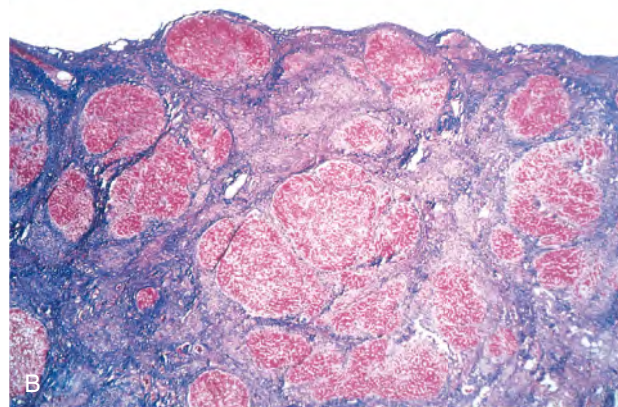


Fig. 18.21 Alcohol-associated cirrhosis. (A) The characteristic diffuse nodularity of the surface is induced by the underlying fibrous scarring. The average nodule size is 3 mm in this close-up view, typical of the “micronodular” cirrhosis of alcohol-associated liver disease. The greenish tint is caused by cholestasis. (B) Microscopically, this cirrhosis is marked by small nodules entrapped in blue-staining fibrous tissue; fatty accumulation is no longer seen in this “burned-out” stage (Masson trichrome stain).

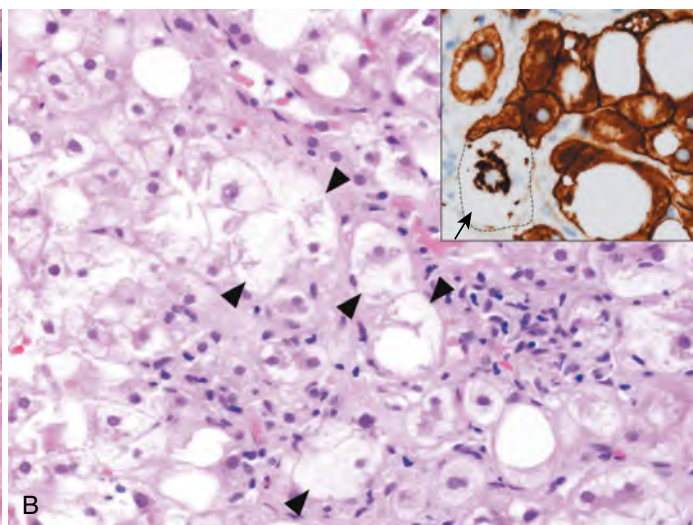
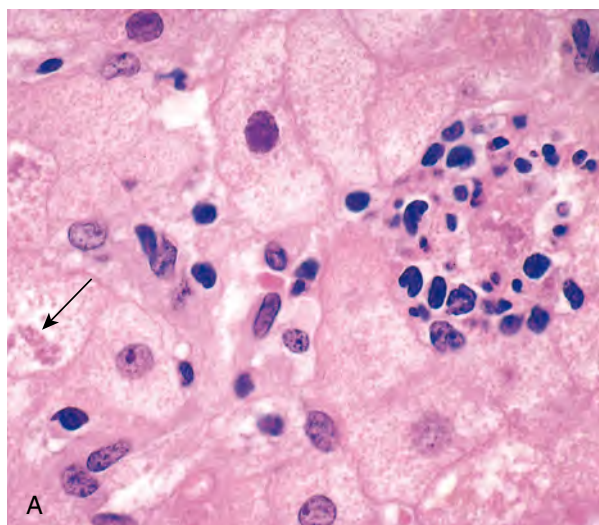


Fig. 18.20 Alcohol-associated liver disease. (A) Steatohepatitis with clustered inflammatory cells marking the site of a necrotic hepatocyte. Mallory hyaline is present in another hepatocyte (arrow). (B) Alcohol-associated steatohepatitis with many ballooned hepatocytes (arrowheads). Clusters of inflammatory cells are also present; inset shows immunostaining for keratins 8 and 18 (brown). Most hepatocytes, including those with fat vacuoles, showing normal cytoplasmic staining, but in the ballooned hepatocyte (arrow) the keratins are collapsed into Mallory hyaline, leaving the cytoplasm “empty.” (Courtesy Dr. Elizabeth Brunt, Washington University, St. Louis.)

Clinical Features

Hepatic steatosis may cause hepatomegaly, with mild elevation of serum bilirubin and alkaline phosphatase levels. Severe hepatic dysfunction is unusual. Alcohol withdrawal and consumption of an adequate diet are sufficient treatment. In contrast, alcohol-associated hepatitis tends to appear acutely, usually following a bout of heavy drinking. Typically, there is malaise, anorexia, weight loss, upper abdominal discomfort, and tender hepatomegaly, accompanied by the laboratory findings of hyperbilirubinemia, elevated serum aminotransferases and alkaline phosphatase, and often a neutrophilic leukocytosis. In contrast to other chronic liver diseases, in which serum ALT tends to be higher than serum AST, in alcohol-associated liver disease serum AST tends to be higher than serum ALT levels by a ratio of 2:1 or greater. This finding can be particularly helpful in the setting of occult alcoholism. On the other end of the spectrum, in severe cases symptoms and laboratory findings may mimic those of acute liver failure. In other instances, an acute cholestatic syndrome picture is seen that resembles large bile duct obstruction.

The outlook is unpredictable; each episode of hepatitis incurs about a 10% to 20% risk of death. With repeated bouts, cirrhosis develops in about one-third of patients within a few years. Alcohol-associated hepatitis also may be superimposed on established cirrhosis. With proper nutrition and total cessation of alcohol consumption, alcohol-associated hepatitis may resolve. However, in some patients, the hepatitis persists despite abstinence, and progresses to cirrhosis.

The manifestations of alcohol-associated cirrhosis are similar to those of other forms of cirrhosis. Laboratory findings reflect hepatic dysfunction and include elevated serum aminotransferases, hyperbilirubinemia, variable elevation of serum alkaline phosphatase, hypoproteinemia (globulins, albumin, and clotting factors), and anemia. In some instances, liver biopsy may be indicated, since in about 10% to 20% of cases of presumed alcohol-associated cirrhosis, another disease process is found. Finally, cirrhosis may be clinically silent, discovered only at autopsy or when a stress such as infection or trauma tips the balance toward hepatic insufficiency.

The long-term outlook for patients with alcohol use disorder and liver disease is variable. The 5-year survival rate approaches 90% in abstainers who are free of jaundice, ascites, or hematemesis; it drops to 50% to 60% in those who continue to imbibe. In advanced disease, common proximate causes of death include (1) hepatic coma, (2) massive gastrointestinal hemorrhage, (3) intercurrent infection (to which these patients are predisposed), (4) hepatorenal syndrome (often following a bout of alcoholic hepatitis), and (5) hepatocellular carcinoma (the risk of developing this tumor in alcoholic cirrhosis is 1% to 6% annually).

KEY CONCEPTS

Alcohol-Associated Liver Disease

- Alcohol-associated liver disease is a chronic disorder that can give rise to steatosis, alcohol-associated hepatitis, progressive fibrosis, and marked derangement of vascular perfusion leading eventually to cirrhosis.
- Consumption of 80 g/day of alcohol is considered to be the threshold for development of alcohol-associated liver disease, but may be lower in females.
- It may take 10 to 15 years of drinking for development of cirrhosis, which occurs only in a small proportion of patients with chronic alcohol use disorder.
- The pathologic effects of alcohol on hepatocytes include changes in lipid metabolism related to altered redox potential, injury caused by ROS generated by metabolism of alcohol by the P450 system, and protein adducts formed by acetaldehyde, a major metabolite of alcohol.

TABLE 18.6 World Health Organization Criteria for the Metabolic Syndrome

One of:	Diabetes mellitus Impaired glucose tolerance Impaired fasting glucose Insulin resistance
And two of:	Blood pressure $\geq 140/90$ mm Hg Dyslipidemia: triglycerides (TG): ≥ 1.695 mmol/L and high-density lipoprotein cholesterol (HDL-C) ≤ 0.9 mmol/L (male), ≤ 1 mmol/L (female) Central obesity: waist-hip ratio >0.90 (male); >0.85 (female), or body mass index >30 kg/m ² Microalbuminuria: urinary albumin excretion rate of ≥ 20 μ g/min or albumin-to-creatinine ratio ≥ 30 mg/g

Metabolic Dysfunction–Associated Steatotic Liver Disease

Metabolic dysfunction–associated steatotic liver disease (MASLD), previously termed nonalcoholic fatty liver disease (NAFLD), is defined as the presence of hepatic steatosis (fatty liver) in individuals who do not consume alcohol or do so in small quantities and who do not have another cause of secondary hepatic injury. (e.g., HCV, Wilson disease, medications) and who have at least one of five cardio-metabolic risk factors (Table 18.6). MASLD is the most common cause of chronic liver disease in the United States and is projected to exceed 30% prevalence in the adult population by 2030. The term *metabolic dysfunction–associated steatohepatitis* (MASH) is reserved for MASLD patients who demonstrate steatohepatitic injury with histologic features similar to those seen with alcohol-associated steatohepatitis. The diagnosis of MASH confers an increased risk of developing advanced fibrotic liver disease, and the rise in MASLD prevalence is expected to produce a concomitant increase in the incidence of MASH and its serious complications (e.g., decompensated cirrhosis, hepatocellular carcinoma).

Pathogenesis

The precise mechanisms underlying MASLD are unknown. The strong association with insulin resistance suggests that this factor is particularly important in disease development. However, even among those with insulin resistance, there is significant variability in the severity of MASLD, and complex associations with genetic variants, diet, and the intestinal microbiome have also been suggested. As in alcohol-associated liver disease, there is an established association between increased gut-derived endotoxin production and liver inflammation and injury. High-fructose diets have also been associated with increased risk of MASLD-related fibrosis, and dietary fat, particularly trans-fat, may have a role in producing liver injury. Obstructive sleep apnea, usually occurring in the setting of obesity, has been associated with disease progression, possibly related to intermittent hypoxia. Fibrosis may be accelerated when injury from another liver disease (e.g., hemochromatosis) is superimposed on MASLD.

In individuals with insulin resistance and metabolic syndrome, visceral adipose tissue not only increases in mass but also becomes dysfunctional. Resistance to insulin leads to increased release of free fatty acids from adipocytes due to overactivity of lipoprotein lipase (Chapter 24). This is associated with reduced production of the hormone adiponectin from adipocytes, which decreases oxidation of free fatty acids by skeletal muscle and increases free fatty acid uptake into hepatocytes (Chapter 9), where the fatty acids are stored as triglycerides. The dysfunctional adipocytes also synthesize proinflammatory cytokines such as TNF- α . At the same time, there is evidence that

hepatocytes in MASLD down-regulate lipolysis, an alteration that may further contribute to the accumulation of lipids. Lipolysis occurs through a mechanism called *lipophagy*, a type of selective autophagy, which you will recall is the process by which cells remove excessive or dysfunctional cellular components and (in times of starvation) generate metabolites for energy production (Chapter 2).

The net effect of these changes is to cause lipid to accumulate in hepatocytes. Fat-laden cells are highly sensitive to lipid peroxidation products generated by oxidative stress, which can damage mitochondrial and plasma membranes, potentially leading to apoptosis or necrosis. These tendencies may be exacerbated by the proinflammatory state that accompanies insulin resistance. Once cell injury becomes established, release of cytokines such as TNF- α and TGF- β locally from Kupffer cells leads to the activation of stellate cells, collagen deposition, and scarring.

MORPHOLOGY

MASH shares many morphologic features with alcohol-associated steatohepatitis; steatosis ($\geq 5\%$ of hepatocytes), lobular inflammation, and ballooned hepatocytes are required for its diagnosis. It is not possible to reliably distinguish alcohol-associated steatohepatitis from MASH based on histologic findings, though alcoholic hepatitis has on average less steatosis and more ballooned hepatocytes, lobular inflammation, Mallory hyaline, neutrophilic infiltrates, cholestasis, and obliterated central veins (Fig. 18.22).

Determination of the extent of fibrosis is important for clinical management. Fibrosis typically develops around the central vein as a fine “spider web” of pericellular collagen deposition (also referred to as a chicken wire pattern), which may only be appreciated with a trichrome stain. Progression of fibrosis usually manifests as periportal fibrosis, followed by bridging fibrosis and cirrhosis. Cirrhosis is often subclinical for years, and, when established, the steatosis or ballooned hepatocytes may be reduced or absent. More than 90% of cases labeled as “cryptogenic cirrhosis” (i.e., cirrhosis of unknown cause) are thought to be due to MASH (i.e., “burned out MASH”) and can be considered cryptogenic SLD if risk factors are present and other etiologies have been excluded or unlikely.

Pediatric MASLD differs significantly from adult MASLD. Typically, children show more diffuse steatosis and portal (rather than central) fibrosis, and ballooned hepatocytes may not be present.

Clinical Features

The varied clinical course of individuals with MASLD is summarized in Fig. 18.23. Individuals with only steatosis are generally asymptomatic. Clinical presentation is often related to other signs and symptoms of the metabolic syndrome, in particular insulin resistance or diabetes mellitus. Imaging studies may reveal fat accumulation in the liver. Liver biopsy is required for diagnosis of MASH and aids in assessment of fibrosis. Viral, autoimmune, and other metabolic diseases of the liver must be excluded, and biopsy may be helpful when there is more than one potential etiology for liver injury. Serum AST and ALT are elevated in most patients with MASH. Despite the enzyme elevations, patients may be asymptomatic. Others have nonspecific symptoms such as fatigue, or complain of right-sided abdominal discomfort caused by hepatomegaly. Because of the association with metabolic syndrome, cardiovascular disease is a frequent cause of death in patients with MASH. MASH also increases the risk of hepatocellular carcinoma, as do other metabolic diseases (discussed later).

The goal of treating individuals with MASH is to reverse the histologic features of disease and prevent or reverse fibrosis by correcting underlying risk factors, such as obesity and hyperlipidemia, and to treat insulin resistance. Weight loss, diet, and exercise can potentially

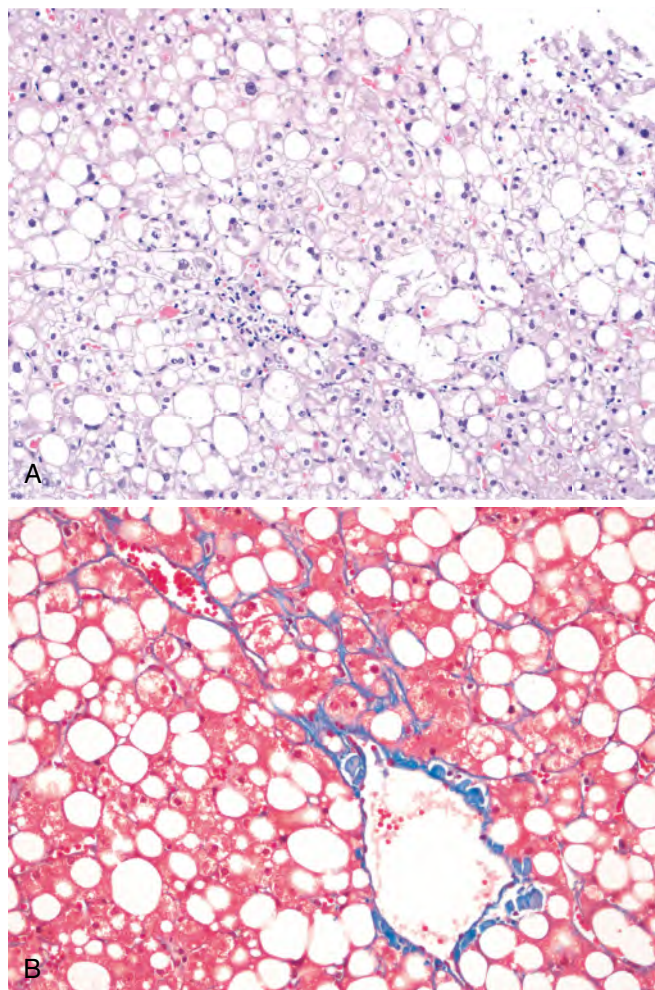


Fig. 18.22 Metabolic dysfunction-associated steatotic liver disease. (A) Liver with mixed small and large fat droplets as well as ballooned hepatocytes. (B) Steatosis and fibrosis extending along sinusoids in a chicken wire pattern in which individual and clustered hepatocytes are surrounded by thin scars (*blue fibers*). Note the resemblance to alcohol-associated steatohepatitis depicted in Fig. 18.19. (Masson trichrome stain.)

reverse the histologic abnormalities in MASH. There are numerous ongoing clinical trials aimed at developing pharmacologic approaches for treating MASH and its complications.

KEY CONCEPTS

Metabolic Dysfunction–Associated Steatotic Liver Disease (MASLD)

- The most common metabolic liver disorder is MASLD, which is associated with metabolic syndrome, obesity, type 2 diabetes mellitus, and hyperlipidemia.
- MASLD may show all the histologic changes associated with alcoholic liver disease (e.g., steatosis, steatohepatitis, and steatofibrosis). The diagnosis of MASH requires biopsy, and MASH cannot be reliably distinguished from alcohol-associated steatohepatitis without clinical history.
- Pediatric MASLD is being increasingly recognized, as the obesity epidemic involves pediatric age groups. Its histologic features differ somewhat from those seen in adults.

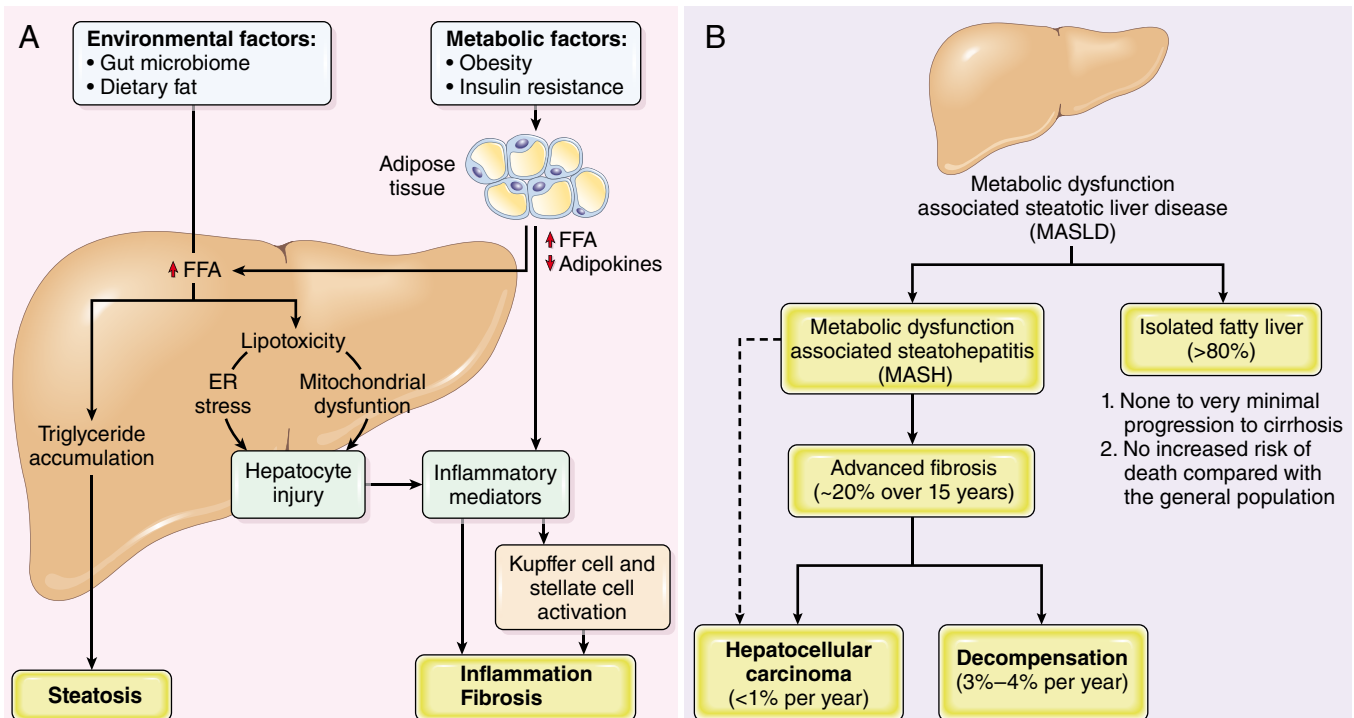


Fig. 18.23 Natural history of metabolic dysfunction–associated steatotic liver disease phenotypes. (A) Environmental and metabolic factors contribute to steatosis, hepatocyte injury, and inflammation/fibrosis. (B) Isolated steatotic liver, shows minimal risk for progression to cirrhosis or increased mortality, and metabolic dysfunction–associated steatohepatitis shows increased overall mortality as well as increased risk for advanced fibrosis (which includes bridging fibrosis and cirrhosis) and hepatocellular carcinoma. *ER*, Endoplasmic reticulum; *FFA*, free fatty acids.

INHERITED LIVER DISEASE

Among inherited metabolic diseases, hemochromatosis, Wilson disease, and α_1 -antitrypsin deficiency are most prominent and are discussed in the following sections.

Hemochromatosis

Hemochromatosis is caused by excessive iron absorption, most of which is deposited in the liver and pancreas, followed by the heart, joints, and endocrine organs. When hemochromatosis results from an inherited disorder, it is referred to as *hereditary hemochromatosis*. When accumulation occurs as a consequence of parenteral administration of iron, usually in the form of transfusions, or other causes, it is called *secondary hemochromatosis*. The classification of the various causes of iron overload is shown in Table 18.7.

As discussed in Chapter 14, the total body iron pool ranges from 2 to 6 g in normal adults; about 0.5 g is stored in the liver, 98% of which is in hepatocytes. In severe hemochromatosis, the total body iron pool may exceed 50 g, more than one-third of which accumulates in the liver. The following features characterize hemochromatosis:

- Fully developed cases exhibit (1) micronodular cirrhosis (all patients); (2) diabetes mellitus (75% to 80% of patients); and (3) abnormal skin pigmentation (75% to 80% of patients).
- Iron accumulation in hereditary forms is lifelong, but the injury caused by excessive iron is slow and progressive; hence symptoms usually first appear in the fourth to fifth decades of life in males and later in females since menstrual bleeding counterbalances the accumulation until menopause.
- Because many females do not accumulate clinically relevant amounts of iron within their lifetime, hereditary hemochromatosis affects more males than females (ratio of 5 to 7:1).

TABLE 18.7 Classification of Iron Overload

Hereditary Hemochromatosis

Mutations of genes encoding HFE, transferrin receptor 2 (TfR2), or ferroportin
Mutations of genes encoding HJV (hemojuvelin or hepcidin: juvenile hemochromatosis)

Hemosiderosis (Secondary Hemochromatosis)

Parenteral iron overload due to red cell transfusions
Severe chronic hemolytic anemias (e.g., sickle cell disease)
Severe forms of thalassemia
Bone marrow failure (e.g., aplastic anemia)
Conditions associated with increased iron uptake
 β -Thalassemia
Myelodysplastic syndrome
Increased oral intake of iron
African iron overload (Bantu siderosis)
Chronic liver disease
Alcohol-associated liver disease
Porphyria cutanea tarda
Neonatal hemochromatosis^a

^aNeonatal hemochromatosis develops in utero but does not appear to be a hereditary condition.

Pathogenesis

In hereditary hemochromatosis, regulation of intestinal absorption of dietary iron is abnormal, leading to net iron accumulation of 0.5 to 1 g/year. The disease typically manifests after 20 g of stored iron has accumulated. Mechanisms of liver injury include (1) lipid peroxidation via iron-catalyzed free radical reactions, (2) stimulation of collagen formation by activation of hepatic stellate cells, and (3) interaction of ROS and iron itself with DNA, leading to lethal cell injury and predisposition to hepatocellular carcinoma.

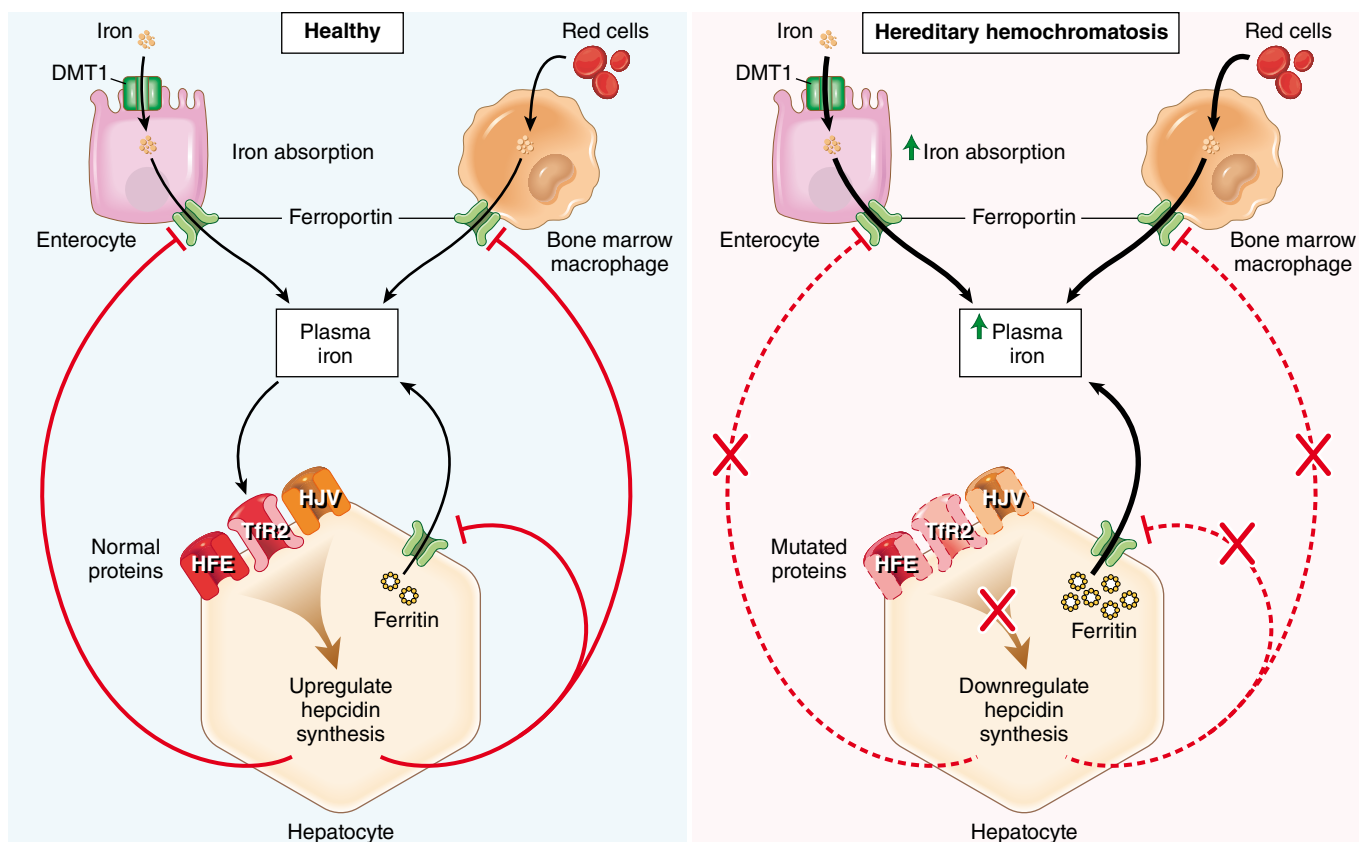


Fig. 18.24 Normal iron metabolism and altered iron metabolism in hereditary hemochromatosis. (A) In normal state, HFE, HJV, and TFR2 regulate hepcidin synthesis by hepatocytes maintaining normal circulating hepcidin levels. Hepcidin binds to ferroportin on enterocytes leading to internalization of the complex and ferroportin degradation. This in turn reduces efflux of iron from enterocytes. Through these regulatory interactions, normal iron absorption is maintained. (B) In hereditary hemochromatosis, *HFE*, *HJV*, or *TFR2* gene mutations reduce hepcidin synthesis, diminishing circulating hepcidin levels. Due to decreased hepcidin-ferroportin interaction, ferroportin activity and iron efflux from enterocytes increases, giving rise to systemic iron overload.

The main regulator of iron absorption is the protein hepcidin, encoded by the *HAMP* gene and produced and secreted by the liver (Fig. 18.24). Transcription of *HAMP* is increased by inflammatory cytokines and iron, and decreased by iron deficiency, hypoxia, and erythroferrone, a hormone produced by bone marrow erythroblasts. These inputs are integrated to regulate the synthesis and plasma levels of hepcidin (see Chapter 14). Hepcidin binds to the cellular iron efflux channel ferroportin, causing its internalization and proteolysis, thereby inhibiting the release of iron from intestinal cells and macrophages. Due to this activity, a deficiency of hepcidin causes iron overload. Decreased hepcidin synthesis or activity may be caused by loss-of-function mutations in the genes *HAMP*, *HJV*, *TFR2*, and *HFE*, all of which are associated with hereditary forms of hemochromatosis. Deficiencies of hepcidin also occur when erythroferrone levels are chronically elevated, as occurs in disorders marked by ineffective hematopoiesis, such as β -thalassemia and myelodysplastic syndrome.

The adult form of hereditary hemochromatosis is usually caused by mutations of *HFE*; mutation of *TFR2* and *SLC40A1*, which encodes ferroportin, is a rare cause. Mutations in *HAMP* and *HJV* genes are also rare and give rise to juvenile forms of hereditary hemochromatosis. *HFE* encodes an HLA class I-like molecule that governs intestinal absorption of dietary iron by regulating hepcidin synthesis. The most common *HFE* mutation produces an inactivating cysteine-to-tyrosine substitution at amino acid 282 (C282Y) and is present in 70% or more

of patients diagnosed with hereditary hemochromatosis. The other common disease-associated *HFE* mutation results in an H63D (histidine at position 63 to aspartate) substitution.

The C282Y mutation of *HFE* is largely confined to populations of European origin, while the H63D mutation has a worldwide distribution. The frequency of C282Y homozygosity in individuals of European descent is 0.45% (1 in 220 persons); however, only about 10% of homozygous males develop evidence of iron overload. The disease penetrance is even lower in female C282Y homozygotes and in individuals who are H63D homozygotes or C282Y/H63D compound heterozygotes.

MORPHOLOGY

Severe hemochromatosis (hereditary or secondary) is characterized by (1) **deposition of hemosiderin** in the following organs (in decreasing order of severity): liver, pancreas, myocardium, pituitary gland, adrenal gland, thyroid and parathyroid glands, joints, and skin; (2) cirrhosis; and (3) pancreatic fibrosis. In the **liver**, iron becomes evident first as golden-yellow hemosiderin granules in the cytoplasm of periportal hepatocytes that stain with Prussian blue (Fig. 18.25). With increasing iron load, there is progressive deposition in the rest of the lobule, bile duct epithelium, and Kupffer cells. Iron is a direct hepatotoxin, and inflammation is characteristically absent. In early stages of disease, the liver is typically slightly larger than normal, dense, and chocolate brown. Fibrous septa

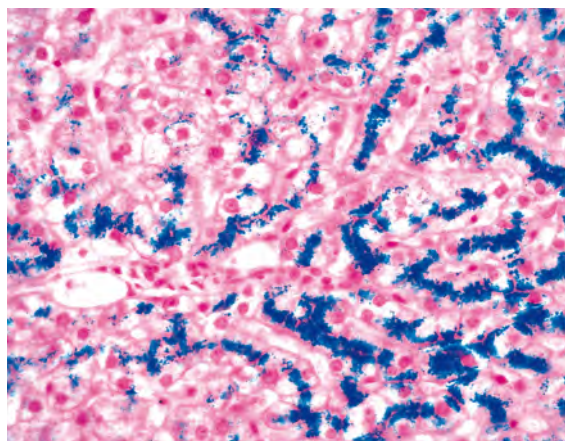


Fig. 18.25 Hereditary hemochromatosis. In this Prussian blue–stained section, hepatocellular iron appears blue. The parenchymal architecture is normal.

develop slowly, leading ultimately to a small, shrunken liver with a micronodular pattern of cirrhosis. The liver parenchyma in later stages is often dark brown to nearly black due to extensive iron accumulation.

Other organs that are particularly susceptible to the toxic effects of iron also show morphologic changes. The **pancreas** becomes intensely pigmented and may undergo parenchymal atrophy associated with interstitial fibrosis. The **heart** is often enlarged and exhibits hemosiderosis, producing a striking brown coloration. Both organs may develop fibrosis. Skin pigmentation predominantly results from increased epidermal melanin production, the mechanism of which is unknown. The combination of these pigments imparts a characteristic slate-gray color to the skin. With hemosiderin deposition in joint linings, **acute synovitis** may develop. Excessive deposition of calcium pyrophosphate damages the articular cartilage, producing a disabling polyarthritis referred to as **pseudogout**. The **testes may be small and atrophic**, secondary to pituitary involvement and reduced gonadotropin and testosterone levels.

Clinical Features

The principal manifestations of hemochromatosis include **hepatomegaly, abdominal pain, abnormal skin pigmentation (particularly in sun-exposed areas), deranged glucose homeostasis or diabetes mellitus (due to destruction of pancreatic islets), cardiac dysfunction (arrhythmias, cardiomyopathy), and atypical arthritis**. In some patients, the presenting complaint is hypogonadism (e.g., amenorrhea in the female, impotence and loss of libido in the male). It is more often a disease of males, for reasons described earlier, and rarely becomes evident before 40 years of age. The classic tetrad of cirrhosis with hepatomegaly, abnormal skin pigmentation, diabetes mellitus, and cardiac dysfunction may not develop until late in the course. Death may result from cirrhosis or cardiac disease.

Another significant cause of death is hepatocellular carcinoma; the risk is 200-fold greater than in the general population. Treatment that reduces iron overload does not fully remove the cancer risk, presumably because DNA damage occurs prior to diagnosis and treatment initiation.

Fortunately, hemochromatosis is often diagnosed before irreversible tissue damage has occurred. Currently most patients with hemochromatosis are identified in the subclinical, precirrhotic stage due to routine serum iron measurements (as part of other diagnostic workups), and the diagnosis is confirmed by DNA sequencing and detection of causative

mutations, usually in *HFE*. Further evaluation includes exclusion of secondary causes of iron overload. A liver biopsy may also provide useful information (e.g., about the presence of fibrosis), but is not done routinely, as genetic testing and imaging have negated the need for quantitative assessment of tissue iron content. Screening of the family members of probands is important, as identification of asymptomatic carriers can prevent disease development.

Prevention of the development or progression of disease in patients with adult-onset hereditary hemochromatosis is remarkably simple, as regular phlebotomy steadily depletes tissue iron stores. With treatment, life expectancy is normal.

The most common causes of secondary (or acquired) hemochromatosis are disorders associated with ineffective erythropoiesis, such as thalassemia (Chapter 14) and myelodysplastic syndrome (Chapter 13). In these disorders, the excess iron results not only from transfusions, but also from increased absorption. Transfusions alone, when given repeatedly over a period of years (e.g., in patients with chronic hemolytic anemias), can also lead to systemic hemosiderosis and parenchymal injury, independent of ineffective hematopoiesis. Finally, other forms of cirrhosis may also diminish hepcidin production due to loss of hepatocyte mass, leading again to increased iron uptake from the gut and increased tissue levels of iron.

Wilson Disease

Wilson disease is an autosomal recessive disorder caused by mutation of the *ATP7B* gene that result in impaired copper excretion into bile and a failure to incorporate copper into ceruloplasmin. This disorder is marked by the accumulation of toxic levels of copper in many tissues and organs, principally the liver, brain, and eye. Free copper is taken up by hepatocytes and incorporated into apoceruloplasmin to form *ceruloplasmin*, which is secreted into the blood. Ceruloplasmin accounts for 90% to 95% of plasma copper. Excess copper within hepatocytes that is not incorporated into ceruloplasmin is sequestered in lysosomes and transported into the bile, from which it is eventually excreted in the feces.

Pathogenesis

Wilson disease results from loss-of-function mutations in *ATP7B*, which encodes a transmembrane copper-transporting ATPase, expressed in hepatocytes. More than 300 sequence variants in the *ATP7B* gene have been identified, but not all cause the disease. The overwhelming majority of patients are compound heterozygotes with different mutations on each *ATP7B* allele. The overall frequency of mutated alleles is 1:100, and the prevalence of the disease is approximately 1:30,000 to 1:50,000 (approximately 9000 patients in the United States). Deficiency of *ATP7B* decreases copper transport into bile, impairs its incorporation into ceruloplasmin, and inhibits ceruloplasmin secretion into the blood (Fig. 18.26). These defects lead to a decrease in circulating ceruloplasmin accompanied by the accumulation of copper in hepatocytes. Total serum copper may be lower than normal due to the deficiency in ceruloplasmin, particularly early in the disease course.

The excess hepatic copper causes toxic injury by three mechanisms: (1) promoting the formation of free radicals by the Fenton reaction (Chapter 2); (2) binding to sulfhydryl groups of cellular proteins; and (3) displacing other metals from hepatic metalloenzymes. Hepatocyte injury causes nonceruloplasmin-bound copper to be spilled into the blood and accumulate in certain tissues, particularly the basal ganglia of the brain and the cornea. Concomitantly, urinary excretion of copper markedly increases, but not to levels sufficient to prevent deposition of copper in tissues.

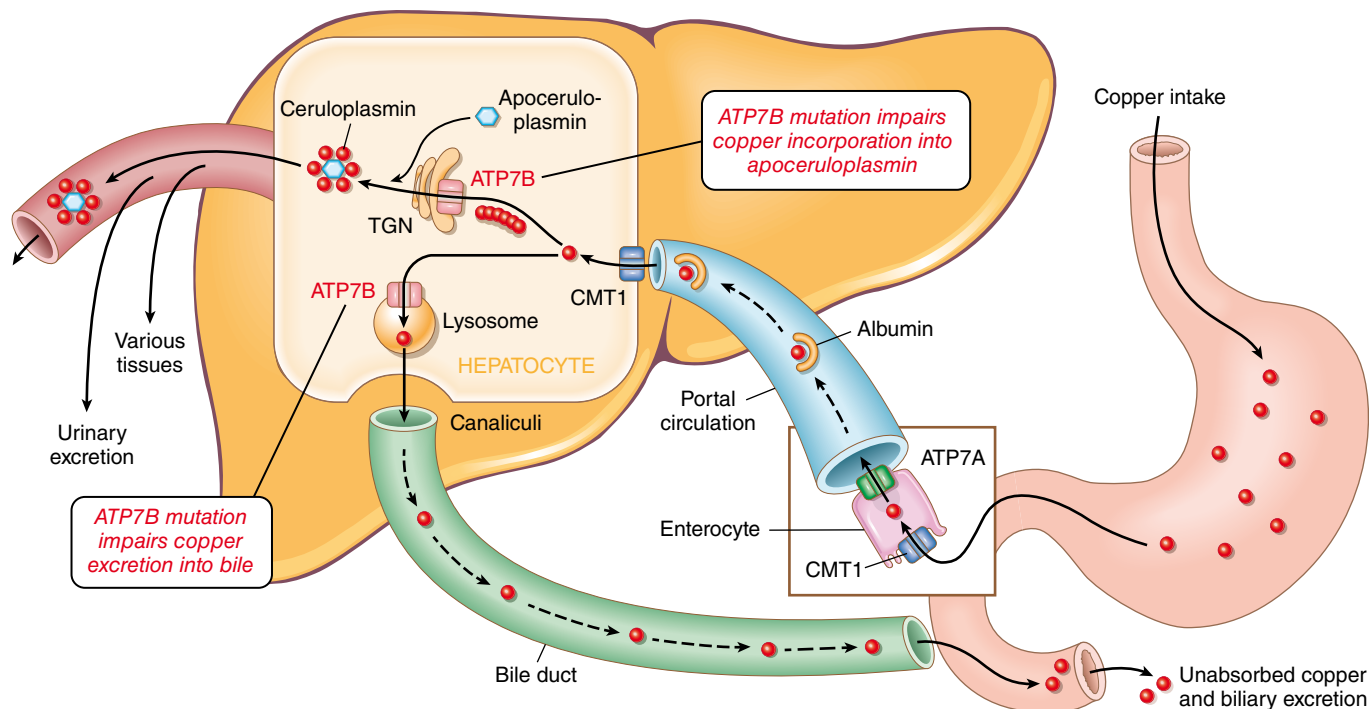


Fig. 18.26 Copper metabolism and *ATP7B* mutation in Wilson disease. Dietary copper enters enterocytes through copper membrane transporter 1 (*CMT1*) and is then transported by *ATP7A* into the portal vein in association with circulating albumin. In hepatocytes, *CMT1* transports copper into the cytoplasm, from which it can either be transported by *ATP7B* into the transgolgi network (TGN), where it is incorporated into ceruloplasmin and secreted, or transported into lysosomes. Lysosomal copper is subsequently transported across the hepatocyte canalicular membrane and excreted in the bile. Loss of *ATP7B* function results in impaired copper incorporation into ceruloplasmin as well as impaired biliary copper excretion in Wilson disease, leading to excessive intrahepatocyte copper accumulation, hepatocellular damage, and finally, increased copper deposition in brain, eyes, lungs, and various other tissues.

MORPHOLOGY

The hepatic changes are variable, ranging from relatively minor to massive damage, and mimic many other disease processes. **Fatty change (steatosis)** may be present with focal hepatocyte necrosis. Acute liver failure can mimic acute viral hepatitis. Chronic hepatitis in Wilson disease typically exhibits moderate to severe portal inflammation and hepatocyte necrosis, admixed with fatty change and features of steatohepatitis (i.e., ballooned hepatocytes, prominent Mallory hyaline, and perisinusoidal fibrosis). Eventually cirrhosis supervenes. Histochemical copper staining is not sensitive or specific for diagnosis of Wilson disease. Toxic brain injury primarily involves the basal ganglia, and nearly all patients with neurologic involvement develop eye lesions called **Kayser-Fleischer rings**, green to brown deposits of copper in Descemet membrane in the limbus of the cornea.

Clinical Features

The age at onset of Wilson disease ranges from 6 to 40 years of age (average age is 11.4 years). Its clinical presentation is extremely variable. Some patients present with acute or chronic liver disease. Neurologic involvement may lead to movement disorders (tremor, poor coordination, chorea, or choreoathetosis) or rigid dystonia (spastic dystonia, masklike facies, rigidity, and gait disturbances). Patients may also have psychiatric symptoms. Hemolytic anemia may occur due to red cell membrane damage caused by oxidants produced by free copper. The

diagnosis of Wilson disease is usually based on the presence of decreased serum ceruloplasmin, increased hepatic copper content (the most sensitive and accurate test), and increased urinary excretion of copper (the most specific test). Sequencing of *ATP7B* is available for cases in which biochemical tests are indeterminate, such as in the setting of liver injury, which may cause serum ceruloplasmin levels to rise into the normal range. Demonstration of Kayser-Fleischer rings by slit examination is also diagnostically helpful. Early recognition and long-term copper chelation therapy (with d-penicillamine or trientine) or zinc-based therapy (which blocks uptake of dietary copper in the gut) is effective. Individuals with hepatitis or unmanageable cirrhosis require liver transplantation, which can be curative.

α_1 -Antitrypsin Deficiency

α_1 -Antitrypsin deficiency is an autosomal recessive disorder of protein folding marked by very low levels of circulating α_1 -antitrypsin (α_1 AT). The major function of this protein is the inhibition of proteases, particularly neutrophil elastase, cathepsin G, and proteinase 3, which are normally released from neutrophils at sites of inflammation. α_1 AT deficiency leads to the development of pulmonary emphysema because excessive activity of neutrophil elastase (Chapter 15). It also causes liver disease because of hepatocellular accumulation of the misfolded protein, an example of a “toxic gain-of-function” mutation.

α_1 AT is a small 394-amino acid plasma glycoprotein synthesized predominantly by hepatocytes. It is a member of the serine protease

inhibitor (serpin) family. The gene is very polymorphic; at least 75 forms of α_1 AT have been identified, denoted alphabetically by their relative migration in an isoelectric gel. The general notation is “Pi” for “protease inhibitor” and an alphabetic letter for the position in the gel; two letters denote the genotype of an individual’s two alleles. The most common genotype is PiMM, occurring in 90% of individuals (the “wild type”).

The most common clinically significant mutation is PiZ; homozygotes for the PiZZ protein have circulating α_1 AT levels that are only 10% of normal. These individuals are at high risk for developing clinical disease. Expression of alleles is autosomal codominant, and, consequently, PiMZ heterozygotes have intermediate plasma levels of α_1 AT. Among people of northern European descent, the PiZZ state affects 1 in 1800 live births. Because of its early presentation with liver disease, α_1 AT deficiency is the most commonly diagnosed inherited hepatic disorder in infants and children.

Pathogenesis

Disease-associated variants show a selective defect in transport of the protein from the endoplasmic reticulum to the Golgi apparatus; this is particularly characteristic of the PiZ polypeptide, resulting from the substitution of lysine for glutamine at amino acid position 342. The mutant polypeptide (α_1 AT-Z) misfolds and aggregates, creating endoplasmic reticulum stress and triggering the unfolded protein response, a signaling cascade that may lead to apoptosis (Chapter 2). All individuals with the PiZZ genotype accumulate α_1 AT-Z in the endoplasmic reticulum of hepatocytes, but only a small subset develops overt liver disease. Other undefined genetic factors or environmental factors are thus posited to play a role in the development of liver disease.

MORPHOLOGY

α_1 AT deficiency is characterized by the presence of round-to-oval **cytoplasmic globular inclusions in hepatocytes**, which are strongly periodic acid–Schiff (PAS)-positive and diastase-resistant (Fig. 18.27). The inclusions appear first in periportal hepatocytes in early and mild forms of the disease and with progression appear in central hepatocytes in more severe disease, such as that associated with the PiZZ variant. However, the number of globule-containing hepatocytes is not tightly correlated with disease severity, and patients may present with neonatal hepatitis prior to the appearance of the globules (generally 12 weeks of age or older for PiZZ-associated disease).

Clinical Features

The clinical findings and course are quite variable. Unusual neonatal presentations tend to be associated with severe disease and rapid progression to cirrhosis. Other patients present in adulthood with chronic hepatitis, cirrhosis, or hepatocellular carcinoma, which develops in 2% to 3% of PiZZ adults, usually (but not always) in the setting of cirrhosis. Liver transplantation is curative of hepatic disease but has no effect on the development and course of pulmonary disease (Chapter 15).

KEY CONCEPTS

Inherited Liver Disease

- Hereditary hemochromatosis is most commonly caused by loss-of-function mutations in the *HFE* gene, whose product regulates intestinal iron uptake by increasing hepcidin synthesis by the liver. It is characterized by increased absorption of dietary iron, and accumulation of iron in the liver and pancreas; injury to these organs results in cirrhosis and diabetes.

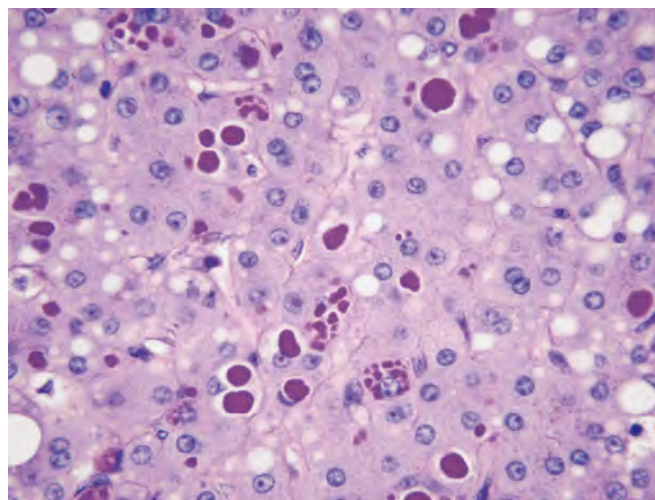


Fig. 18.27 Periodic acid–Schiff stain after diastase digestion of the liver, highlighting the characteristic magenta cytoplasmic granules of α_1 -antitrypsin deficiency.

- Wilson disease is caused by mutations that abolish the function of the metal ion transporter ATP7B, which results in accumulation of copper in the liver, brain (particularly basal ganglia), and eyes (Kayser–Fleisher rings).
- α_1 -Antitrypsin deficiency is a disease of protein misfolding that results in impaired secretion of α_1 -antitrypsin into the serum and creates endoplasmic reticulum stress, leading to hepatocyte injury through the unfolded protein response pathway. The main consequence of α_1 -antitrypsin deficiency is pulmonary emphysema because of unchecked elastase activity.

CHOLESTATIC DISEASE

Bile Formation and Secretion

Bile plays a critical role in elimination of bilirubin, excess cholesterol, xenobiotics, and trace metals such as copper, arsenic, selenium, and zinc, and its detergent action emulsifies dietary fat in the intestinal lumen, enabling its absorption by the gut. The major components of bile are bilirubin, bile salts, cholesterol, and phospholipids (mainly phosphatidylcholine).

Bilirubin is a toxic end product of heme degradation that is processed by the liver and excreted in the bile (Fig. 18.28). Hepatic handling of bilirubin involves uptake from the circulation, intracellular storage, conjugation with glucuronic acid, and excretion into bile. Most of the bilirubin produced daily (0.2 to 0.3 g, 85%) is derived from breakdown of effete red cells by macrophages in the spleen, liver, and bone marrow. The remaining bilirubin is produced by the turnover of hepatic proteins containing heme groups (e.g., P-450 cytochromes). Heme is converted to bilirubin by the action of several phagocyte enzymes and released into the blood, where it binds albumin, a necessary step for transport, as bilirubin is insoluble at physiologic pH. Uptake by hepatocytes at the sinusoidal membrane is followed by conjugation of bilirubin with one or two molecules of glucuronic acid in the endoplasmic reticulum, and excretion of the water soluble, nontoxic bilirubin glucuronides into bile. Most bilirubin glucuronides are deconjugated in the gut lumen by bacterial β -glucuronidases and degraded to colorless urobilinogens. The urobilinogens and the residue of intact pigment are largely excreted in feces. Approximately 20% of the urobilinogens formed are reabsorbed in the ileum and colon, returned to the liver, and re-excreted into bile. A small amount of reabsorbed urobilinogen is excreted in the urine.

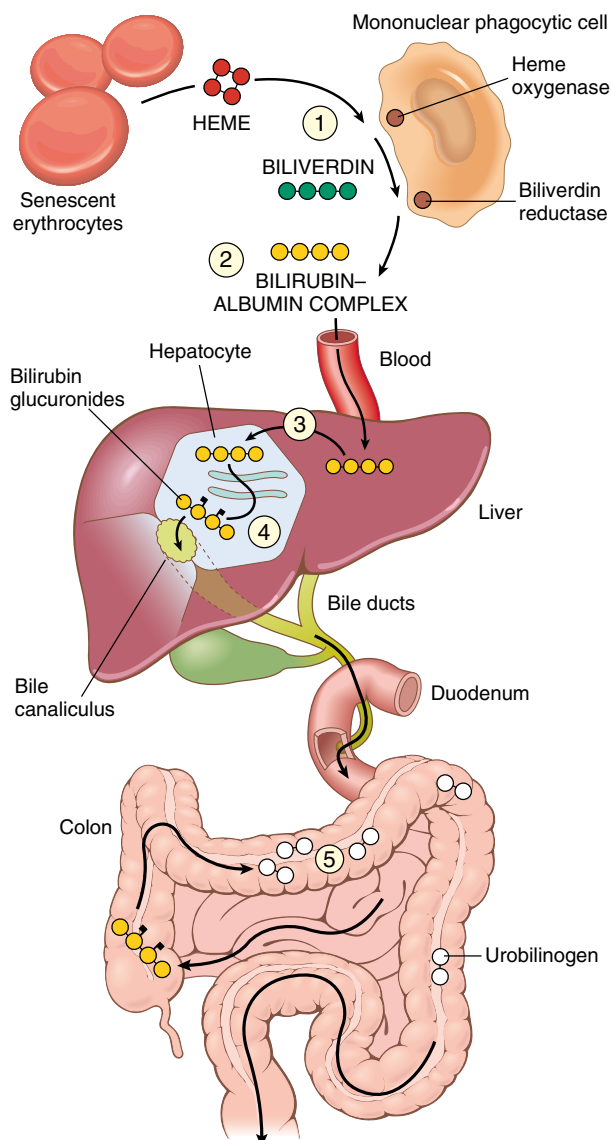


Fig. 18.28 Bilirubin metabolism and elimination. (1) Normal bilirubin production from heme (0.2 to 0.3 g per day) is derived primarily from the breakdown of aging red cells. (2) Extrahepatic bilirubin is bound to serum albumin and delivered to the liver. (3) Hepatocellular uptake and (4) glucuronidation in the endoplasmic reticulum generates bilirubin monoglucuronides and diglucuronides, which are water soluble and readily excreted into bile. (5) Gut bacteria deconjugate the bilirubin and degrade it to colorless urobilinogens. The urobilinogens and the residue of intact pigments are excreted in the feces, with some reabsorption and excretion into urine.

Bile salts are formed by the conjugation of bile acids with taurine or glycine. The predominant bile acids in humans are cholic acid and chenodeoxycholic acid, both of which are highly effective detergents. Bile acids combine with cholesterol and phospholipids to form micelles that solubilize cholesterol and reduce the toxic effect of bile acids on biliary epithelium. Most of the secreted bile acids, conjugated or unconjugated, are reabsorbed from the intestine and recirculate to the liver (enterohepatic circulation), where they are taken up by hepatocytes, thus helping to maintain the endogenous pool of bile acids.

The constituents of bile are ferried across the canalicular membrane of hepatocytes by a variety of transporter proteins. Some of the important ones are MRP2 (multidrug-resistant protein-2) for conjugated bilirubin, bile salt export pump (BSEP) for bile salts, MDR3 (multidrug resistance-3) for phosphatidylcholine, and sterolin-1 and -2 for cholesterol.

Pathophysiology of Hyperbilirubinemia

Serum bilirubin levels in the normal adult vary between 0.3 and 1.2 mg/dL, and the rate of bilirubin production is equal to the rate of hepatic uptake, conjugation, and biliary excretion. Jaundice becomes evident when the serum bilirubin levels rise above 2 to 2.5 mg/dL. Depending on the underlying etiology (summarized in Table 18.8), the elevation can predominantly involve unconjugated (indirect) or conjugated (direct) bilirubin. Testing for conjugated and unconjugated plasma bilirubin helps determine the cause of hyperbilirubinemia. Excess bilirubin production (e.g., due to hemolytic anemia or ineffective erythropoiesis) or defective conjugation (due to immaturity or hereditary causes) leads to the accumulation of unconjugated bilirubin. This form is largely insoluble and cannot be excreted in the urine. Although most unconjugated bilirubin is tightly bound to albumin in the blood, at excessive levels the unbound fraction rises and may diffuse into tissues, particularly the brain in infants, and produce neurologic damage (kernicterus). Conjugated hyperbilirubinemia most often results from hepatocellular disease, bile duct injury, and biliary obstruction. Since this form is water soluble and loosely bound to serum albumin, it can be excreted in the urine. Hepatobiliary disorders that lead to hyperbilirubinemia often show cholestasis, which refers to the retention of bilirubin and other solutes eliminated in bile due to impaired bile formation or obstruction of flow.

MORPHOLOGY

The hallmark of cholestasis is accumulation of green-brown plugs of bile pigment in hepatocytes and dilated canaliculi (Fig. 18.29). Rupture of canaliculi can lead to extravasation of bile, which is phagocytosed by Kupffer cells. Accumulation of bile salts in hepatocytes results in a swollen, foamy appearance of the cytoplasm ("feathery degeneration").

TABLE 18.8 Causes of Jaundice

Predominantly Unconjugated Hyperbilirubinemia

- Excess production of bilirubin
 - Hemolytic anemias
 - Resorption of blood from internal hemorrhage (e.g., alimentary tract bleeding, hematoma)
 - Ineffective erythropoiesis (e.g., pernicious anemia, thalassemia)
- Reduced hepatic uptake
 - Drug interference with membrane carrier systems
 - Some cases of Gilbert syndrome
- Impaired bilirubin conjugation
 - Physiologic jaundice of the newborn (decreased UGT1A1 activity, decreased excretion)
 - Breast milk jaundice (β -glucuronidases in milk)
 - Genetic deficiency of UGT1A1 activity (Crigler-Najjar syndrome types I and II, some cases of Gilbert syndrome)

Predominantly Conjugated Hyperbilirubinemia

- Deficiency of canalicular membrane transporters (Dubin-Johnson syndrome, Rotor syndrome)
- Hepatocellular disease (e.g., viral or drug-induced hepatitis, cirrhosis)
- Impaired bile flow from duct obstruction or autoimmune cholangiopathies

UGT1A1, Uridine diphosphate-glucuronyltransferase family, peptide A1.

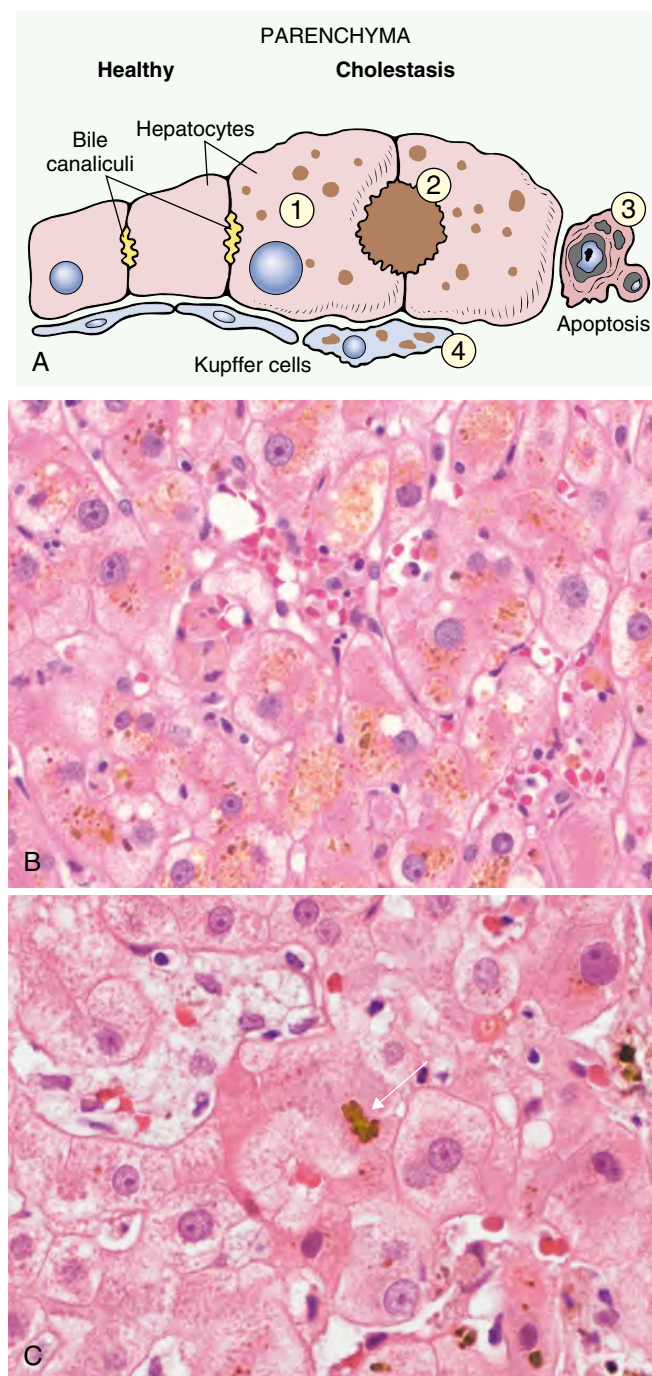


Fig. 18.29 Cholestasis. (A) Morphologic features of cholestasis (right) and comparison with normal liver (left). Cholestatic hepatocytes (1) are enlarged with dilated canalicular spaces (2). Apoptotic cells (3) may be seen, and Kupffer cells (4) frequently contain regurgitated bile pigments. (B) Intracellular cholestasis showing the bile pigments in the cytoplasm. (C) Bile plug (arrow) showing the expansion of bile canaliculus by bile.

Clinical Features

Elevated bilirubin becomes clinically evident as yellow discoloration of the skin (jaundice) and conjunctiva (icterus). Other manifestations include pruritus, skin xanthomas (focal accumulation of cholesterol), or symptoms related to intestinal malabsorption, including deficiencies of fat-soluble vitamins (vitamins A, D, and K). Elevated serum

alkaline phosphatase and γ -glutamyl transpeptidase (GGT), enzymes present on the apical (canalicular) membranes of hepatocytes and bile duct epithelial cells, are characteristic laboratory findings of cholestatic disease.

Physiologic Jaundice of the Newborn

Levels of UGT1A1 (uridine diphosphate-glucuronyltransferase family, peptide A1), the enzyme that is responsible for bilirubin glucuronidation, are low at birth and do not reach adult levels until 3 to 4 months of age. Hence, transient and mild unconjugated hyperbilirubinemia is nearly universal in the first week. Breastfeeding may exacerbate unconjugated hyperbilirubinemia, possibly because of the presence of bilirubin-deconjugating enzymes in breast milk. In most infants, phototherapy with blue light (which converts bilirubin to a soluble isomer that is readily excreted in the urine) is sufficient to keep the levels of unconjugated bilirubin within a safe range until the hepatic machinery for conjugation matures sufficiently.

Hereditary Hyperbilirubinemia

Genetic mutations can result in impaired uptake, conjugation, or secretion of bilirubin. *Crigler-Najjar syndrome type 1* is caused by severe UGT1A1 deficiency and is fatal around birth; in *Crigler-Najjar type 2* and *Gilbert syndrome*, there is residual UGT1A1 activity, leading to a milder phenotype. In contrast, *Dubin-Johnson syndrome* and *Rotor syndrome* are associated with conjugated hyperbilirubinemia. Both are autosomal recessive disorders and clinically innocuous. Dubin-Johnson syndrome is caused by mutation in the *MRP2* gene (multidrug-resistant protein 2), which is required for transport of non-bile salt organic anions at the canalicular membranes. Deposition of brown-black melanin-like pigment in the hepatocytes is a striking feature in this disease and can lead to blackening of the liver.

Large Bile Duct Obstruction

In adults, the common causes of obstruction of the large bile ducts are stones (choledocholithiasis), malignant neoplasms of the biliary tree or head of pancreas (usually adenocarcinoma), and strictures resulting from previous surgical procedures or ischemic injury. Primary sclerosing cholangitis (described later) may lead to an obstructive picture due to inflammatory injury to the large intrahepatic or extrahepatic bile ducts. In the pediatric setting, the common culprits are biliary atresia, cystic fibrosis, and choledochal cysts. The resulting cholestatic changes are reversible if the obstruction is corrected early in the disease course, but persistent obstruction can lead to fibrosis and biliary cirrhosis. Biliary obstruction also predisposes to ascending cholangitis, a bacterial infection of the biliary tree most commonly caused by enteric organisms such as coliforms and enterococci. Cholangitis usually presents with fever, chills, abdominal pain, and jaundice. Severe cases can result in abscess formation, sepsis, and death.

MORPHOLOGY

Obstruction of extrahepatic or large intrahepatic bile ducts leads to proximal duct dilation. The hallmark on liver biopsies is portal expansion due to edema, prominent ductular reaction at the portal-parenchymal interface, and infiltrating neutrophils associated with ductules ("**pericholangitis**") (Fig. 18.30). In ascending cholangitis, the neutrophils also involve the bile duct epithelium and lumens (Fig. 18.31). Persistent obstruction leads to fibrosis, which can eventually proceed to biliary cirrhosis (Fig. 18.32). Swelling of periportal hepatocytes (**feathery degeneration**), bile pigment, and Mallory hyaline is seen in periportal hepatocytes in advanced disease. Superimposed ascending cholangitis in advanced disease can precipitate acute-on-chronic liver failure.

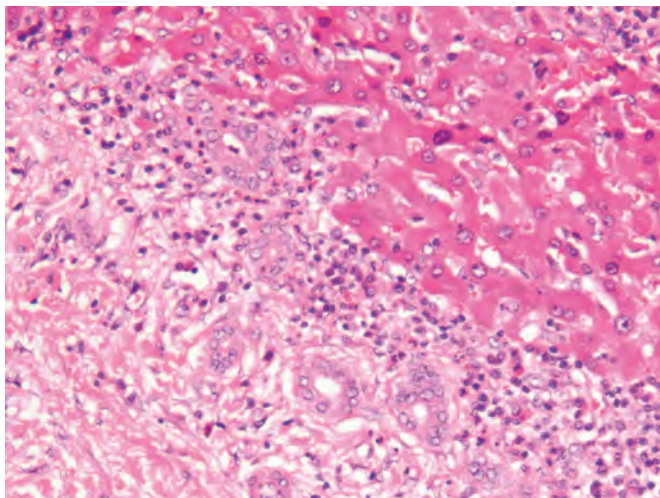


Fig. 18.30 Acute large duct obstruction. There is marked edema of the portal tract stroma indicated by empty spaces between portal collagen fibers and a ductular reaction with admixed neutrophils at the interface between portal tract and hepatocellular parenchyma.

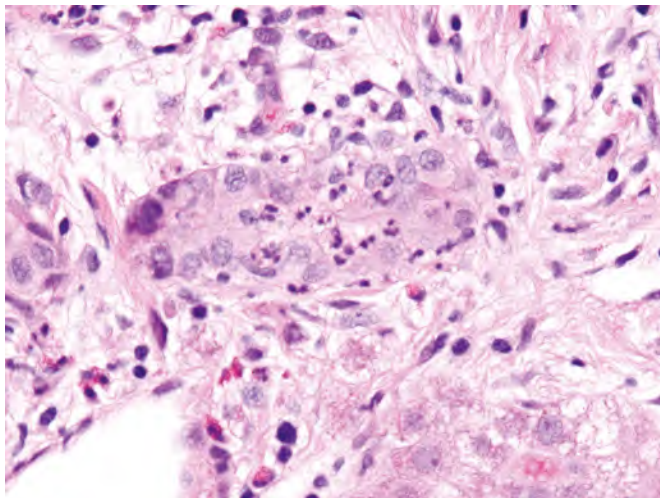


Fig. 18.31 Ascending cholangitis. Neutrophils are seen within the bile duct epithelial lining and within the lumen.

Cholestasis of Sepsis

Sepsis may affect the liver by several mechanisms: (1) through direct effects of intrahepatic bacterial infection (e.g., abscess formation or bacterial cholangitis); (2) ischemia relating to hypotension caused by sepsis (particularly when the liver is cirrhotic); or (3) in response to circulating microbial products. The latter effect is most likely to lead to the cholestasis of sepsis, particularly when the systemic infection is caused by gram-negative organisms. Characteristic morphologic findings in the setting of severe sepsis include canalicular cholestasis and bile plugs within dilated canals of Hering and bile ductules at the portal-parenchymal interface (“ductular” or “cholangiolar cholestasis”) (Fig. 18.33). Inflammation and hepatocellular injury are typically mild.

Primary Hepatolithiasis

At one time called “recurrent pyogenic cholangitis,” hepatolithiasis refers to stones in the intrahepatic bile ducts that can lead to repeated bouts of ascending cholangitis and progressive inflammatory destruction of

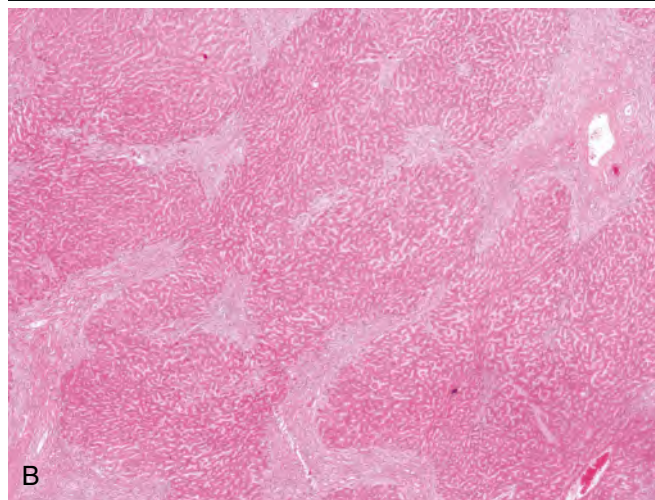


Fig. 18.32 Biliary cirrhosis. (A) Sagittal section through the liver demonstrates the nodularity (most prominent at the right) and bile staining of end-stage biliary cirrhosis. (B) Unlike other forms of cirrhosis, nodules of liver cells in biliary cirrhosis are often not round but irregular, like jigsaw puzzle shapes.

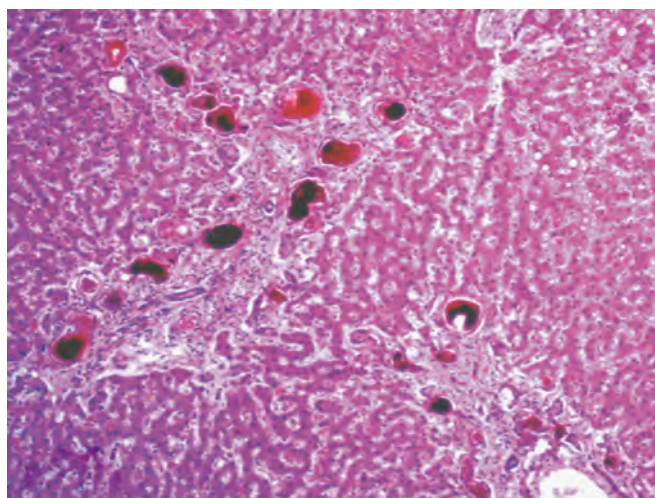


Fig. 18.33 Ductular cholestasis of sepsis. Large, dark bile concretions within markedly dilated canals of Hering and ductules at the portal-parenchymal interface. (Courtesy Dr. Jay Lefkowitz, Columbia University College of Physicians and Surgeons, New York.)

hepatic parenchyma. The disease is highly prevalent in East Asia, but is rare in other parts of the world. Its cause is uncertain; congenital duct abnormalities, diet, and chronic infection with bacteria or parasites have all been suggested as possible etiologies. Chronic inflammatory injury associated with hepatolithiasis is a risk factor for cholangiocarcinoma, particularly in Taiwan and to a lesser extent in Japan.

MORPHOLOGY

Pigmented calcium bilirubinate stones are present in distended intrahepatic bile ducts (Fig. 18.34). The ducts show chronic inflammation, mural fibrosis, and peribiliary gland hyperplasia. Obstruction of extrahepatic ducts is not present. Repeated bouts of inflammation, parenchymal collapse, and fibrosis can lead to a mass-like lesion, which can mimic a tumor on imaging.

Neonatal Cholestasis

Physiologic jaundice of the newborn resolves by 2 weeks, and infants who have jaundice beyond 14 to 21 days after birth must be evaluated for neonatal cholestasis. The major causes can be grouped under two broad categories: (1) obstructive biliary disease such as biliary atresia; and (2) nonobstructive etiologies that include paucity of bile ducts, infectious/metabolic diseases, bile transporter defects, and idiopathic neonatal hepatitis.

Extrahepatic Biliary Atresia

Extrahepatic biliary atresia is characterized by complete or partial obstruction of the extrahepatic biliary tree that usually presents within 2 to 6 weeks after birth. It accounts for one-third of neonatal cholestasis cases and for 50% to 60% of children referred for liver transplantation. In its most common perinatal form (80% of cases), the biliary tree is normally formed, and the disease onset is after birth. Infections, toxic agents, and autoimmune injury have been invoked, but the cause is unknown. The less common fetal form likely results from aberrant development of the extrahepatic biliary tree. Infants with extrahepatic biliary atresia present with jaundice, dark urine, light or acholic stools, and hepatomegaly.

MORPHOLOGY

Inflammation and fibrosis of the hepatic or common bile ducts is the hallmark of the disease and can extend to involve the intrahepatic ducts. Typical features of biliary obstruction are seen on liver biopsy, including portal edema, ductular reaction, and neutrophil infiltrates. If left uncorrected, cirrhosis can develop by 3 to 6 months of age.



Fig. 18.34 Hepatolithiasis. A resected, atrophic right hepatic lobe with characteristic findings including markedly dilated and distorted bile ducts containing large pigment stones and broad areas of collapsed liver parenchyma. (Courtesy Dr. Wilson M.S. Tsui, Caritas Medical Centre, Hong Kong.)

Clinical Features

Since extrahepatic biliary atresia requires surgical intervention (portoenterostomy or Kasai procedure), its differentiation from nonobstructive neonatal cholestasis is critical. Other etiologies of obstructive biliary diseases like cystic fibrosis must also be excluded. The combination of clinical presentation, imaging (ultrasound and hepatospecific iminodiacetic-acid [HIDA] scans), and biopsy can confirm the diagnosis in a vast majority of cases. Ultrasound findings of a small or absent gallbladder and fibrosis at the porta hepatis support the diagnosis of extrahepatic biliary atresia, as do HIDA scans with technetium-99m (^{99m}Tc), which is excreted in bile. In extrahepatic biliary atresia, there is total lack of secretion of ^{99m}Tc into the bile, and the biliary tree is not visualized. Involvement of ducts proximal to the porta hepatis, intrahepatic progression of disease, and ascending cholangitis are impediments to successful surgical therapy. Liver transplantation is the only option when surgical intervention is not feasible.

Nonobstructive Neonatal Cholestasis

A diverse group of disorders is associated with nonobstructive neonatal cholestasis. These can be divided into those in which cholestasis appears as part of a syndrome or as an isolated liver abnormality.

- *Alagille syndrome* is an autosomal dominant disorder that is associated with cholestasis and paucity of bile ducts as well as other abnormalities such as dysmorphic facies, butterfly-shaped vertebra, eye defects, and cardiac defects. It is caused by loss-of-function mutations in the Notch pathway involving either the genes encoding the ligand JAG1 or the receptor NOTCH2, both of which are required for normal development of the biliary tree.
- *Certain inborn errors of metabolism*, particularly galactosemia (Chapter 10) and Niemann-Pick disease (Chapter 5), can present with nonobstructive cholestasis.
- *Nonsyndromic causes* of nonobstructive cholestasis include α_1 -antitrypsin deficiency (discussed earlier) and diverse disorders of bile acid synthesis and bile transport. The latter are referred to collectively as progressive familial intrahepatic cholestasis (PFIC) and include FIC1 deficiency (Familial intrahepatic cholestasis 1, type 1 or Byler disease; *ATP8B1* mutation), BSEP deficiency (Bile salt export protein, type 2; *ABCB11* mutation) and MDR3 deficiency (Multidrug resistance protein 3, type 3; *ABCB4* mutation).

Neonatal hepatitis refers to a broad group of conditions that lead to a hepatitis and neonatal cholestasis. The main entities in this group are infections (including TORCH infections), hypopituitarism and genetic diseases like α_1 -antitrypsin deficiency and BSEP deficiency.

With recent advances, the underlying etiology can be determined in 85% to 90% of cases (summarized in Table 18.9).

MORPHOLOGY

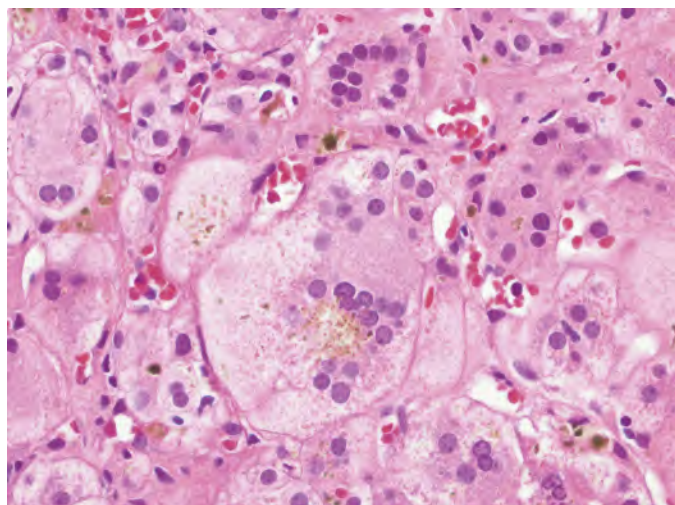
The most striking features are hepatocanalicular cholestasis and giant-cell or syncytial cell change with multinucleated hepatocytes (Fig. 18.35). Reactive Kupffer cells and extramedullary hematopoiesis are usually present. Decreased numbers of bile ducts within portal regions profiles is the most characteristic and constant feature of Alagille syndrome. Cases with hepatitis are characterized by lobular inflammation and hepatocellular apoptosis/necrosis.

Autoimmune Cholangiopathies

This section discusses the two main autoimmune disorders of bile ducts: primary biliary cholangitis and primary sclerosing cholangitis. The features of these two conditions are contrasted in Table 18.10.

TABLE 18.9 Major Causes of Neonatal Cholestasis

Obstructive Biliary Disease
Extrahepatic biliary atresia
Neonatal Infection
Cytomegalovirus
Bacterial sepsis
Urinary tract infection
Syphilis
Genetic Disorders
Metabolic diseases: Tyrosinemia, galactosemia
Lipid storage diseases: Niemann-Pick disease
Bile synthesis defects: Bile transport abnormalities (progressive intrahepatic cholestasis), bile acid synthesis defects
Cystic fibrosis
α_1 -Antitrypsin deficiency
Alagille syndrome (syndromic paucity of bile ducts)
Miscellaneous
Shock/hypoperfusion, drugs, total parenteral nutrition, hypopituitarism
Idiopathic neonatal hepatitis

**Fig. 18.35** Neonatal hepatitis. Note the multinucleated giant hepatocytes.

Primary Biliary Cholangitis

Primary biliary cholangitis (PBC) is an autoimmune disease characterized by inflammatory destruction of small- and medium-sized intrahepatic bile ducts. Large intrahepatic ducts and the extrahepatic biliary tree are not involved. Most patients are diagnosed in the early stages of disease, and hence the former name “primary biliary cirrhosis” is no longer used. PBC has a striking female predilection of 9:1, with a peak incidence between 40 and 50 years of age. The disease is most common in the United States and Northern Europe, and the incidence is low in Africa and the Indian subcontinent. Family members of PBC patients have an increased risk for development of the disease.

Pathogenesis

PBC is thought to be an autoimmune disorder resulting from a T cell-mediated attack on small interlobular bile ducts. The trigger for the attack is not known but may involve exposure to environmental factors like infections and toxic chemicals in genetically susceptible individuals. This may lead to expression of “autoantigens” on bile duct epithelial cells and their resultant destruction by T lymphocytes. The retention of bile salts due to bile duct injury leads to secondary hepatocellular injury which can eventually lead to cirrhosis.

Antimitochondrial antibody (AMA) directed against the E2 component of the pyruvate dehydrogenase complex (PDC-E2) is the most characteristic finding in PBC. PDC-E2-specific T cells are also present in PBC, further supporting the notion of an immune-mediated process. The role of AMA in the pathogenesis of PBC is unclear, as 5% of patients with otherwise typical PBC are AMA-negative. Moreover, antibody titers do not correlate with disease severity or disease progression, and they are not predictive of response to therapy. Autoantibodies against nuclear pore proteins and centromeric proteins may also be present.

MORPHOLOGY

The hallmark of PBC is lymphocytic infiltration and epithelial injury involving the small interlobular bile ducts. Poorly formed epithelioid granulomas are often present in the portal tracts and may be centered on bile ducts. **The histologic picture of lymphocytic and/or granulomatous bile duct destruction (florid duct lesion) is highly characteristic of PBC (Fig. 18.36).** Portal lymphoplasmacytic inflammation and ductular reaction are often present. The bile duct involvement has a patchy distribution, with involvement of a minority of portal tracts being a common feature in early disease. Disease progression

TABLE 18.10 Main Features of Primary Biliary Cholangitis and Primary Sclerosing Cholangitis

Parameter	Primary Biliary Cholangitis	Primary Sclerosing Cholangitis
Age	Median age 50 years	Median age 30 years
Sex	90% female	70% male
Associated conditions	Sjögren syndrome (70%), thyroid disease, scleroderma	Inflammatory bowel disease (70%)
Serology	95% AMA-positive, 40%–50% ANA positive	65% ANCA-positive; ANA variable, AMA typically negative 6% ANA positive
Radiology	Normal	Strictures and beading of large bile ducts; pruning of smaller ducts
Duct lesion	Florid duct lesions; loss of small ducts	Inflammatory destruction of extrahepatic and large intrahepatic ducts; fibrotic obliteration of medium and small intrahepatic ducts; ductular reaction in smaller portal tracts

AMA, Antimitochondrial antibody; ANA, antinuclear antibody; ANCA, antineutrophil cytoplasmic antibody.

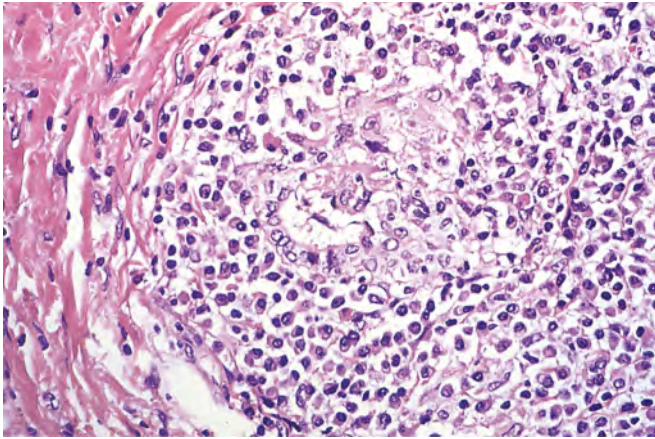


Fig. 18.36 Primary biliary cholangitis. A portal tract is markedly expanded by an infiltrate of lymphocytes and plasma cells surrounding a destructive granulomatous reaction centered on a bile duct ("florid duct lesion").

leads to loss of small intrahepatic bile ducts ("ductopenia"). Unlike obstructive, drug-induced, or sepsis-associated cholestasis, the bile accumulation in PBC is not centrilobular, but is seen in periportal/periseptal regions. Stasis of bile salts leads to swelling of periportal hepatocytes, which have clear cytoplasm with granular strands (**feathery degeneration**) and may develop Mallory hyaline. These periportal changes are referred to as cholate stasis and can be seen in any chronic biliary disease. End-stage disease culminates in cirrhosis. Like other

biliary diseases, cirrhotic nodules in PBC tend to be elongated ("garland shaped"), in contrast to the round nodules in hepatitic diseases. Some patients develop portal hypertension due to regenerative nodules without fibrosis, a feature called nodular regenerative hyperplasia. The basis of this phenomenon in PBC is not known.

Clinical Features

Symptomatic PBC usually presents with fatigue and pruritus, which increase slowly over time. Hypercholesterolemia is common. Splenomegaly and jaundice are seen in advanced disease. Other features that may be seen include skin hyperpigmentation, xanthelasmas, steatorrhea, and vitamin D malabsorption–related osteomalacia and/or osteoporosis. Other autoimmune diseases, such as Sjögren syndrome, systemic sclerosis, thyroiditis, rheumatoid arthritis, Raynaud phenomenon, and celiac disease, may also be present in affected patients.

An increasing number of cases are being detected in asymptomatic patients with elevated serum alkaline phosphatase levels. AMA are present in 90% to 95% of patients at all stages of the disease process and are highly characteristic. For uncertain reasons, many patients also have elevated serum levels of IgM. AMA can be detected by indirect immunofluorescence or enzyme-linked immunosorbent assay (ELISA); the latter has higher sensitivity and specificity as it selectively detects antibodies directed against the E2 subunit of the pyruvate dehydrogenase complex. The diagnosis can be established if two of the following are present: elevated alkaline phosphatase for more than 6 months; positive test for antimitochondrial antibodies; and characteristic histologic findings. The AMA-negative form of the disease has only two of these findings and is sometimes termed "autoimmune cholangiopathy" or "autoimmune cholangitis." Its clinical presentation, pathology, and natural history are otherwise similar.

PBC is slowly progressive and end-stage liver disease appears in 20% to 25% of patients over 15 to 20 years. Treatment with oral ursodeoxycholic acid, a naturally occurring bile acid, slows disease progression in most patients, but the response is inadequate in up to

40% of those affected; other medical therapies, such as with obeticholic acid (a synthetic bile acid) are used in cases that are refractory to ursodeoxycholic acid. Liver transplantation yields excellent results for cases with end-stage disease, with survival of more than 70% at 7 years. Like many chronic forms of hepatic injury, PBC is associated with an increased risk of hepatocellular carcinoma.

Primary Sclerosing Cholangitis

Primary sclerosing cholangitis (PSC) is characterized by inflammation and obliterative fibrosis of extrahepatic and large intrahepatic ducts and dilation of preserved segments. PSC tends to occur in the third through fifth decades and has a 2:1 male predominance.

Pathogenesis

The presence of circulating autoantibodies and T cells in the periductal stroma and associations with HLA-B8, other MHC alleles, and ulcerative colitis all support the idea that PSC is an immunologically mediated process. A combination of environmental and genetic factors is thought to trigger the inflammatory injury of bile ducts. First-degree relatives of patients with PSC have an increased risk of developing the disease, suggesting a genetic component. It has been proposed that T cells, activated in the damaged mucosa of patients with ulcerative colitis, migrate to the liver, where they recognize a cross-reacting bile duct antigen. Infections or changes in the intestinal microbiome have also been postulated to lead to alterations in cholangiocytes that incite inflammatory injury. Unlike in PBC, the autoantibody profile in PSC is not characteristic, but perinuclear antineutrophil cytoplasmic antibodies (pANCA) targeting a nuclear envelope protein are found in approximately 65% of patients.

MORPHOLOGY

The large intrahepatic bile ducts and extrahepatic ducts show epithelial injury and neutrophilic infiltrates superimposed on chronic inflammation. The edema and inflammation cause narrowing of the lumen, leading to fibrosis and strictures. Lithiasis can develop in the dilated ducts. The inflammatory injury may result in circumferential "onion skin" fibrosis (periductal fibrosis/sclerosis) around an increasingly atrophic duct lumen (Fig. 18.37), eventually leading to obliteration by a "tombstone" scar. The smaller intrahepatic bile ducts, which are more commonly sampled on biopsy, are not directly involved by inflammation, but may exhibit mild injury and prominent ductular reaction due to cholestasis. Progression of cholestasis and fibrosis culminates in biliary cirrhosis. Biliary intraepithelial neoplasia can develop and eventually lead to cholangiocarcinoma, a dreaded complication of primary sclerosing cholangitis.

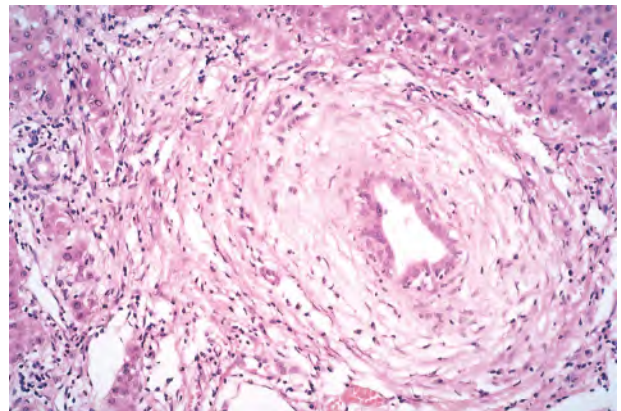


Fig. 18.37 Primary sclerosing cholangitis. A degenerating bile duct is entrapped in a dense, "onion skin" concentric scar.

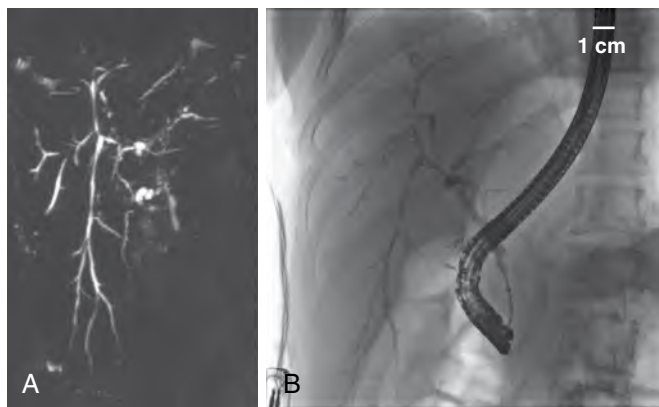


Fig. 18.38 Imaging studies of a patient with primary sclerosing cholangitis. (A) Magnetic resonance cholangiography shows focal dilation in some bile ducts (*bright, broad areas*) and strictures in others (*thinning or absence*). (B) Endoscopic retrograde cholangiography of the same patient shows nearly identical features as in A. The endoscope is visible, giving a sense of scale. (Courtesy Dr. M. Edwyn Harrison, MD, Mayo Clinic, Scottsdale, AZ.)

Clinical Features

The most common presenting symptoms are fatigue, pruritus, and jaundice. Nearly one-half of patients are asymptomatic at presentation and come to attention because of persistent elevation of serum alkaline phosphatase, particularly in ulcerative colitis patients who are routinely screened. Ascending cholangitis can also be the initial presentation. Chronic pancreatitis and chronic cholecystitis, from involvement of pancreatic ducts and gallbladder, are also seen. Fibrosis and eventually cirrhosis can occur.

The gold standard for diagnosis of PSC is the characteristic “beading” seen in the large intrahepatic and extrahepatic biliary tree by endoscopic/magnetic retrograde cholangiopancreatography (ERCP/MRCP), a pattern attributable to irregular biliary strictures and dilations (Fig. 18.38). Inflammatory bowel disease, particularly ulcerative colitis, affects approximately 70% of individuals with PSC, whereas 8% of inflammatory bowel disease patients develop PSC. In a small subset of cases, mostly in association with inflammatory bowel disease, there is involvement of smaller bile ducts only and the ERCP/MRCP examination is normal. This is referred to as small-duct PSC and can progress to typical large-duct PSC. Classic PSC needs to be distinguished from cholangitis that occurs in IgG₄-related sclerosing disease (Chapter 6), which is steroid-responsive and often associated with pancreatitis. In 5% to 10% of cases, autoimmune hepatitis is present along with PSC. Other causes that lead to a similar picture (secondary sclerosing cholangitis) include longstanding biliary obstruction (such as stricture), ischemic injury to the biliary tree, drugs (such as ketamine), late onset bile transporter defects (such as MDR3 deficiency) and infections (such as CMV, HIV, COVID-19).

The disease typically follows a protracted course leading to cirrhosis over 10 to 15 years. The lifetime risk of developing cholangiocarcinoma is 20%. There is no established medical therapy for PSC; clinical trials of ursodeoxycholic acid are ongoing. Cholestyramine, a bile acid-binding resin, is used to alleviate pruritus, and endoscopic dilation or stenting is used to relieve biliary obstruction. Liver transplantation is the treatment of choice for end-stage liver disease.

Vanishing Bile Duct Syndrome

This term has been used to describe acquired bile duct loss leading to ductopenia (loss of bile ducts in more than 50% of portal tracts). Vanishing bile duct syndrome is most commonly encountered with

drug-induced liver injury, but can also be observed with other etiologies of bile duct injury such as autoimmune biliary diseases (primary biliary cholangitis, primary sclerosing cholangitis), infections (CMV, EBV, COVID-19, HIV), ischemic injury, sarcoidosis, hematolymphoid neoplasms (especially Hodgkin lymphoma), graft vs. host disease (usually in patients with bone marrow transplant) and chronic ductopenic rejection (in liver allografts).

KEY CONCEPTS

Cholestatic Diseases

- Cholestasis occurs with impaired bile flow, leading to accumulation of bile pigment in the hepatic parenchyma. Hepatic etiologies include hepatocellular injury, metabolic defects in bile formation or secretion and inflammatory injury to bile ducts, while posthepatic etiologies include mechanical obstruction or inflammatory destruction of the extrahepatic bile ducts.
- Large bile duct obstruction is most commonly associated with gallstones and carcinomas involving the head of the pancreas. Ascending cholangitis may develop. Chronic obstruction can lead to secondary sclerosing cholangitis and cirrhosis.
- Cholestasis in sepsis may result from direct effects of intrahepatic bacterial infection, circulating microbial products, or ischemia relating to hypotension.
- Primary hepatolithiasis is a disorder of intrahepatic gallstone formation that is most common in East Asia and leads to repeated bouts of ascending cholangitis and inflammatory parenchymal destruction. It predisposes to cholangiocarcinoma.
- PBC is an autoimmune disease with inflammatory and often granulomatous destruction of small- to medium-sized intrahepatic bile ducts. It occurs most often in middle-aged females and is typically associated with AMA and with other autoimmune diseases (e.g., Sjögren syndrome and Hashimoto thyroiditis).
- PSC is an autoimmune disease that is more common in males and is strongly associated with inflammatory bowel disease, particularly ulcerative colitis. The diagnosis is established by visualization of the biliary tree. Histologically, the typical features are inflammation, fibrosis, and strictures involving large intrahepatic and extrahepatic bile ducts. Patients are at risk for developing cholangiocarcinoma.

STRUCTURAL ANOMALIES OF THE BILIARY TREE

Choledochal Cyst

Choledochal cysts are congenital dilations of the common bile duct. They present most often in children before age 10 as jaundice and/or recurrent abdominal pain, symptoms that are typical of biliary colic. Approximately 20% of cases become symptomatic only in adulthood. In some cases, choledochal cysts occur in conjunction with cystic dilation of the intrahepatic biliary tree (*Caroli disease*, described later). The female-to-male ratio is 3:1 to 4:1. These uncommon cysts may take the form of segmental or cylindric dilation of the common bile duct, diverticula of the extrahepatic ducts, or choledochoceles, which are cystic lesions that protrude into the duodenal lumen. Choledochal cysts predispose to stone formation, stenosis and stricture, pancreatitis, and obstructive biliary complications within the liver. There is also an increased risk of developing adenocarcinoma.

Fibropolycystic Disease

Fibropolycystic disease of the liver is a heterogeneous group of lesions in which the primary abnormality is congenital malformation of the biliary tree. Lesions may be found incidentally during radiographic studies, surgery, or autopsy. The most severe forms may manifest as hepatosplenomegaly or portal hypertension in the absence of hepatic dysfunction, starting in late childhood or adolescence. These lesions are part of

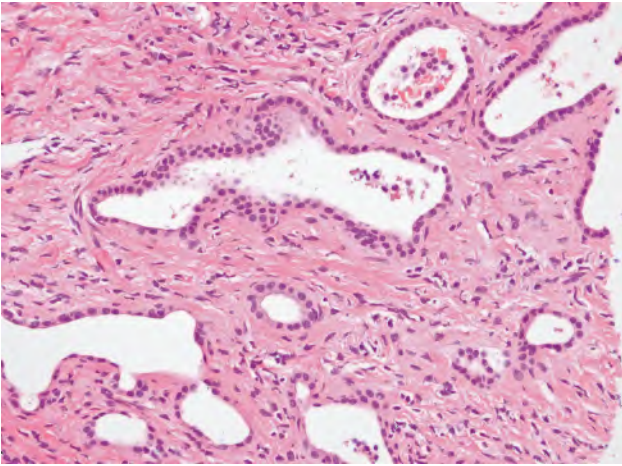


Fig. 18.39 von Meyenburg complex (bile duct hamartoma). Dilated irregular bile ducts with curvilinear outlines thought to represent ductal plate formation.

the spectrum of developmental ductal plate malformations, which result from persistence of the fetal periportal ductal plates. The caliber of involved portal tracts determines the size, morphology, and distributions of the lesions. Fibropolycystic liver disease often occurs in association with autosomal recessive polycystic renal disease (Chapter 20). These patients are at increased risk for cholangiocarcinoma.

Three sets of pathologic findings may be seen with polyfibrocystic disease, sometimes overlapping with each other:

- *von Meyenburg complexes* are small bile duct hamartomas (Fig. 18.39). Occasional von Meyenburg complexes are common in otherwise normal individuals. When they are diffuse, they signal the presence of underlying, clinically important fibropolycystic disease.
- *Single or multiple intrahepatic or extrahepatic biliary cysts*. When present in isolation, these may be symptomatic due to ascending cholangitis. Multifocal cystic dilation of the large intrahepatic bile ducts is referred to as *Caroli disease*. When the cystic dilation of the biliary tree occurs along with congenital hepatic fibrosis, the term *Caroli syndrome* is applied (Fig. 18.40). Ducts may be cystically dilated, but true cysts are also present. These may be intrahepatic cysts or choledochal cysts, as described earlier.
- *Congenital hepatic fibrosis* is characterized by portal tracts enlarged by broad bands of collagenous tissue forming septa that divide the liver into irregular islands (Fig. 18.41). Variable numbers of abnormally shaped bile ducts are embedded in the fibrous tissue, although they remain in continuity with the biliary tree. As with cirrhosis, individuals with congenital hepatic fibrosis are at risk for developing portal hypertension and its complications, particularly bleeding varices.

CIRCULATORY DISORDERS

Given the large volume of blood that flows through the liver, it is not surprising that circulatory disturbances have considerable impact on the organ. These disorders can be grouped according to whether blood flow into, through, or from the liver is impaired (Fig. 18.42).

Impaired Blood Flow Into the Liver

Hepatic Artery Compromise

While the liver's dual blood supply mitigates against infarction, obstruction of an intrahepatic branch of the hepatic artery by emboli,

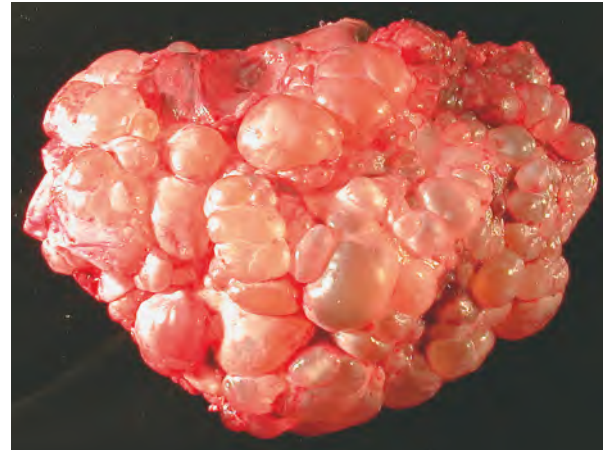


Fig. 18.40 Congenital hepatic fibrosis with multiple biliary cysts.

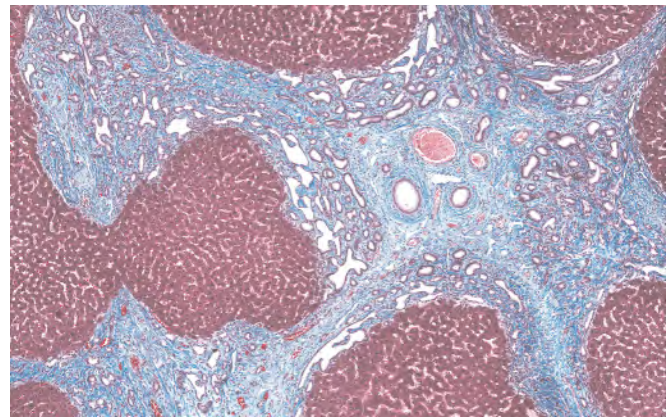


Fig. 18.41 Congenital hepatic fibrosis. Broad bands of fibrosis with dilated remnants of ductal plates. The intervening parenchyma is relatively normal (Masson Trichrome stain).

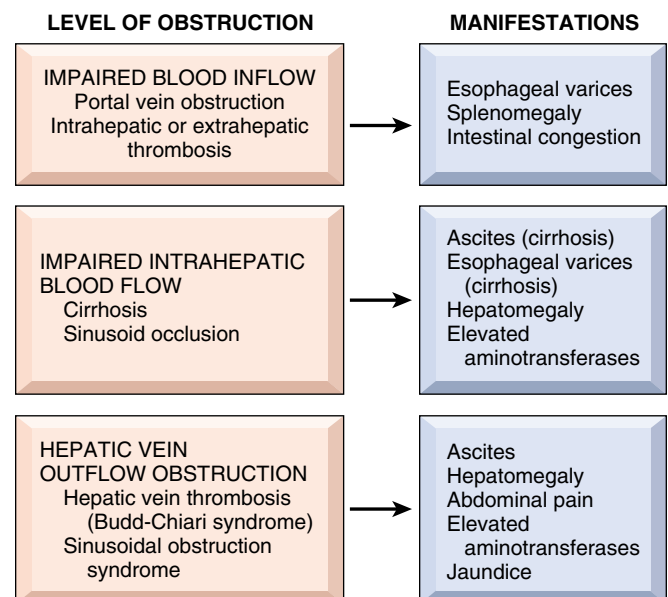


Fig. 18.42 Levels of obstruction and clinical manifestations of hepatic circulatory disorders.



Fig. 18.43 Liver infarct. A thrombus is lodged in a peripheral branch of the hepatic artery (arrow) and compresses the adjacent portal vein; the distal necrotic infarcted tissue has pale margins and multifocal areas of hemorrhage.

thrombi, or compression may result in a localized infarct that is either pale and anemic, or hemorrhagic if there is suffusion with portal blood (Fig. 18.43). Underlying causes include neoplasia, polyarteritis nodosa (Chapter 11), and sepsis. Interruption of blood flow through the main hepatic artery does not always produce ischemic necrosis of the organ, particularly if the liver is otherwise normal, as retrograde arterial flow through accessory vessels, coupled with portal venous supply, is usually sufficient to sustain the liver parenchyma.

Several conditions increase the vulnerability of the liver to infarction in the setting of hepatic artery obstruction. Hepatic artery thrombosis in a transplanted liver generally leads to infarction of the major ducts of the biliary tree, since their blood supply is entirely arterial, and may also lead to hepatic injury and regenerative changes. Large infarcts may occur with combined portal vein and hepatic artery thrombosis, which eventually result in subcapsular scarring.

Portal Vein Obstruction and Thrombosis

Blockage of the portal vein and its tributaries may be insidious and well tolerated, or may be a catastrophic, potentially lethal event; most cases fall somewhere in between. Occlusive disease of the portal vein or its major branches typically produces abdominal pain and, in most instances, manifestations of portal hypertension, principally esophageal varices that are prone to rupture.

The obstruction may occur at the level of the extrahepatic portal vein, intrahepatic portal vein radicles, or small portal vein branches. Extrahepatic portal vein obstruction may be idiopathic (approximately one-third of cases) or may arise from the following conditions:

- *Neonatal umbilical sepsis or umbilical vein catheterization*, often producing subclinical occlusion of the portal vein that presents as variceal bleeding and ascites years later
- *Intraabdominal infection* caused by acute diverticulitis or appendicitis, leading to pyelephlebitis in the splanchnic circulation
- *Inherited or acquired hypercoagulable states*, including myeloproliferative neoplasms such as polycythemia vera (Chapter 13), in which platelet abnormalities predispose to portal vein thrombosis
- *Trauma*, surgical or otherwise
- *Pancreatitis- and pancreatic cancer-associated splenic vein thrombosis*, which propagates into the portal vein
- *Invasion of the portal vein by hepatocellular carcinoma*

- *Cirrhosis*, which is associated with portal vein thrombosis in about 25% of patients. Many such patients have an underlying thrombophilic genotype (e.g., Factor V Leiden, Chapter 4).

Intrahepatic portal vein radicles are most likely to be obstructed by acute thrombosis, often in the setting of a malignancy or some other hypercoagulable state. By contrast, small portal vein branch obstruction may be seen in a variety of pathogenetically distinct conditions characterized by noncirrhotic portal hypertension:

- *The most common cause of small portal vein branch obstruction is schistosomiasis*; the eggs of the parasites and the associated granulomatous inflammatory response obstruct the smallest portal vein branches.
- *Other diseases associated with obstruction of small portal vein branches are collectively referred to as idiopathic noncirrhotic portal hypertension*. The pathogenesis is unknown; associations with prothrombotic states, infections (e.g., HIV), drugs, toxins, immune deficiencies, chronic biliary obstruction, and autoimmune diseases have been noted. Geographic variation in the incidence of noncirrhotic portal fibrosis and hypertension has been reported. It is particularly common in India, although the incidence appears to be declining. Patients often present with upper gastrointestinal bleeding. In East Asia, particularly Japan, there is a female predominance and patients present with splenomegaly, often in association with rheumatologic diseases. The disease is seen in untreated HIV infection and in those being treated with antiretroviral therapy, in whom it may represent a complication of treatment. Liver transplantation may be necessary to avoid fatal sequelae of portal hypertension in all these settings.

Impaired Blood Flow Through the Liver

The most common intrahepatic cause of blood flow impairment is cirrhosis, as described earlier. In addition, physical occlusion of sinusoidal blood flow occurs in a small group of diseases, including:

- *Sickle cell disease* (Fig. 18.44), due to obstruction by sickled red cells
- *Disseminated intravascular coagulation*, due to myriad small thrombi
- *Eclampsia* (discussed later)
- *Diffuse intrasinusoidal metastatic tumor*, due to physical obstruction by plugs of tumor, sometimes with superimposed thrombosis

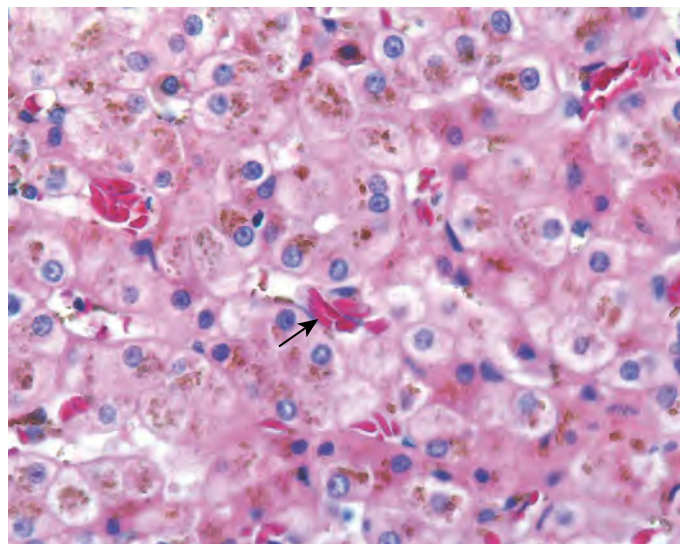


Fig. 18.44 Sickle cell crisis in liver. The photomicrograph shows several sinusoids containing "sickled" red cells (arrow).

In all of these scenarios, obstruction of blood flow may lead to massive necrosis of hepatocytes and acute liver failure.

Peliosis hepatis is a peculiar form of sinusoidal dilation that occurs in any condition in which efflux of hepatic blood is impeded. The liver contains blood-filled cystic spaces, either unlined or lined with sinusoidal endothelial cells. The pathogenesis is unknown. *Bartonella* species have been seen in the sinusoidal endothelial cells in AIDS-associated peliosis, but peliosis is also seen in cancer and with other infections, such as tuberculosis. Sex hormone administration (e.g., anabolic steroids, oral contraceptives, danazol) sometimes causes peliosis as well. While clinical signs are generally absent, potentially fatal intraabdominal hemorrhage or hepatic failure may occur. Lesions usually disappear after correction of the underlying cause.

Hepatic Venous Outflow Obstruction

Hepatic Vein Thrombosis

The obstruction of major hepatic veins produces liver enlargement, pain, and ascites, a condition known as **Budd-Chiari syndrome**. Hepatic damage is the consequence of increased intrahepatic blood pressure. Hepatic vein thrombosis is associated with myeloproliferative neoplasms such as polycythemia vera (Chapter 13), inherited disorders of coagulation (Chapter 4), antiphospholipid antibody syndrome, paroxysmal nocturnal hemoglobinuria (Chapter 14), and intraabdominal cancers, particularly hepatocellular carcinoma. In pregnancy or with oral contraceptive use, it occurs through interaction with an underlying thrombophilic disorder. Obstruction of a single main hepatic vein by thrombosis is clinically silent.

MORPHOLOGY

In Budd-Chiari syndrome, the liver is swollen and red-purple and has a tense capsule (Fig. 18.45). There may be areas of hemorrhagic collapse alternating with preserved or regenerating parenchyma, with the patterns being dependent on which small and large hepatic veins are obstructed. Microscopically, the affected hepatic parenchyma reveals severe centrilobular congestion and necrosis. Pericentral/sinusoidal fibrosis develops in instances in which thrombosis develops more slowly. The major veins contain thrombi showing various degrees of organization.

Clinical Features

The mortality of untreated acute hepatic vein thrombosis is high. Prompt surgery to create a portosystemic venous shunt permits reverse flow through the portal vein and improves the prognosis. The chronic form is far less lethal, and more than two-thirds of patients are alive after 5 years.

Sinusoidal Obstruction Syndrome

Originally described in Jamaican drinkers of pyrrolizidine alkaloid-containing bush tea and named *veno-occlusive disease*, sinusoidal



Fig. 18.45 Budd-Chiari syndrome. Thrombosis of the major hepatic veins has caused hemorrhagic liver necrosis.

obstruction syndrome now occurs primarily in two settings: (1) following allogeneic (or less commonly, autologous) hematopoietic stem cell transplantation, usually within the first 3 weeks (though changes in conditioning regimens have lowered the incidence); and (2) in cancer patients receiving certain forms of chemotherapy. The mortality rate is up to 80% in severe disease.

Pathogenesis

Sinusoidal obstruction syndrome arises from toxic injury to the sinusoidal endothelium. Injured, sloughed endothelium obstructs sinusoidal blood flow, and associated debris accumulates in the terminal hepatic vein. Red cells enter the space of Disse in the disrupted sinusoids, and the cessation of blood flow often leads to necrosis of perivenular hepatocytes.

MORPHOLOGY

Sinusoidal obstruction syndrome is characterized by obliteration of the terminal hepatic venules by swollen or necrotic endothelium, edema, and eventual collagen deposition. In acute disease, there is centrilobular congestion, hepatocellular necrosis, and accumulation of hemosiderin-laden macrophages. In advanced lesions, the obliterated lumens of venules can be highlighted by special stains for connective tissue (Fig. 18.46). In chronic or healed sinusoidal obstruction syndrome, complete fibrous obliteration of the venule may follow.

Clinical Features

Although histology is the gold standard for diagnosis, liver biopsy is often risky in these patients. As a result, the diagnosis is typically made on clinical grounds based on the findings of tender hepatomegaly, ascites, weight gain, and jaundice, as well as reversed or attenuated hepatic venous flow on Doppler ultrasonography. Early results suggest that treatment with anticoagulants and ursodeoxycholate may lower the incidence and severity of sinusoidal obstruction syndrome following hematopoietic stem cell transplantation.

Passive Congestion and Centrilobular Necrosis

These hepatic manifestations of systemic circulatory compromise—passive congestion and centrilobular necrosis—are considered together because they represent a morphologic continuum. Both changes are commonly seen at autopsy because there is an element of preterminal circulatory failure in virtually every nontraumatic death.

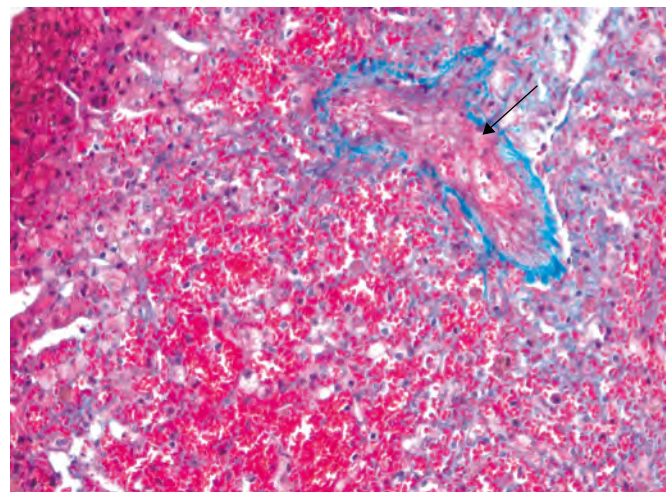


Fig. 18.46 Sinusoidal obstruction syndrome. Collagen stain reveals marked sinusoidal congestion, hepatocyte atrophy and loss, and organizing thrombus within the vein lumen (arrow). (Masson trichrome stain.)

MORPHOLOGY

Right-sided cardiac decompensation leads to passive congestion of the liver. The liver is slightly enlarged, tense, and cyanotic, with rounded edges. Microscopically, there is **congestion of dilated centrilobular sinusoids**. With time, centrilobular hepatocytes become atrophic, resulting in markedly attenuated liver cell plates.

Left-sided cardiac failure or shock may lead to hepatic hypoperfusion and hypoxia of hepatocytes around central veins. The combination of hypoperfusion and retrograde congestion acts synergistically to cause **centrilobular hemorrhagic necrosis**. The liver takes on a variegated mottled appearance, reflecting hemorrhage and necrosis in the centrilobular regions (Fig. 18.47A). This finding is known as **nutmeg liver** due to its resemblance to the cut surface of a nutmeg. In most instances, the only clinical evidence of centrilobular necrosis or its variants is transient elevation of serum aminotransferases, but the parenchymal damage may be sufficient to induce mild to moderate jaundice.

By microscopy, there is a sharp demarcation of viable periportal and necrotic or atrophic pericentral hepatocytes, with suffusion of blood through the centrilobular region (Fig. 18.47B). With sustained chronic severe congestive heart failure, **cardiac sclerosis** develops with centrilobular pericellular fibrosis, sometimes with bridging fibrous septa.

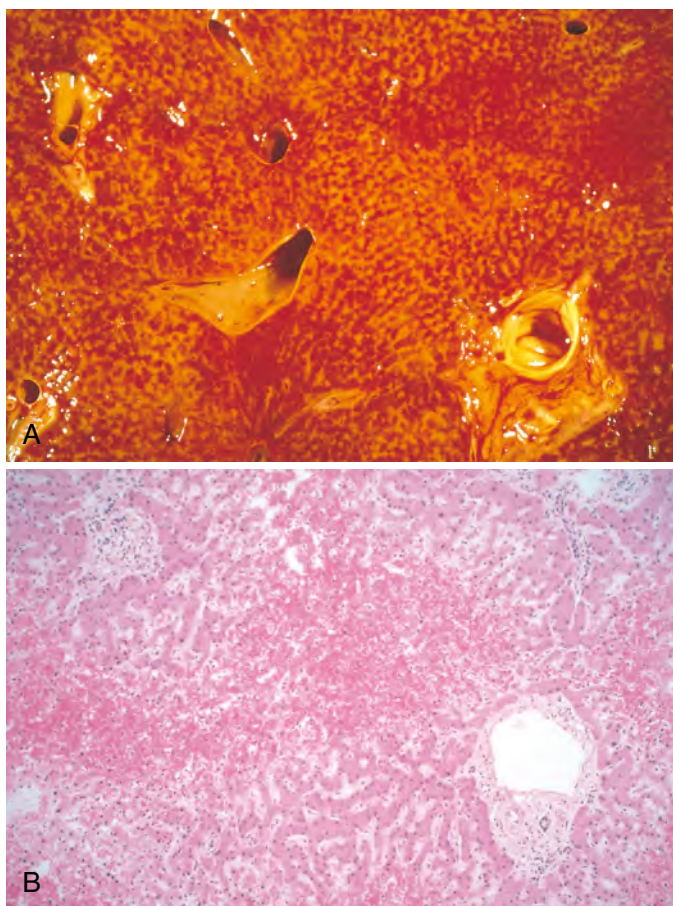


Fig. 18.47 Acute passive congestion (“nutmeg liver”). (A) The cut surface of the liver has a variegated mottled red appearance, representing congestion and hemorrhage in the centrilobular regions of the parenchyma. (B) On microscopic examination, the centrilobular region is suffused with red cells, and atrophied hepatocytes are not easily seen. Portal tracts and the periportal parenchyma are intact.

KEY CONCEPTS

Circulatory Disorders

- Circulatory disorders of the liver can be caused by impaired blood inflow, defects in intrahepatic blood flow, and obstruction of blood outflow.
- Portal vein obstruction by intrahepatic or extrahepatic thrombosis may cause portal hypertension, esophageal varices, and ascites.
- The most common cause of impaired intrahepatic blood flow is cirrhosis.
- Obstructions of blood outflow include hepatic vein thrombosis (Budd-Chiari syndrome) and sinusoidal obstruction syndrome, previously known as veno-occlusive disease.

HEPATIC DISEASE ASSOCIATED WITH PREGNANCY

Some forms of hepatic disease may be exacerbated by pregnancy. Viral hepatitis (HAV, HBV, HCV, or HBV+HDV) is the most common cause of jaundice in pregnancy. Among these infections, pregnancy only alters the course of HEV infection, which, for unknown reasons, has a fatality rate approaching 20% in pregnant patients. The liver may also be secondarily involved by other infections during pregnancy. These include hepatitis caused by herpes simplex virus, a rare cause of acute liver failure in pregnancy, and liver abscess caused by *Listeria monocytogenes*, an organism that thrives in placental tissue, from where it may seed the liver.

Abnormal liver tests occur in 3% to 5% of pregnancies, but in most cases, these are of no clinical importance. In a very small subgroup of pregnant females (0.1%), more serious hepatic complications develop. These disorders include preeclampsia and eclampsia, acute fatty liver of pregnancy, and intrahepatic cholestasis of pregnancy. In extreme cases, eclampsia and acute fatty liver of pregnancy may be fatal.

Preeclampsia and Eclampsia

Preeclampsia affects up to 10% of pregnancies and is characterized by maternal hypertension, proteinuria, peripheral edema, and coagulation abnormalities (see Chapter 22 for detailed discussion). When hyperreflexia and convulsions occur, the condition is called *eclampsia* and may be life-threatening. Alternatively, subclinical hepatic disease may be the primary manifestation of preeclampsia, as part of a syndrome of hemolysis, elevated liver enzymes, and low platelets, dubbed the *HELLP syndrome*. Here we focus on the hepatic pathology of these entities.

MORPHOLOGY

In preeclampsia, **periportal sinusoids contain fibrin deposits associated with hemorrhage into the space of Disse**, leading to periportal hepatocellular coagulative necrosis. In eclampsia, blood under pressure may coalesce and expand to form a hepatic hematoma; dissection of blood under Glisson capsule may lead to catastrophic hepatic rupture (Fig. 18.48).

Clinical Features

Patients with hepatic involvement in preeclampsia may show modest to severe elevation of serum aminotransferases and mild elevation of serum bilirubin. Hepatic dysfunction sufficient to cause a coagulopathy signifies advanced and potentially lethal disease. Mild cases may be managed conservatively. Termination of pregnancy is required in severe cases. Females who survive mild or severe preeclampsia recover without sequelae.

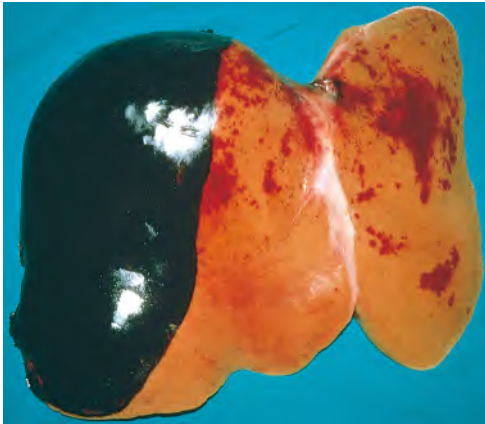


Fig. 18.48 Eclampsia. Subcapsular hematoma dissecting under Glisson capsule in a fatal case. (Courtesy Dr. Brian Blackbourne, Office of the Medical Examiner, San Diego, CA.)

Acute Fatty Liver of Pregnancy

Acute fatty liver of pregnancy presents with a spectrum of disease ranging from subclinical or modest hepatic dysfunction (evidenced by elevated serum aminotransferase levels) to hepatic failure, coma, and death. It is a rare disease affecting approximately 1 in 16,000 pregnant females.

Pathogenesis

The pathogenesis of this disease in some cases involves an inherited disorder of metabolism. In a subset of patients, both mother and father carry a mutation in the gene encoding mitochondrial long-chain 3-hydroxyacyl coenzyme A (CoA) dehydrogenase. Offspring who inherit both mutant alleles are deficient in the enzyme. The fetus and placenta fare well, but this deficiency results in hepatic dysfunction in the mother, because long-chain 3-hydroxyacyl metabolites produced by the fetus, or placenta, enter the maternal circulation and cause hepatic toxicity. Thus, this is a rare instance of the fetus causing metabolic disease in the mother.

MORPHOLOGY

The diagnosis of acute fatty liver of pregnancy rests on biopsy identification of characteristic **diffuse microvesicular steatosis** in hepatocytes. In severe cases, there may be lobular disarray with hepatocyte dropout, reticulin collapse, and portal tract inflammation, making distinction from viral hepatitis difficult. Cholestasis may also be present.

Clinical Features

While this condition usually runs a mild course, females with acute fatty liver of pregnancy can progress within days to hepatic failure and death. The primary treatment is termination of pregnancy. Affected females present in the latter half of pregnancy, usually in the third trimester. Symptoms are directly attributable to incipient hepatic failure, including bleeding, nausea and vomiting, jaundice, and coma. In 20% to 40% of cases, the presenting symptoms may be those of coexistent preeclampsia.

Intrahepatic Cholestasis of Pregnancy

The onset of pruritus in the second or third trimester, followed in some cases (10% to 25%) by darkening of the urine, and occasionally light stools and jaundice, heralds the development of this enigmatic syndrome, which resolves within 2 to 3 weeks of delivery. Serum bilirubin

(mostly conjugated) rarely exceeds 5 mg/dL; alkaline phosphatase may be slightly elevated. The level of bile salts is increased greatly. The altered hormonal state of pregnancy likely combines with environmental factors and transporter protein defects to engender canalicular cholestasis. There is a modest risk of fetal loss, and the condition may recur in subsequent pregnancies. The pruritus resulting from retention of bile salts can be extremely distressing for the pregnant mother.

NODULES AND TUMORS

Hepatic mass lesions include non-neoplastic processes like focal nodular hyperplasia, regenerative nodules, and abscess or other infectious processes. Neoplastic mass lesions include benign hepatocellular adenoma; malignant primary liver epithelial neoplasms, such as hepatocellular carcinoma or intrahepatic cholangiocarcinoma; and nonepithelial malignancies, such as angiosarcoma and metastatic tumors.

Nonneoplastic Mass Lesions

Focal Nodular Hyperplasia

Focal nodular hyperplasia (FNH) is a benign, nonneoplastic lesion that is most common in adult females.

Pathogenesis

FNH is thought to result from altered blood flow that leads to hyperplastic changes in hepatocytes. FNH-like nodules can occur adjacent to other tumors and in vascular conditions such as hemangiomas and Budd-Chiari syndrome. Multifocal FNH can occur in association with hepatic or extrahepatic hemangioma, vascular malformations, or brain tumors, such as meningioma and astrocytoma.

MORPHOLOGY

FNH is well circumscribed, typically lacks a capsule, and has a **central stellate scar** in up to 80% of cases (Fig. 18.49A). Most lesions are 5 cm or less in diameter. Microscopically, fibrous septa radiate from the central scar dividing the hepatic parenchyma into nodules. The fibrous septa and central scar contain thick-walled arteries with intimal and fibromuscular hyperplasia and poorly formed elastic lamina (Fig. 18.49B). A prominent ductular reaction is often present, but interlobular bile ducts are absent. The hepatocytes in the nodules are arranged in one- to two-cell-thick plates; cytologic and architectural atypia is not present. Immunohistochemistry for glutamine synthetase (GS), an enzyme normally present in centrilobular hepatocytes, is very useful for diagnosis. In FNH, a highly characteristic **map-like pattern of strong cytoplasmic GS staining** is seen within anastomosing groups of hepatocytes.

Clinical Features

Most cases are asymptomatic and come to clinical attention during imaging or surgery. The presence of a central scar is a highly characteristic finding and helps in establishing the diagnosis on imaging. FNH is a benign lesion and does not need any treatment. Resection is performed for large symptomatic lesions.

Other Nonneoplastic Mass Lesions

Nonneoplastic mass forming lesions like abscesses, granulomas, and inflammatory pseudotumors can mimic true neoplasms. Large regenerative nodules can occur in noncirrhotic liver in vascular conditions such as Budd-Chiari syndrome, hemangioma, and vascular malformations. In some conditions, the entire liver becomes nodular without significant fibrosis, a condition referred to as *nodular regenerative hyperplasia*, which can lead to portal hypertension and mimic cirrhosis on imaging.

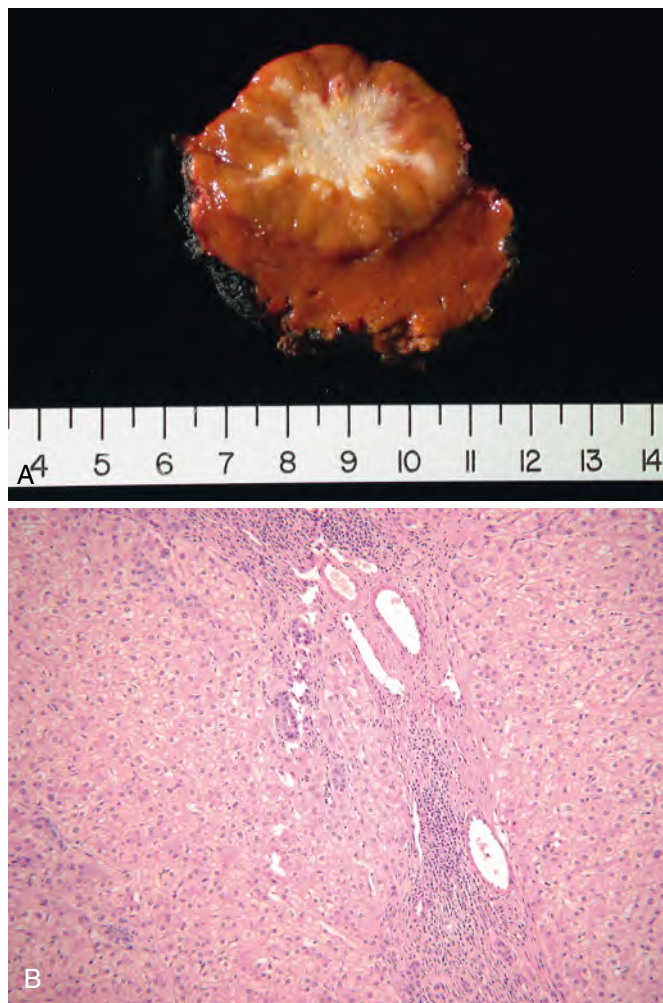


Fig. 18.49 Focal nodular hyperplasia. (A) Resected specimen showing lobulated contours and a central stellate scar. (B) Low-power micrograph showing a broad fibrous scar with thick-walled arteries and ductular reaction, but no interlobular bile ducts.

Benign biliary proliferations include bile duct adenoma and biliary hamartoma (von Meyenburg complex, described earlier). Both are asymptomatic and generally less than 2 cm. These lesions come to clinical attention as tiny nodules or cysts on imaging, or as small subcapsular lesions observed during abdominal surgery when they can be mistaken for metastatic disease. Both lesions show haphazard bile ducts in a fibrous stroma and lack atypical cytologic or architectural features. The ductular profiles in bile duct adenoma tend to be round and non-dilated, while those in biliary hamartoma are dilated, curvilinear, and often show inspissated bile or secretions. The latter are thought to be a remnant of ductal plate malformation. The recent identification of pathogenic acquired *BRAF* mutations in most bile duct adenomas indicates that this is a benign neoplasm, settling a longstanding debate.

Benign Neoplasms

Cavernous Hemangioma

Cavernous hemangioma is the most common benign liver tumor. It appears as a discrete red-blue, soft nodule, usually less than 2 cm in diameter, generally in a subcapsular location. Histologically, the tumor consists of dilated thin-walled vascular channels (Fig. 18.50). Most are asymptomatic and are detected incidentally by imaging.

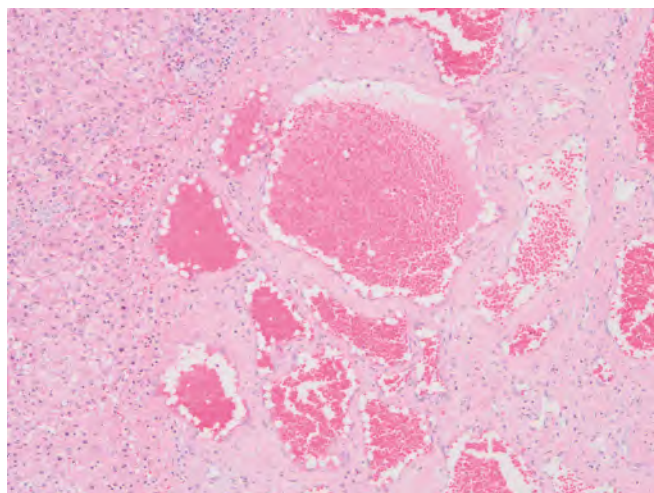


Fig. 18.50 Cavernous hemangioma. Blood-filled vascular channels separated by a dense fibrous stroma.

Hepatocellular Adenoma

Hepatocellular adenoma is a benign neoplasm that typically occurs in young females and is strongly associated with use of oral contraceptives and anabolic steroids. The incidence has increased in the past decade, presumably related to obesity and metabolic syndrome. Multiple hepatocellular adenomas (termed *hepatic adenomatosis* when 10 or more tumors are present) can occur both in familial and acquired settings.

Pathogenesis

Four molecular subtypes have been defined, each associated with distinct clinicopathologic features and varying risk of transformation into hepatocellular carcinoma.

- *Hepatocyte nuclear factor 1-alpha (HNF1α)*-inactivated hepatocellular adenoma. These tumors have loss-of-function mutations in the *HNF1A* gene, which encodes a transcription factor that regulates many genes in hepatocytes. Heterozygous germline mutations are responsible for autosomal dominant maturity onset diabetes of the young, type 3, and are also associated with 10% of *HNF1α*-inactivated hepatocellular adenomas. This subtype accounts for 30% to 35% of cases, has a strong female predilection, and is associated with minimal risk of transformation to hepatocellular carcinoma.
- *Inflammatory hepatocellular adenoma* is a subtype that results from acquired activating mutations in gp130, a coreceptor for IL-6, that lead to constitutive activation of JAK-STAT signaling. This subtype accounts for 35% to 50% of cases, is more common in females, and is associated with obesity and metabolic syndrome. β -catenin-activating mutations are also present in 10% of inflammatory hepatocellular adenomas, and these tumors have a higher risk of malignant transformation.
- *β -catenin-activated hepatocellular adenoma* is defined by activating mutations in the β -catenin gene (*CTNNB1*) or in other components of the Wnt pathway (such as *APC*, *AXIN2*). Mutations involving exon 3 of *CTNNB1* are associated with high levels of β -catenin activation and these tumors are at high risk for malignant transformation to hepatocellular carcinoma. Anabolic steroid use is a well-known risk factor for these tumors. Nearly 40% occur in males. Mutations involving exon 7 or 8 of *CTNNB1* are associated with low level of β -catenin activation and have a low association (<5%) with hepatocellular carcinoma.
- *Sonic hedgehog hepatocellular adenoma* is a recently described subtype driven by *GLI1* fusion genes leading to activation of the sonic hedgehog pathway. These tumors have a high propensity for hemorrhage.

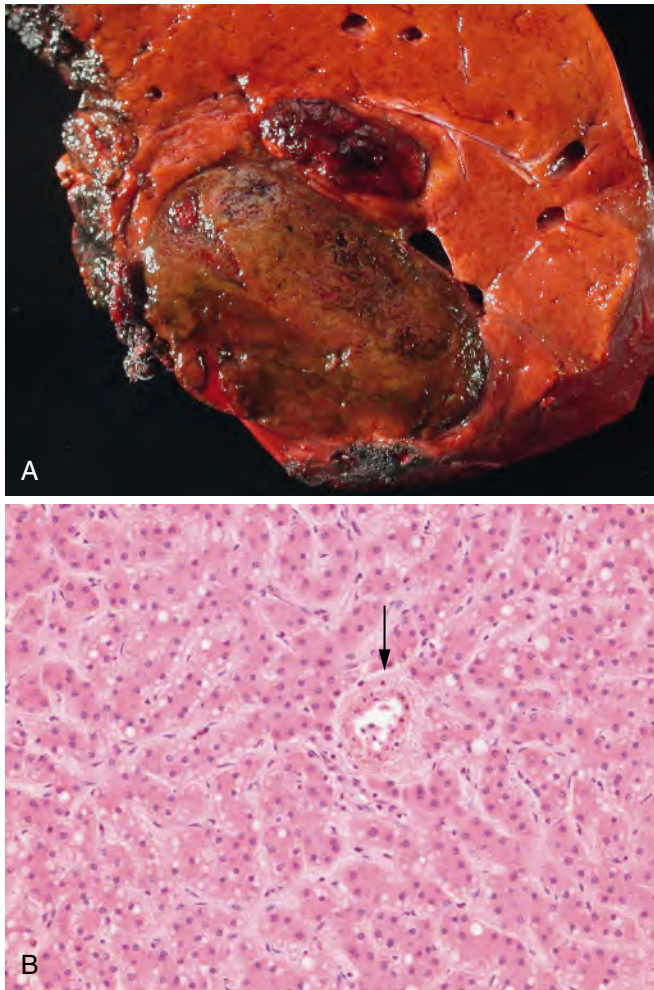


Fig. 18.51 Hepatocellular adenoma. (A) Resected specimen showing a well-defined tan mass in the liver. (B) Microscopic view showing thin cords of hepatocytes, with an arterial vascular supply (*arrow*) and no portal tracts.

MORPHOLOGY

Hepatocellular adenomas appear as discrete masses, sometimes with associated hemorrhage, composed of hepatocytes arranged in one- to two-cell-thick plates (Fig. 18.51). Large arteries are present within a small amount of fibrous stroma, while interlobular bile ducts and normal portal tracts are absent. *HNF1α*-mutated tumors typically show prominent fat in the lesional hepatocytes and loss of expression of liver fatty acid-binding protein (LFABP), a protein regulated by *HNF1α*, (Fig. 18.52A). β -Catenin–mutated tumors often show atypical cytologic or architectural features, and areas showing overt transformation to hepatocellular carcinoma may be present. Due to β -catenin mutations or other mutations that activate Wnt signaling, strong nuclear localization of β -catenin can be seen in the tumor cells (see Fig. 18.52B). Inflammatory hepatocellular adenomas typically show sinusoidal dilation and may have features mimicking focal nodular hyperplasia, such as fibrous septa and ductular reaction. Because of hyperactive JAK-STAT signaling, these tumors overexpress acute-phase reactants, such as C-reactive protein and serum amyloid A (Fig. 18.52C).

Clinical Features

Hepatocellular adenoma may be detected incidentally by imaging or come to attention due to symptom-related abdominal pain or hemorrhagic necrosis. Rupture of hepatocellular adenomas can cause massive abdominal bleeding and represents a surgical emergency. Resection is

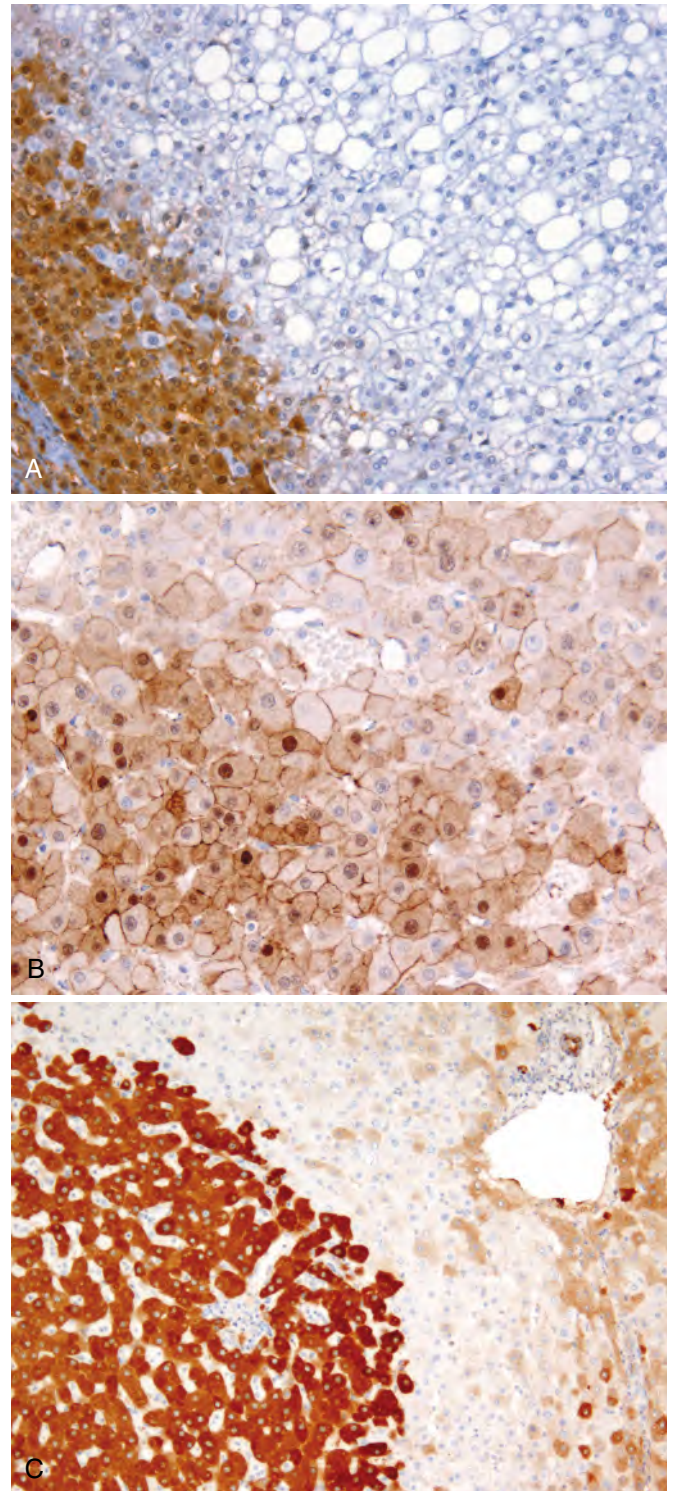


Fig. 18.52 Molecular subtypes of hepatocellular adenoma. (A) *HNF1α*-inactivated hepatocellular adenoma. Liver fatty acid-binding protein (LFABP, expression of which depends on *HNF1α*) is absent in the tumor by immunostaining and is present in adjacent normal hepatocytes (*lower left*). (B) Hepatocellular adenoma with β -catenin activation. Note nuclear immunostaining in some neoplastic hepatocytes (compared to other tumor hepatocytes that maintain membranous staining). (C) Inflammatory hepatocellular adenoma. There is marked up-regulation of C-reactive protein in neoplastic hepatocytes, compared to the highly variable and low-level expression in adjacent hepatic parenchyma. (Immunostain with DAB [brown] and hematoxylin counterstain.) (A, Courtesy Dr. Valerie Paradis, Beaujon Hospital, Paris.)

recommended in male patients (irrespective of other high-risk features) for β -catenin-activated tumors as well as tumors 5 cm or larger due to the risk of hemorrhage and malignant transformation. For smaller tumors without β -catenin activation, close follow-up and cessation of exposure to oral contraceptive or anabolic steroids may suffice.

Primary Malignant Neoplasms

Malignant tumors in the liver can be primary or metastatic. Among primary epithelial tumors, the most common are hepatocellular carcinoma and intrahepatic cholangiocarcinoma. Hepatoblastoma is a rare hepatocellular tumor that occurs in children. Nonepithelial tumors, such as angiosarcoma, are exceedingly rare.

Hepatoblastoma

Hepatoblastoma is the most common liver tumor of early childhood. It rarely occurs over the age of 3 years, and its incidence is increasing. Tumors with overlapping features of hepatoblastoma and hepatocellular carcinoma can occur in older children and are currently referred to as hepatocellular neoplasm, not otherwise specified. Activation of the Wnt signaling pathway is characteristic of hepatoblastoma, leading to nuclear β -catenin and increased expression of Wnt target genes such as glutamine synthetase in nearly all cases. These tumors are associated with several syndromes, including *familial adenomatous polyposis* (caused by germline loss-of-function mutations in *APC*; Chapter 17) and *Beckwith-Wiedemann syndrome*. The latter is a disorder associated with congenital growth abnormalities caused by abnormal epigenetic imprinting of a region of chromosome 11 that leads to overexpression of insulin-like growth factor-2 and loss of expression of *CDKN1C*, a tumor suppressor gene that encodes the cyclin-dependent kinase inhibitor p57.

MORPHOLOGY

There are a variety of histologic subtypes of hepatoblastoma, which can be broadly grouped into two categories: (1) **epithelial type**, composed of small polygonal fetal cells or smaller embryonal cells forming trabeculae, acini, tubules, or papillary structures vaguely recapitulating the developing liver (Fig. 18.53); and (2) **mixed epithelial and mesenchymal type**, which contains additional foci of mesenchymal differentiation that may consist of primitive mesenchyme, osteoid, cartilage, or striated muscle.

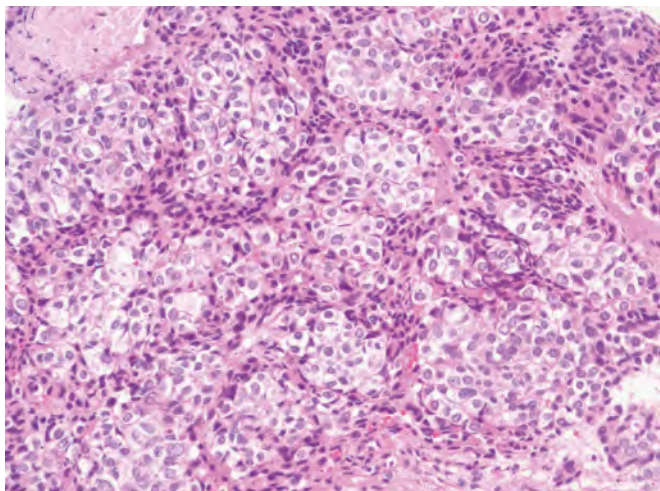


Fig. 18.53 Hepatoblastoma. The tumor cells are arranged in sheets and resemble embryonal and fetal hepatocytes.

Clinical Features

Hepatoblastoma usually comes to clinical attention due to abdominal swelling in an asymptomatic infant or child. Symptoms related to liver dysfunction (jaundice, pruritis) are seen in a subset of patients, and about 20% of tumors will have metastasized to the lungs by the time of diagnosis. The tumor is treated with surgical resection and chemotherapy, which has improved outcomes. The overall 5-year survival rate is approximately 80%.

Hepatocellular Carcinoma

Hepatocellular carcinoma (HCC) accounts for approximately 5.4% of all cancers worldwide and is one of the most common cancers in geographic regions with high rates of hepatitis B infection. More than 85% of cases occur in countries in Asia (southeast China, Korea, Taiwan) and sub-Saharan Africa, where chronic HBV infection is common. The peak incidence of HCC in these areas is in young adults between 20 and 40 years of age who acquired hepatitis B virus by maternal-fetal transmission. Encouragingly, the incidence of HCC is decreasing in Asia due to hepatitis B vaccination, but at the same time the incidence is increasing in Western countries owing to rising rates of metabolic syndrome. For unclear reasons, there is a pronounced male predominance, as high as 8:1, in high-incidence areas.

Pathogenesis

Most HCCs occur in the setting of chronic liver disease with cirrhosis, while 15% to 20% arise in noncirrhotic livers (Fig. 18.54). The most common underlying diseases are chronic viral hepatitis (B and C), metabolic diseases such as hereditary hemochromatosis and α_1 -antitrypsin deficiency, and alcoholic liver disease. MASLD also increases the risk of HCC, even in the absence of cirrhosis. It is believed that the chronic injury, inflammation, and hepatocyte regeneration that are seen in these disorders contribute to the acquisition of driver mutations that lead to HCC development. Part of the risk in Africa and Asia appears to be related to contamination of crops by *aflatoxin*, a mycotoxin produced by *Aspergillus* species that acts synergistically with alcohol and hepatitis B. The risk for HCC in cirrhosis related to other etiologies, like Wilson disease and chronic biliary diseases, is somewhat lower but still elevated above the population average.

As with other cancers, HCC is associated with complementary sets of driver mutations that lead to the acquisition of cancer hallmarks (Chapter 7). Among the most common are activating mutations in the β -catenin gene (40% of tumors), mutations in the *TERT* (telomerase transcriptase) gene promoter that up-regulate telomerase activity (50% to 60% of tumors), and inactivating mutations in *TP53* (up to 60% of tumors). One unusual histologic subtype that often occurs in adolescents and young adults in the absence of preexisting liver disease, *fibrolamellar HCC*, is strongly associated with formation of a *DNAJB1::PRKACA* fusion gene that leads to excessive activity of protein kinase A, an enzyme that participates in a signaling pathway regulated by cAMP.

MORPHOLOGY

Several precursor lesions for HCC have been described. In noncirrhotic liver, HCC can arise in hepatocellular adenoma, especially those with β -catenin-activating mutations. In chronic liver disease, the earliest morphologic alterations that appear to correlate with the presence of “at-risk” hepatocytes are called “large cell change” and “small cell change” (Fig. 18.55). **Large cell change** refers to hepatocytes that are larger than normal and often have enlarged, multiple, pleomorphic nuclei, without an increase in nuclear-to-cytoplasmic ratio (see Fig. 18.55A). In **small cell change**, the hepatocytes have a high nuclear-to-cytoplasmic ratio and mild nuclear hyperchromasia and/or pleomorphism (see Fig. 18.55B). More ominous are nodular lesions in

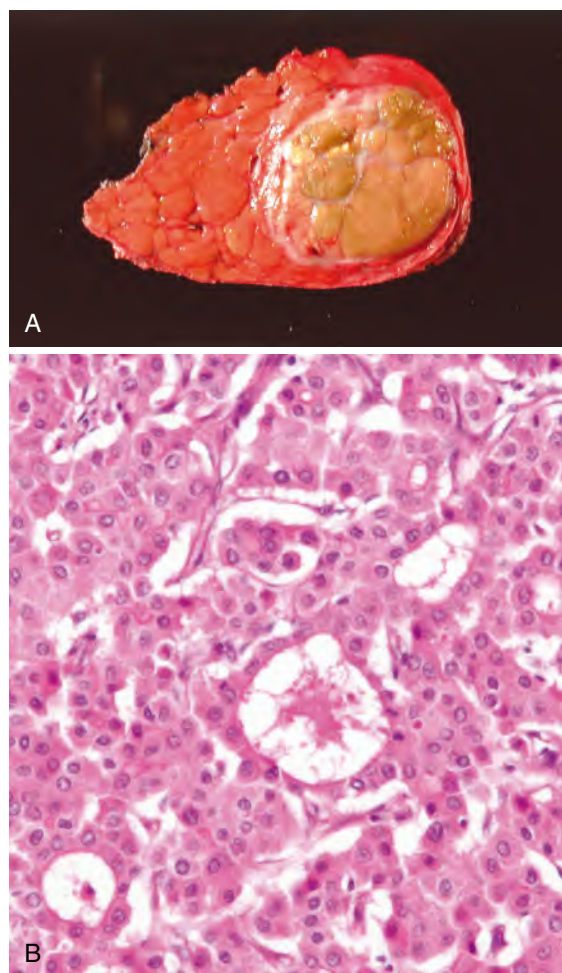


Fig. 18.54 Hepatocellular carcinoma. (A) Liver removed at autopsy showing a unifocal neoplasm replacing most of the right hepatic lobe. (B) Malignant hepatocytes growing in distorted versions of normal architecture, including large pseudoacinar spaces (malformed, dilated bile canaliculi) and thickened hepatocyte trabeculae.

cirrhotic liver referred to as **dysplastic nodules**, which are often associated with small cell change. Dysplastic nodules differ from adjacent cirrhotic nodules in size, color, and vascularization, show varying degrees of dysplasia, and have clonal aberrations associated with full-blown HCC. Small areas of HCC may sometimes be seen in high-grade dysplastic nodules ("nodule-in-nodule appearance") (Fig. 18.56).

HCC may form a single mass or multiple discrete masses, or it may diffusely infiltrate the liver. Masses may be pale and yellow due to fatty change or green due to cholestasis. Tumors larger than 2 cm are more likely to be associated with vascular invasion and intrahepatic metastases. Invasion of veins with extension into the portal vein, inferior vena cava, and even the right side of the heart may occur.

Microscopically, well and moderately differentiated HCC are composed of cells that resemble normal hepatocytes, while poorly differentiated tumors show marked cytologic atypia. The tumor cells grow in thick plates or trabeculae, pseudoglandular structures with bile plugs, or sheets (see Fig. 18.54B). The distinctive fibrolamellar variant shows a characteristic triad of features: large polygonal cells with granular (oncocyctic) cytoplasm due to abundant mitochondria; vesicular nuclei with a prominent nucleolus; and parallel lamellae of dense collagen bundles (Fig. 18.57).

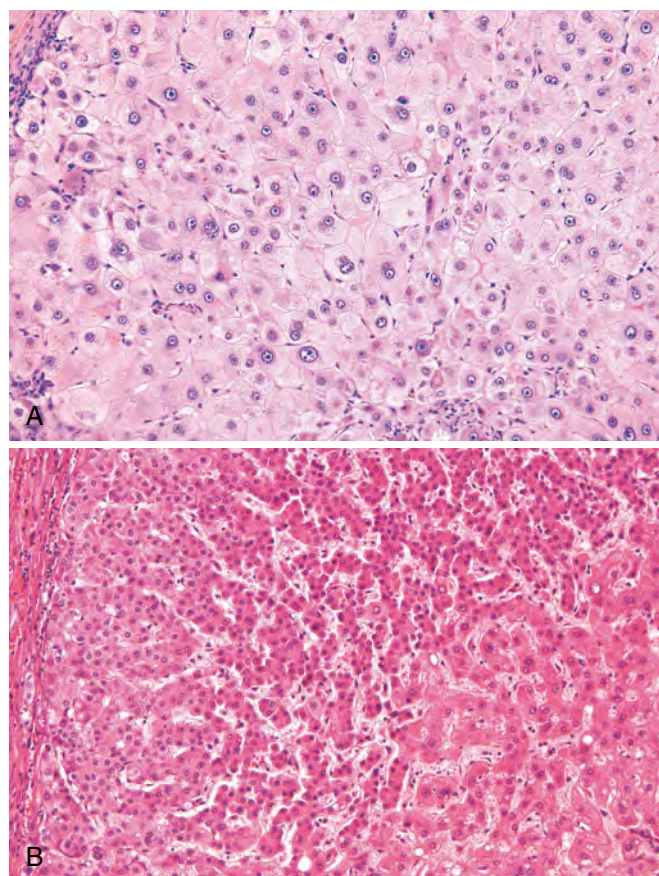


Fig. 18.55 Premalignant changes in hepatocytes. (A) Large cell change. Large hepatocytes with large, often atypical nuclei are scattered among normal-sized hepatocytes with round, typical nuclei. (B) Small cell change. The abnormal cells have a high nuclear-to-cytoplasmic ratio and are separated by thickened plates. Normal-appearing hepatocytes are in the lower-right corner. (Courtesy Dr. Young Nyun Park, Yonsei Medical College, Seoul, South Korea.)

Clinical Features

The clinical manifestations of HCC are nonspecific and include abdominal pain, malaise, fatigue, weight loss, and hepatomegaly. Features of underlying chronic liver disease can be present. Elevated levels of serum α -fetoprotein is a frequent finding in advanced disease, but it is not sensitive as a screening test for early tumors and is not observed in the fibrolamellar variant. Ultrasonography is used for screening high-risk patients such as those with cirrhosis. Contrast-enhanced computed tomography and magnetic resonance imaging shows early enhancement of the tumor due to contrast uptake in the arterial phase, followed by rapid venous washout. These findings are highly characteristic of HCC.

Surgical resection when possible is the treatment of choice for tumors in noncirrhotic livers and in cirrhotic livers with adequate function. Liver transplantation is considered for HCC in the setting of advanced cirrhosis. Image-guided tumor ablation with alcohol or radiofrequency waves can be done for unresectable tumors or those that do not meet criteria for transplantation. Hematogenous metastases, especially to the lung, tend to occur late in the disease. Lymph node metastases occur in <5% of cases.

Overall outcomes in HCC are poor due to underlying liver disease and the intrinsic resistance of HCC to conventional chemotherapy. The overall 5-year survival rate is 30% for tumor confined to the liver and

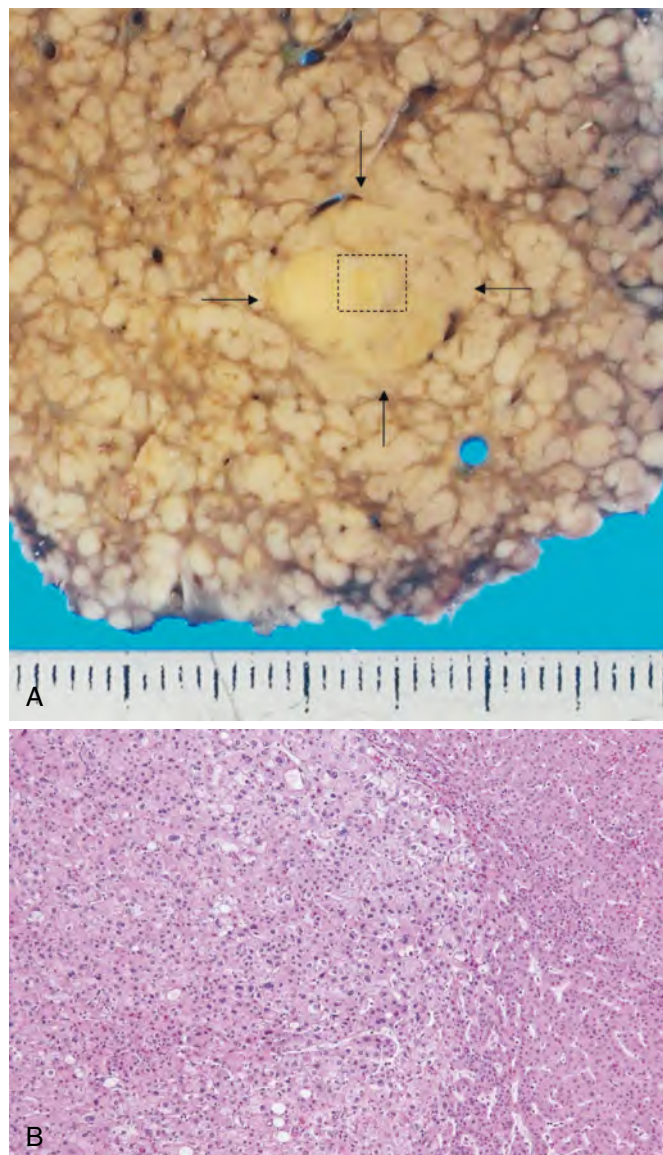


Fig. 18.56 (A) Hepatitis C–related cirrhosis with a distinctively large nodule (arrows). Nodule-in-nodule growth suggests an evolving cancer. (B) Histologically, the region within the box in A shows a well-differentiated hepatocellular carcinoma (HCC) (right side) and a subnodule of moderately differentiated HCC within it (center, left). (Courtesy Dr. Masamichi Kojiro, Kurume University, Kurume, Japan.)

only 5% to 10% for cases with extrahepatic spread. Outcomes are better for HCC in noncirrhotic liver and the fibrolamellar variant, with up to 40% of patients with the latter surviving 10 years or longer. This is in large part because, in the absence of underlying liver disease, extensive surgical resection is possible due to the remarkable regenerative capacity of the remaining liver.

Malignant Biliary Tumors

Adenocarcinomas arising from the intrahepatic biliary tree are referred to as *intrahepatic cholangiocarcinoma*, while similar tumors arising from the extrahepatic bile ducts are referred to as *extrahepatic cholangiocarcinoma* (*extrahepatic biliary adenocarcinoma*). Intrahepatic cholangiocarcinoma is the most common primary malignant tumor of the liver after HCC. Its incidence is rising in the United States. It accounts for 7.6% of cancer deaths worldwide and 3% of

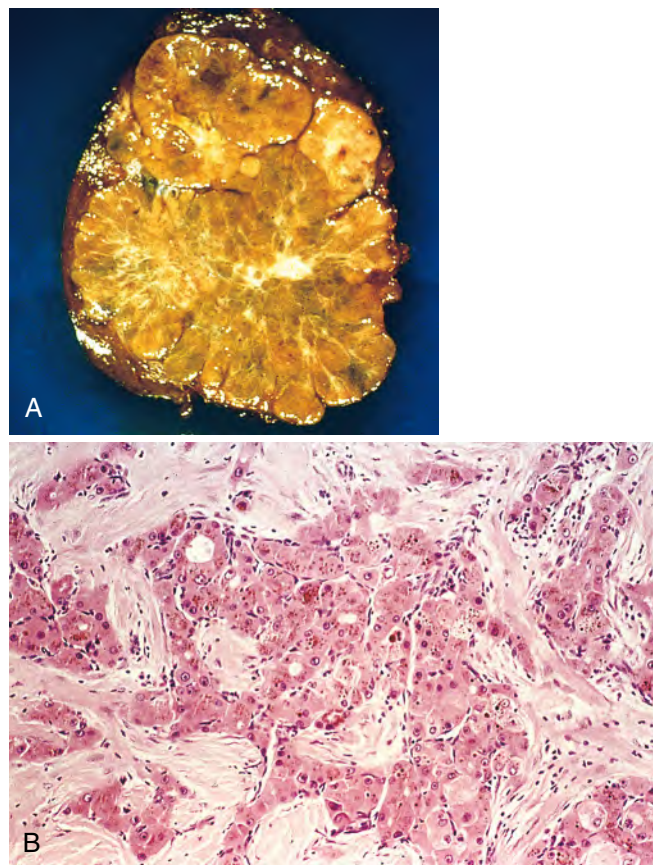


Fig. 18.57 Fibrolamellar carcinoma. (A) Resected specimen showing a well-demarcated nodule. (B) Microscopic view showing nests and cords of malignant-appearing, oncocytic hepatocytes separated by dense bundles of collagen.

cancer deaths in the United States. It is very common in Southeast Asian countries such as Thailand, Laos, and Cambodia, where liver fluke infestation is endemic.

Pathogenesis

Developmental disorders, like fibropolycystic liver disease, and chronic inflammatory conditions involving the bile ducts, including primary sclerosing cholangitis, infestation by liver flukes (particularly *Opisthorchis* and *Clonorchis* species), and hepatolithiasis are risk factors for biliary tract neoplasms. Chronic liver diseases that predispose to HCC, such as hepatitis B, hepatitis C, and MASLD, also increase the risk for intrahepatic cholangiocarcinoma. As with HCC, chronic injury, inflammation, and regeneration of biliary epithelium in these conditions may set the stage for acquisition of driver mutations leading to cancer.

Driver mutations in intrahepatic cholangiocarcinoma and extrahepatic biliary adenocarcinoma only partially overlap. *KRAS* mutations are common in both tumors. In addition, subsets of intrahepatic cholangiocarcinoma have distinctive driver mutations in *IDH1* and *IDH2* (isocitrate dehydrogenase) that generate “oncometabolites” (Chapter 7), and in chromatin modifier genes like *BAP1* and *PBRM1*. Fusion genes involving *FGFR2* (fibroblast growth factor receptor 2) are also common. By contrast, in addition to *KRAS* mutations, extrahepatic biliary adenocarcinomas are most likely to have mutations in *TP53* and *SMAD4*, genetic features that are similar to those of pancreatic adenocarcinoma.

MORPHOLOGY

Intrahepatic cholangiocarcinoma usually occurs in noncirrhotic liver (Fig. 18.58) and typically forms a firm gray-white mass. Extrahepatic biliary adenocarcinomas include perihilar tumors known as **Klatskin tumors**, which are located at the junction of the right and left hepatic ducts. These tumors comprise 60% to 70% of extrahepatic biliary adenocarcinomas, while the remaining 30% to 40% involve the common bile duct. Extrahepatic biliary adenocarcinomas are generally small at diagnosis, as they are detected early due to obstructive features. These tumors may form a firm nodule in the bile duct wall, may be diffusely infiltrative, or may form papillary lesions.

Microscopically, both intrahepatic and extrahepatic tumors show features of adenocarcinomas. Most are well to moderately differentiated and are arranged in clearly defined glandular/tubular structures lined by malignant epithelial cells embedded in an abundant fibrous stroma (see Fig. 18.58B). Lymphovascular and perineural invasion (see Fig. 18.58C) are common. These tumors are thought to arise from premalignant lesions termed biliary intraepithelial neoplasia. Other less common precursor lesions are mucinous cystic neoplasms and intraductal papillary neoplasms.

Clinical Features

Intrahepatic cholangiocarcinoma may be detected incidentally on imaging or may present with a cholestatic picture or symptomatic liver mass, whereas extrahepatic adenocarcinoma typically presents with symptoms related to biliary obstruction. Tumor resection with negative margins and absence of lymph node involvement are favorable factors. Lymph node metastasis is present in 50% to 60% of patients at presentation. Adjuvant chemotherapy is commonly given after resection, but tumor responses are usually transient. The overall prognosis is poor, as recurrences are common, and the 5-year survival rate is 20% to 40% after surgical resection.

Other Primary Hepatic Malignant Tumors

Several other cancers may arise within the liver. Among the most common or pathogenically interesting are:

- **Combined hepatocellular-cholangiocarcinoma** comprises components of both HCC and intrahepatic cholangiocarcinoma. These tumors often have the same risk factors as HCC.
- **Angiosarcoma** is a malignant vascular tumor that is historically associated with exposure to vinyl chloride, arsenic, or the imaging agent thorotrast (Chapters 9 and 11). This tumor has become rare with reduced exposures to these agents.
- **Epithelioid hemangioendothelioma** is another vascular tumor with intermediate malignant potential and a somewhat better prognosis than angiosarcoma, which is almost uniformly fatal.
- **Primary hepatic lymphoma** is a rare type of liver malignancy. The most common subtype is *diffuse large B-cell lymphoma*, a form of non-Hodgkin lymphoma that frequently occurs at extranodal sites (Chapter 13). Another rare subtype is *hepatosplenic T-cell lymphoma*, most common in young adult males, which has a predilection for growth within the sinusoids of the liver, spleen, and bone marrow.

Metastasis

Involvement of the liver by metastatic malignancy is far more common than primary hepatic neoplasia. Although the most common primary sources are the colon, breast, lung, and pancreas, virtually any primary tumor may spread to the liver. Typically, multiple nodular metastases are found that often cause striking hepatomegaly and replace much of the normal liver parenchyma. The liver weight can exceed several kilograms. Metastasis may also appear as a single nodule, in which case it may be

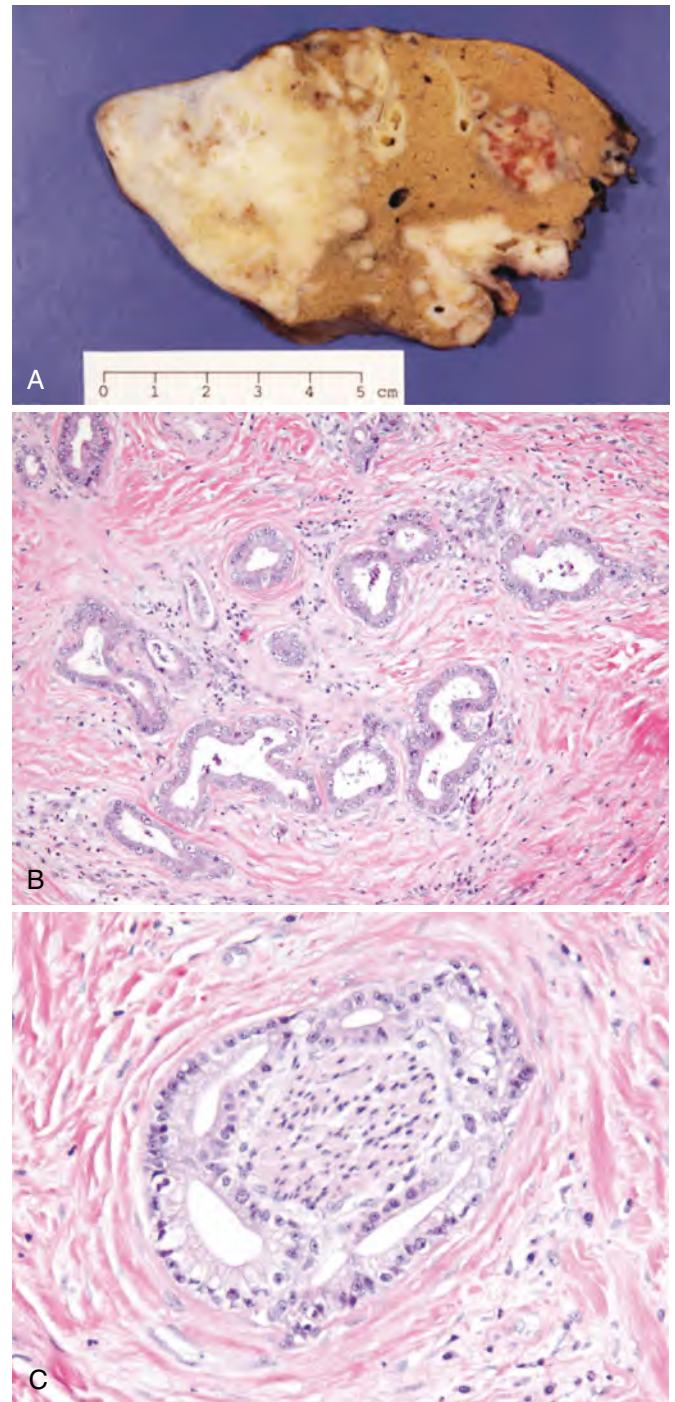


Fig. 18.58 Cholangiocarcinoma. (A) Multifocal cholangiocarcinoma in a liver from a patient with infestation by the liver fluke *Clonorchis sinensis*. (B) Invasive malignant glands in a reactive, sclerotic stroma. (C) Perineural invasion by malignant glands, forming a wreathlike pattern around the central, trapped nerve. (A, Courtesy Dr. Wilson M.S. Tsui, Caritas Medical Centre, Hong Kong.)

resected surgically. Always surprising is the extent of metastatic involvement that may be present in the absence of clinical or laboratory evidence of hepatic insufficiency; often the only telltale sign is hepatomegaly. However, with massive destruction of hepatocytes or direct obstruction of major bile ducts, jaundice and other evidence of liver dysfunction inevitably appears.

KEY CONCEPTS

Liver Tumors

- The liver is the most common site of metastatic cancers from primary tumors of the colon, lung, and breast.
- Hepatocellular adenomas are benign tumors of neoplastic hepatocytes, which can be subclassified based on different sets of driver mutations.
- HCC is the most common primary hepatic malignant neoplasm, often occurring in the setting of cirrhosis related to chronic liver disease such as chronic hepatitis B and C, autoimmune hepatitis, alcoholic/ metabolic dysfunction–associated steatotic liver disease, and hemochromatosis.
- Intrahepatic cholangiocarcinoma and extrahepatic biliary adenocarcinoma show similar histologic features and share risk factors like primary sclerosing cholangitis and infestation by liver flukes such as *Opisthorchis* and *Clonorchis*. These tumors have a poor prognosis.

GALLBLADDER

As much as 1 L of bile is secreted by the liver per day. Between meals, bile is stored in the gallbladder, where it is concentrated. The adult gallbladder has a capacity of about 50 mL. The organ is not essential for biliary function since humans do not experience indigestion or malabsorption of fat after cholecystectomy.

CONGENITAL ANOMALIES

The gallbladder may be congenitally absent, or there may be gallbladder duplication with conjoined or independent cystic ducts. A longitudinal or transverse septum may create a bilobed gallbladder. Aberrant locations of the gallbladder occur in 5% to 10% of the population, most commonly with partial or complete embedding in the liver substance. A folded fundus is the most common anomaly, creating a *phrygian cap* (Fig. 18.59). *Agensis* of all or any portion of the hepatic or common bile ducts and hypoplastic narrowing of biliary channels (true “biliary atresia”) may also occur. *Choledochal cysts*, described earlier, may be isolated findings in the gallbladder or associated with other cysts in the extrahepatic biliary tree or with fibropolycystic disease.

CHOLELITHIASIS (GALLSTONES)

More than 95% of biliary tract disease is attributable to gallstones. Gallstones afflict 10% to 20% of adult populations in high resource



Fig. 18.59 Phrygian cap of the gallbladder; the fundus is folded inward.

TABLE 18.11 Risk Factors for Gallstones

Cholesterol Stones

Demography: northern Europeans, North and South Americans, Native Americans
 Advancing age
 Female sex hormones
 Female sex
 Oral contraceptives
 Pregnancy
 Obesity and metabolic syndrome
 Rapid weight reduction
 Gallbladder stasis
 Inborn disorders of bile acid metabolism
 Hyperlipidemia syndromes

Pigment Stones

Demography: rural more than urban
 Chronic hemolytic anemias
 Biliary infection
 Gastrointestinal disorders: ileal disease (e.g., Crohn's disease), ileal resection or bypass, cystic fibrosis with pancreatic insufficiency

countries. It is estimated that more than 20 million persons in the United States have gallstones, totaling some 25 to 50 tons in weight, leading to more than 700,000 cholecystectomies performed annually at a cost of approximately \$6 billion.

Epidemiology

There are two general classes of gallstones: cholesterol stones, containing more than 50% of crystalline cholesterol monohydrate, and pigment stones composed predominantly of bilirubin calcium salts, each with different risk factors. Cholesterol gallstones are more prevalent in the United States and western Europe (90%) and uncommon in low-resource countries. The prevalence rates of cholesterol gallstones approach 75% in Native Americans of the Pima, Hopi, and Navajo groups, while pigment stones are rare in these populations. Pigment gallstones arise primarily in the setting of bacterial infections or parasitic infestations of the biliary tree, and as well as in individuals with diseases that lead to chronic red cell hemolysis.

The major risk factors associated with the development of gallstones are listed in Table 18.11 and are briefly described here:

- *Age and sex.* The prevalence of cholesterol gallstones increases throughout life, but they predominantly affect individuals of middle to older age. Prevalence is higher in females in any region or ethnicity. Hypersecretion of biliary cholesterol seems to play the major role in both age and sex differences. Significant associations are also seen with metabolic syndrome and obesity.
- *Environmental factors.* Estrogen exposure, including through oral contraceptive use and during pregnancy, increases expression of hepatic lipoprotein receptors and stimulates hepatic HMG-CoA reductase activity, enhancing both cholesterol uptake and biosynthesis, respectively. The net result is excess biliary secretion of cholesterol. Obesity and rapid weight loss are also strongly associated with increased biliary cholesterol secretion.
- *Acquired disorders.* Gallbladder stasis, either neurogenic or hormonal, fosters a local environment that is favorable for both cholesterol and pigment gallstone formation.
- *Hereditary factors.* Genes encoding hepatocyte proteins that transport biliary lipids, known as ATP-binding cassette (ABC) transporters, have associations with gallstone formation. In particular, a common variant of the sterol transporter encoded by the *ABCG8* gene is associated with an increased risk of cholesterol gallstones.

Pathogenesis of Cholesterol Stones

Cholesterol is rendered soluble in bile by forming micelles with bile salts and lecithins, both of which act as detergents. When cholesterol concentrations exceed the solubilizing capacity of bile (supersaturation), cholesterol can no longer remain dispersed and nucleates into solid cholesterol monohydrate crystals. Four conditions appear to contribute to formation of cholesterol gallstones: (1) supersaturation of bile with cholesterol; (2) hypomotility of the gallbladder; (3) accelerated cholesterol crystal nucleation; and (4) hypersecretion of mucus in the gallbladder, which traps the nucleated crystals, leading to accretion of more cholesterol and the appearance of macroscopic stones.

Pathogenesis of Pigment Stones

Pigment gallstones are complex mixtures of insoluble calcium salts of unconjugated bilirubin and inorganic calcium salts. Disorders that are associated with elevated levels of unconjugated bilirubin in bile increase the risk of developing pigment stones. These include chronic hemolytic anemia, severe ileal dysfunction or bypass, and bacterial contamination of the biliary tree. Unconjugated bilirubin is normally a minor component of bile, but it increases when infection of the biliary tract leads to release of microbial β -glucuronidases, which hydrolyze bilirubin glucuronides. Thus, infection of the biliary tract with *Escherichia coli*, *Ascaris lumbricoides*, or the liver fluke *C. sinensis* increases the likelihood of pigment stone formation. In hemolytic anemias, the secretion of conjugated bilirubin into bile increases. About 1% of bilirubin glucuronides are deconjugated in the biliary tree, and in the setting of chronically increased secretion of conjugated bilirubin, a sufficiently large amount of deconjugated bilirubin is generated to allow pigment stones to form.

MORPHOLOGY

Cholesterol stones arise exclusively in the gallbladder and range from 100% pure (which is rare) down to around 50% cholesterol. Pure cholesterol stones are pale yellow, round to ovoid, and have a finely granular, hard external surface (Fig. 18.60), which on transection reveals a glistening radiating crystalline palisade. With increasing proportions of calcium carbonate, phosphates, and bilirubin, the stones take on a gray-white to black color and may be lamellated. Multiple stones are usually present that range up to several centimeters in diameter. Rarely, a very large stone may virtually fill the fundus. Surfaces of stones may be rounded or faceted because of tight apposition to adjacent stones. Stones composed largely of cholesterol are radiolucent; sufficient calcium carbonate is found in 10% to 20% of cholesterol stones to render them radiopaque. The supersaturated cholesterol in bile that forms cholesterol stones may also diffuse into the mucosa and manifest as *cholesterolosis* (Fig. 18.61).

Pigment gallstones are brown to black. In general, black pigment stones are found in sterile gallbladder bile, and brown stones are found in infected large bile ducts. Black stones contain oxidized polymers of calcium salts of unconjugated bilirubin, small amounts of calcium carbonate, calcium phosphate, mucin glycoprotein, and some cholesterol monohydrate crystals. Brown stones contain similar compounds along with some cholesterol and calcium salts of palmitate and stearate. Black stones are rarely greater than 1.5 cm in diameter, are almost invariably present in great number (with an inverse relationship between size and number; Fig. 18.62) and are quite friable. Their contours are usually spiculated and molded. Brown stones tend to be laminated and soft and may have a soap-like or greasy consistency. Approximately 50% to 75% of black stones are radiopaque due to calcium salts, while brown stones, which contain calcium soaps, are radiolucent. Mucin glycoproteins constitute the scaffolding and interparticle cement of all types of stones.



Fig. 18.60 Cholesterol gallstones. The wall of the gallbladder is thickened and fibrotic due to chronic cholecystitis.

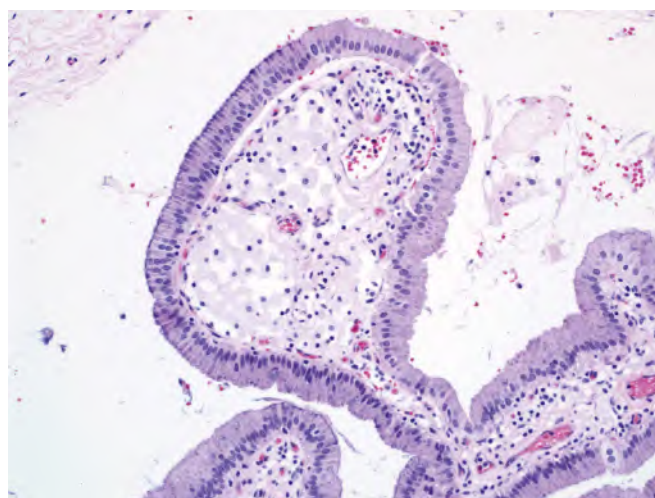


Fig. 18.61 Cholesterolosis. Gallbladder mucosa demonstrates lamina propria distended by foamy macrophages.



Fig. 18.62 Pigment gallstones. Several faceted black gallstones are present in this otherwise unremarkable gallbladder from a patient with a mechanical mitral valve prosthesis, leading to chronic intravascular hemolysis.

Clinical Features

Gallstones may be present for decades before symptoms develop, and 70% to 80% of patients remain asymptomatic throughout their lives. Asymptomatic individuals convert to being symptomatic at an average rate of up to 4% per year, although the risk diminishes with time. Symptomatic patients with biliary “colic” experience excruciating pain, although this name is a misnomer, as the pain is usually constant, not colicky. Pain often follows a fatty meal that induces gallbladder contraction, which presses a stone against the gall bladder outlet, leading to increased pressure and eventually pain. The pain is localized to the right upper quadrant or epigastrium and may radiate to the right shoulder or the back. Inflammation of the gallbladder (cholecystitis, discussed later) in association with stones also generates pain. More severe complications include empyema, perforation, fistula, inflammation of the biliary tree (cholangitis), obstructive cholestasis, and pancreatitis. The larger the calculi, the less likely they are to enter the cystic or common ducts to produce obstruction; it is very small stones, or “gravel,” that are most dangerous. Occasionally a large stone may erode directly into an adjacent loop of small bowel, generating intestinal obstruction (“gallstone ileus” or *Bouveret syndrome*). Gallstones also are associated with an increased risk of gallbladder carcinoma (discussed later).

CHOLECYSTITIS

Inflammation of the gallbladder may be acute, chronic, or acute superimposed on chronic. It almost always occurs in association with gallstones. In the United States, cholecystitis is one of the most common indications for abdominal surgery. Its epidemiology closely parallels that of gallstones.

Acute Cholecystitis

Acute cholecystitis is precipitated in 90% of cases by obstruction of the neck of the gallbladder or the cystic duct by a stone. Acute calculous cholecystitis is the primary complication of gallstones and the most common reason for emergency cholecystectomy. Cholecystitis without gallstones (*acalculous cholecystitis*) may also occur in severely ill patients and accounts for the remaining 10% of cases.

Pathogenesis

Acute calculous cholecystitis results from chemical irritation and inflammation of a gallbladder obstructed by stones. The action of mucosal phospholipases hydrolyzes luminal lecithins to toxic lysolecithins. The normally protective glycoprotein mucus layer is disrupted, exposing the mucosal epithelium to the direct detergent action of bile salts. Prostaglandins released within the wall of the distended gallbladder contribute to mucosal and mural inflammation; distention and increased intraluminal pressure compromise blood flow to the mucosa. These events initially occur in the absence of bacterial infection, but later in the course bacterial infection may be superimposed and exacerbate the inflammatory process. Acute calculous cholecystitis is particularly common in diabetic patients who have symptomatic gallstones.

Acute acalculous cholecystitis is thought to result from ischemia.

The cystic artery is an end artery without a collateral circulation. Contributing factors may include inflammation and edema of the wall (compromising blood flow) and gallbladder stasis due to accumulation of microcrystals of cholesterol (biliary sludge), viscous bile, and mucus, causing cystic duct obstruction in the absence of stones. It usually occurs in acutely ill patients who are hospitalized for unrelated conditions. Risk factors for acute acalculous cholecystitis include (1) sepsis with hypotension and multisystem organ failure; (2) immunosuppression; (3) major trauma and burns; (4) diabetes mellitus; and (5) infections.

MORPHOLOGY

In **acute cholecystitis** the gallbladder is usually enlarged and tense, and it may assume a bright red or blotchy, violaceous to green-black discoloration, imparted by subserosal hemorrhages. The serosa is frequently covered by a fibrinous exudate that may be fibrinopurulent in severe cases. There are no specific morphologic differences between acute acalculous and calculous cholecystitis, save the absence of stones in the acalculous form. In **calculous cholecystitis**, an obstructing stone is usually present in the neck of the gallbladder or the cystic duct. The gallbladder lumen contains one or more stones and is filled with cloudy or turbid bile mixed with fibrin, pus, and hemorrhage. When the exudate is virtually pure pus, the condition is referred to as **gallbladder empyema**. In mild cases, the gallbladder wall is thickened, edematous, and hyperemic. In more severe cases, it is transformed into a green-black necrotic organ, termed **gangrenous cholecystitis**, with small-to-large perforations. The invasion of gas-forming organisms, notably clostridia and coliforms, may cause an **acute “emphysematous” cholecystitis**. Histologically, early changes of acute cholecystitis include edema, congestion, and mucosal erosion. Neutrophils are typically sparse, unless there is superimposed infection.

Clinical Features

Individuals with acute calculous cholecystitis usually (but not always) have experienced previous episodes of pain related to the gallbladder. An attack begins with progressive right upper quadrant or epigastric pain that lasts for more than 6 hours. It is frequently associated with mild fever, anorexia, tachycardia, sweating, nausea, and vomiting. Most patients are free of jaundice; the presence of hyperbilirubinemia suggests obstruction of the common bile duct. Mild to moderate leukocytosis may be accompanied by modestly elevated serum alkaline phosphatase. Acute calculous cholecystitis may appear with remarkable suddenness and constitute an acute surgical emergency, or it may present with mild symptoms that resolve without medical intervention. In the absence of medical attention, the attack usually subsides in 7 to 10 days and frequently within 24 hours. However, as many as 25% of patients develop progressively more severe symptoms and require immediate surgical intervention. Recurrence is common in patients who recover without surgery.

Clinical symptoms of acute acalculous cholecystitis tend to be more insidious, since they are obscured by the underlying conditions precipitating the attack. A higher proportion of patients have no symptoms referable to the gallbladder; diagnosis therefore rests on a high index of suspicion. In the severely ill patient, early recognition of the condition is crucial since failure to do so almost ensures a fatal outcome. As a result of either delay in diagnosis or the disease itself, the incidence of gangrene and perforation is much higher in acalculous cholecystitis than in calculous cholecystitis. In rare instances, primary bacterial infection by agents such as *S. typhi* and staphylococci can give rise to acute acalculous cholecystitis. A more indolent form of acute acalculous cholecystitis may occur in the setting of systemic vasculitis, severe atherosclerotic ischemic disease in the elderly, AIDS (usually related to *Cryptosporidium* infection), or ascending biliary tract infection.

Chronic Cholecystitis

Chronic cholecystitis may be a sequel to repeated bouts of mild to severe acute cholecystitis, but in many instances it develops in the apparent absence of antecedent acute attacks. Since it is associated with cholelithiasis in more than 90% of cases, the at-risk patient population is the same as that for gallstones. The evolution of chronic cholecystitis is obscure; it is not clear that gallstones play a direct role in the initiation of inflammation or the development of pain, particularly since chronic acalculous cholecystitis shows symptoms and histology

similar to those of the calculous form. Rather, supersaturation of bile predisposes to both chronic inflammation and, in most instances, stone formation. Microorganisms, usually *E. coli* and enterococci, are cultured from the bile in about one-third of cases. Unlike acute calculous cholecystitis, obstruction of gallbladder outflow is not a prerequisite.

MORPHOLOGY

The morphologic changes in chronic cholecystitis are extremely variable and sometimes minimal. The serosa is usually smooth and glistening but may be dulled by **subserosal fibrosis**. Dense fibrous adhesions may be present that represent the sequelae of prior acute inflammation. On sectioning, the wall is variably thickened and has an opaque gray-white appearance. In uncomplicated cases, the lumen contains green-yellow, mucoid bile and usually stones. The mucosa itself is generally preserved.

On histologic examination, the degree of inflammation is variable. In the mildest cases, only scattered lymphocytes, plasma cells, and macrophages are found in the mucosa and in the subserosal fibrous tissue (Fig. 18.63A). In more advanced cases, there is marked subepithelial and subserosal fibrosis, accompanied by mononuclear cell infiltration. Reactive proliferation of the mucosa and fusion of the mucosal folds may give rise to buried crypts of epithelium within the gallbladder wall. Outpouchings of the mucosal epithelium through the wall (**Rokitansky-Aschoff sinuses**) may be quite prominent (Fig. 18.63B).

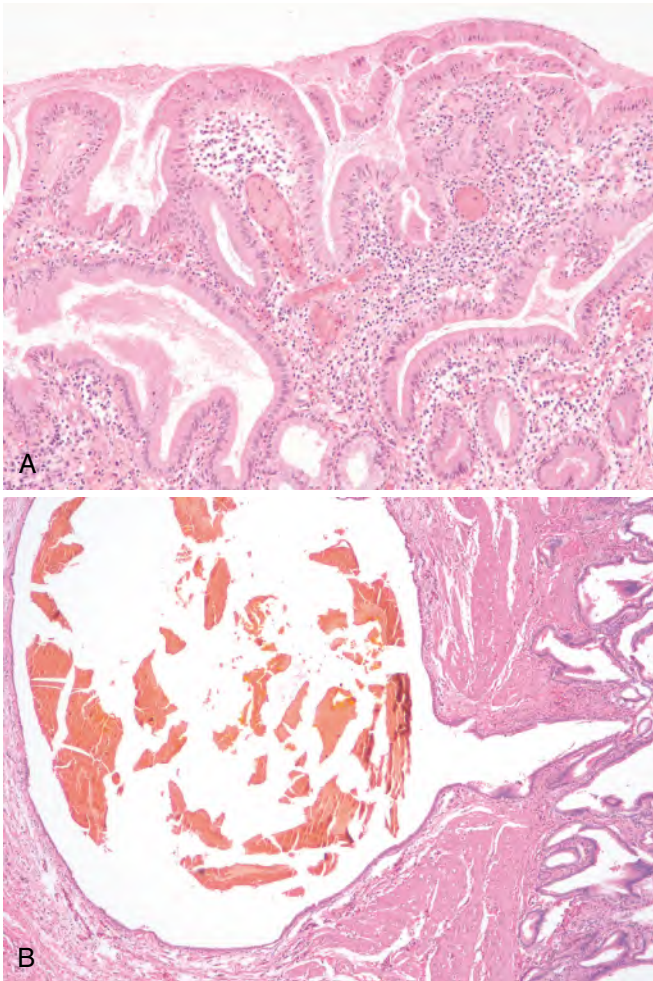


Fig. 18.63 Chronic cholecystitis. (A) The gallbladder mucosa is infiltrated by inflammatory cells. (B) Outpouching of the mucosa through the wall forms Rokitansky-Aschoff sinus (contains bile).

In rare instances, there is extensive dystrophic calcification of the gallbladder wall (**porcelain gallbladder**). More often, there is complete replacement of the gallbladder wall and mucosa by dense fibrosis (**hyalinizing cholecystitis**), with or without calcification. This change is notable for being associated with an increased incidence of gallbladder carcinoma (discussed later). In **xanthogranulomatous cholecystitis**, the gallbladder has a massively thickened wall and is shrunken, nodular, and chronically inflamed with foci of necrosis and hemorrhage. It is triggered by rupture of Rokitansky-Aschoff sinuses into the wall of the gallbladder followed by an accumulation of macrophages that have ingested biliary phospholipids. Such lipid-containing cells with foamy cytoplasm are called xanthoma cells, hence the name of this condition. Finally, an atrophic, chronically obstructed, often dilated gallbladder may contain only clear secretions, a condition known as **hydrops of the gallbladder**. Other rare forms of chronic cholecystitis include IgG4 related sclerosing cholecystitis, another manifestation of IgG4 related fibrosing disease.

Clinical Features

Chronic cholecystitis does not have the striking manifestations of the acute forms and is usually characterized by recurrent attacks of either steady epigastric or right upper quadrant pain. Nausea, vomiting, and intolerance for fatty foods are frequent accompaniments.

Diagnosis of both acute and chronic cholecystitis is important because of the following complications:

- *Bacterial superinfection* with cholangitis or sepsis
- *Gallbladder perforation* and local abscess formation
- *Gallbladder rupture* with diffuse peritonitis
- *Biliary enteric (cholecystenteric) fistula*, with drainage of bile into adjacent organs, entry of air and bacteria into the biliary tree, and potentially, gallstone-induced intestinal obstruction (ileus)
- *Aggravation of preexisting medical illness*, with cardiac, pulmonary, renal, or liver decompensation

GALLBLADDER CARCINOMA

Carcinoma of the gallbladder is the most common malignancy of the extrahepatic biliary tract. Approximately 6000 new cases of gallbladder cancer are diagnosed each year in the United States. There are wide variations in the incidence of gallbladder cancer worldwide, with regions such as Chile, Bolivia, and Northern India harboring the highest numbers of cases. Even within the United States, areas with large Native American or Hispanic populations, such as the southwest, have a higher incidence of gallbladder cancer than the rest of the country. Gallbladder cancer is at least twice as common in females than in males; this sex disparity can be several-fold greater in regions of highest incidence. The overwhelming majority of patients are diagnosed at an advanced, surgically unresectable, stage, and the mean 5-year survival rate for affected patients is less than 10%.

Pathogenesis

The most important risk factor for gallbladder cancer (besides sex and ethnicity) is gallstones, which are present in 95% of cases. However, it should be noted that only 1% to 2% of patients with gallstones develop gallbladder cancer. In Asia, chronic bacterial or parasitic infections have been implicated as risk factors, and the coexistence of gallstones with gallbladder cancer is much lower. Nonetheless, the common thread that ties gallstones or chronic infections together with gallbladder cancer is chronic inflammation. Relatively common driver mutations include gain-of-function aberrations affecting members of the EGF receptor gene family (including *HER2*) and genes encoding downstream signaling components such as *RAS*, and loss-of-function mutations in the *TP53* tumor suppressor gene, which are present in up

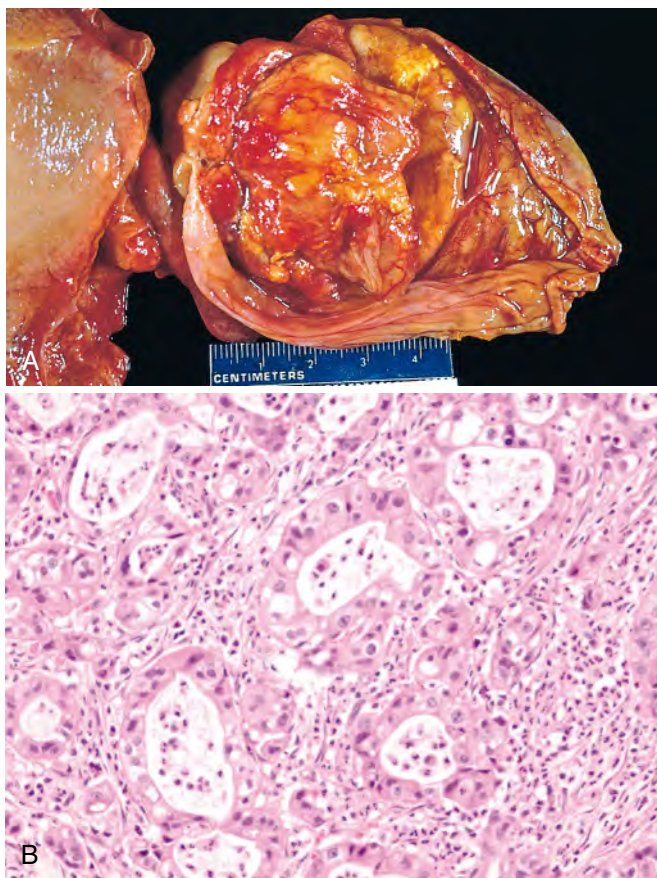


Fig. 18.64 Gallbladder adenocarcinoma. (A) The opened gallbladder contains a large, exophytic tumor that virtually fills the lumen. (B) Malignant glands are seen infiltrating a densely fibrotic gallbladder wall.

to one-half of tumors. As is typical of *TP53*-mutated cancers, most gallbladder carcinomas exhibit aneuploidy.

MORPHOLOGY

Several precursor lesions of gallbladder carcinoma have been described, indicating that (as is typical of carcinoma generally) full-blown cancer likely arises from a lengthy stepwise process. These precursors include flat in situ lesions with varying degrees of dysplasia, mass-forming adenoma-like lesions termed intracholecystic papillary tubular neoplasm, and intestinal metaplasia. Gallbladder cancers are mainly adenocarcinomas and are most often detected in the fundus (Fig. 18.64). They may produce a firm, poorly circumscribed mass or may diffusely infiltrate the gallbladder wall, simulating the gross appearance of chronic cholecystitis. Microscopically, they are usually characterized by the presence of glands embedded in desmoplastic stroma, but in some cases cytologic atypia and stromal response are minimal; in these instances, identification of perineural and vascular invasion help establish the diagnosis. With progression, direct extension into other organs, fistula formation, peritoneal and biliary spread, and metastasis to the liver and portahepatic lymph nodes occur. Rarely, other types of malignant tumors arise primarily within the gallbladder, including neuroendocrine tumors, squamous cell carcinoma, and sarcomas.

Clinical Features

Preoperative diagnosis of carcinoma of the gallbladder is the exception rather than the rule, occurring in fewer than 20% of patients. Presenting

symptoms are insidious and typically indistinguishable from those associated with cholelithiasis, including abdominal pain, jaundice, anorexia, and nausea and vomiting. If the tumor is detected prior to invasion and spread, such as when carcinoma is an incidental finding during surgery for symptomatic gallstones or acute cholecystitis, cure by surgical resection is possible. However, most patients have advanced disease at diagnosis, and the overall prognosis for such patients is poor, even with apparently complete resection of evident tumor. Adjuvant chemotherapy is often offered to such patients but is not curative, and most patients eventually succumb to their disease.

KEY CONCEPTS

Diseases of the Gallbladder

- Gallbladder diseases include cholelithiasis, acute and chronic cholecystitis, and gallbladder cancer.
- Gallstones are common in Western countries. The great majority are cholesterol stones. Pigmented stones containing bilirubin and calcium are most common in Asian countries.
- Risk factors for the development of cholesterol stones are advancing age, female sex, estrogen use, obesity, and heredity.
- Cholecystitis almost always occurs in association with cholelithiasis, although in about 10% of cases it occurs in the absence of gallstones. Gallstones are also a risk factor for gallbladder cancer.
- Acute calculous cholecystitis is the most common reason for emergency cholecystectomy.
- Gallbladder cancers are associated with gallstones in most cases. Typically, they are detected late because of nonspecific symptoms and hence carry a poor prognosis.

SELECTED READINGS

- Bernal W, Wendon J. Acute liver failure. *N Engl J Med*. 2013;369:2525. [An excellent clinical review.]
- Chun LJ, Tong MJ, Busuttil RW, et al. Acetaminophen hepatotoxicity and acute liver failure. *J Clin Gastroenterol*. 2009;43:342. [About the most common cause of acute liver failure leading to transplantation.]
- Ghany MG, Morgan TR, AASLD-IDS A Hepatitis C Guidance Panel. Hepatitis C Guidance 2019 Update: American Association for the Study of Liver Diseases-Infectious Diseases Society of America Recommendations for Testing, Managing, and Treating Hepatitis C Virus Infection. *Hepatology*. 2020;71(2):686-721. [Review of diagnosis, screening, and treatment guidelines for hepatitis C.]
- Gill RM, Theise ND, Rappaport, Glisson, Hering, and Mall-champions of liver microanatomy: microscopic and ultramicroscopic anatomy of the liver into the modern age. *Clin Liver Dis (Hoboken)*. 2021;18(Suppl 1):76. [A review of liver microarchitecture which discusses historical contributions by Frances Glisson, Ewald Hering, Franklin Mall, and Aron Rappaport as well as recent refinements to their discoveries.]
- Gouw ASW, Clouston AD, Theise ND. Ductular reactions in human livers: diversity at the interface. *Hepatology*. 2011;54:1853. [A review of ductular reactions, the stem cell response of human livers in all liver diseases, that are related to mechanisms of regeneration, fibrogenesis, and neoplasia.]
- Guido M, Alves VAF, Balabaud C et al: Histology of portal vascular changes associated with idiopathic non-cirrhotic portal hypertension: nomenclature and definition, *Histopathology* 74(2):219–226, 2019. [Provides a review of the spectrum of histologic changes and a unifying nomenclature.]
- Kleiner DE: Drug-induced liver injury: the hepatic pathologist's approach, *Gastroenterol Clin North Am* 46(2):273–296, 2017. [Review of the manner in which morphologic changes are evaluated in the setting of suspected drug-induced liver injury.]
- Kocabayoglu P, Friedman SL. Cellular basis of hepatic fibrosis and its role in inflammation and cancer. *Front Biosci (Schol Ed)*. 2013;5:217. [Interweaving

- what is known about hepatic stellate cells and other myofibroblastic cells of the liver with inflammatory, fibrosing, and neoplastic disease processes.]*
- Koyama Y, Brenner DA. Liver inflammation and fibrosis. *J Clin Invest.* 2017; 127(1):55-64. [A recent review of the complex interplay between the immune system and hepatic fibrosis.]
- Lamontagne RJ, Bagga S, Bouchard MJ: Hepatitis B virus molecular biology and pathogenesis, *Hepatoma Res* 2:163-186, 2016. [Review of hepatitis B virus infection.]
- Lohse AW, Sebode M, Bhathal PS, et al. Consensus recommendations for histological criteria of autoimmune hepatitis from the International AIH Pathology Group: Results of a workshop on AIH histology hosted by the European Reference Network on Hepatological Diseases and the European Society of Pathology: Results of a workshop on AIH histology hosted by the European Reference Network on Hepatological Diseases and the European Society of Pathology. *Liver Int.* 2022;42(5):1058-1069. [Updated criteria for diagnosis of autoimmune hepatitis.]
- Perlmutter DH, Silverman GA: Hepatic fibrosis and carcinogenesis in α_1 -antitrypsin deficiency: a prototype for chronic tissue damage in gain-of-function disorders, *Cold Spring Harb Perspect Biol* 1:3, 2011. [A broader perspective on some chronic hereditary diseases using α_1 -antitrypsin as the paradigmatic example.]
- Pietrangelo A: Genetics, genetic testing, and management of hemochromatosis: 15 years since hepcidin, *Gastroenterology* 149(5):1240-1251, 2015. [Comprehensive review of genetics of hemochromatosis with emphasis on hepcidin in mechanism of disease.]
- Quaglia A, Alves VA, Balabaud C, et al. International Liver Pathology Study Group. Role of aetiology in the progression, regression, and parenchymal remodelling of liver disease: implications for liver biopsy interpretation. *Histopathology.* 2016;68(7):953-967. [This paper challenges pathologists to evolve past use of the term cirrhosis to communicate more useful information about fibrosis in relation to disease etiology.]
- Rinella ME, Lazarus J, Ratziu V, et al. A multi-society Delphi consensus statement on new fatty liver disease nomenclature. *Hepatology.* 2023;78(6): 1966-1986. [This paper details the process and rationale behind the international consensus to change NAFLD terminology.]
- Singal AG, Llovet JM, Yarchoan M, et al. AASLD Practice Guidance on prevention, diagnosis, and treatment of hepatocellular carcinoma. *Hepatology.* 2023;78(6):1922-1965. [Recent guidelines focused on HCC]
- Umetsu SE, Kakar S. Evaluating liver biopsies with well-differentiated hepatocellular lesions. *Surg Pathol Clin.* 2023;16(3):581-598. [Review of pathologic and molecular aspects of adenomas and atypical tumors with borderline HCC features.]
- Wanless IR, Nakashima E, Sherman M. Regression of human cirrhosis. Morphologic features and the genesis of incomplete septal cirrhosis. *Arch Pathol Lab Med.* 2000;124:1599. [A revolutionary paper that began the reassessment of whether fibrosis/cirrhosis could regress; take a look also, in the same volume of the journal, at the critiques by the original reviewers of the paper (published despite their objections) and the authors' response to these. Pathology history happening before your eyes.]